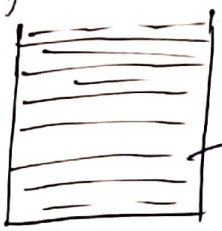
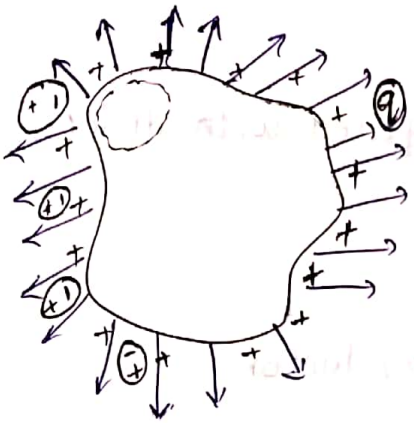
Capacitors

28/04/25 | Monday

* Capacitor is a device used to store charge and energy.



* capacity depends on volume of container but not on quanity of liquid / liquid level.



Break down strength – max value of electric field which can break molecules into ions.

$E_{max} = 3 \times 10^6$ v/m (for air)

* we have,

$q \propto V$

$q = CV$

water ↔ charge
coatellevel ↔ potential

C = capacitance of a conductor

$C = \frac{q}{V}$ *

* It is scalar quantity.

* unit → C/volt (or) Faraday (F).

$1F = \frac{1C}{1V}$

D.F → $[M^{-1} L^{-2} T^4 I^2]$

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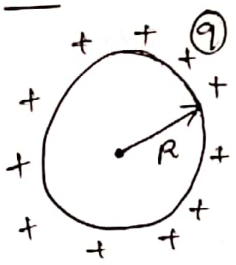
Capacity of a spherical conductor:-



where, $V = \frac{q}{4\pi\varepsilon_0 R}$

$$C = \frac{q}{V}$$

$$\boxed{C = 4\pi\varepsilon_0 R}$$

* If surrounding medium is filled with dielectric (K)

then $\boxed{C = 4\pi\varepsilon_0 K R}$

* C depends on 1) size of conductor

2) surrounding medium.

Points:

① For $C = 1F \Rightarrow 1 = 4\pi\varepsilon_0 R$.

$$R = \frac{1}{4\pi\varepsilon_0}$$

$$\boxed{R = 9 \times 10^9 m}$$

* $1\mu F \longrightarrow 10^{-6} F$
* $1mF \longrightarrow 10^{-3} F$
* $1nF \longrightarrow 10^{-9} F$
* $1pF \longrightarrow 10^{-12} F$

② For earth,

$$R = 6400 km$$

$$= 6.4 \times 10^6 m$$

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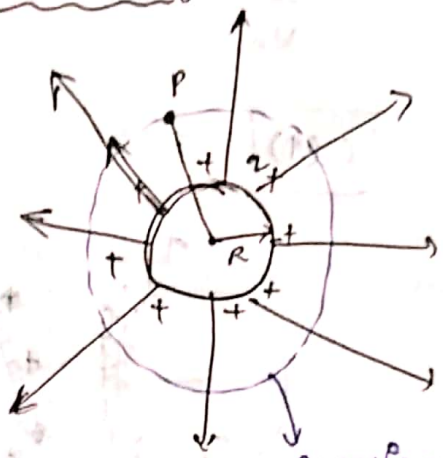
$C = 4\pi\varepsilon_0 R$

$$C = \frac{6.4\times10^6}{9\times10^9}$$

$$\boxed{C = 711\,\mu F}$$

③ Energy stored in a conductor:



Energy stored,

$$U = \int_{r=R}^{r=\infty} \frac{1}{2}\varepsilon_0 E^2 . dv$$

$$= \int_{r=R}^{r=\infty} \frac{1}{2}\varepsilon_0 \frac{q^2}{(4\pi\varepsilon_0)^2 r^4}(4\pi r^2).dr$$

$$E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

$$\boxed{E = \frac{q}{4\pi\varepsilon_0 r^2}}$$

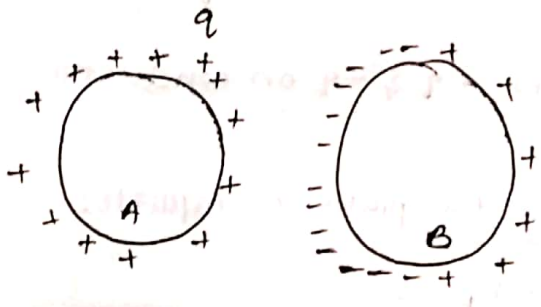
$$U = \frac{q^2}{8\pi\varepsilon_0 R} = \frac{q^2}{2\times4\pi\varepsilon_0 R}$$

$$\boxed{U = \frac{q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}qV}$$ *

Capacitor:

* For Isolated conductor, $C = \frac{q}{V}$

*



now, potential of conductor B,

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$$V_A = V - |V_-| + |V_+|$$

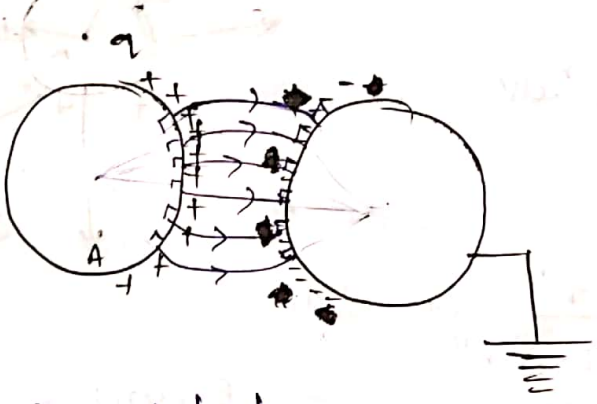
but $|V_{-ve}| > |V_{+ve}|$

$\Rightarrow V_A$ (↓)

$\therefore \quad C = \frac{q}{V_A}$

$\boxed{C\ (\uparrow)}$

* now,



Here,

$V_A = V - |V_-|$

(↓)

$\boxed{C = \frac{q}{V_A}\ (\uparrow)}$

finally, $\boxed{C = q/\Delta V}$

* symbol of capacitor- —||— or) —|(—

Calculation of Capacity of Capacitor

Step-1: Assign charges $+q$ & $-q$ on conductors.

Step-2: Find ΔV b/w the plates of capacitor.

M-1 $\quad \Delta V = V_+ - V_-$

V_+, V_- are absolute potentials of both

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the conductors.

M-II :

$$\int_{V_i}^{V_f} dV = \int_{i}^{f} \bar{E} \cdot d\bar{r}$$

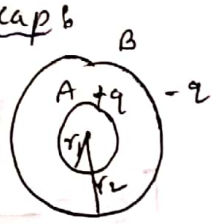
use gauss law to find E.

Step-3 :

Finally,

$$C = \frac{q}{\Delta V}$$

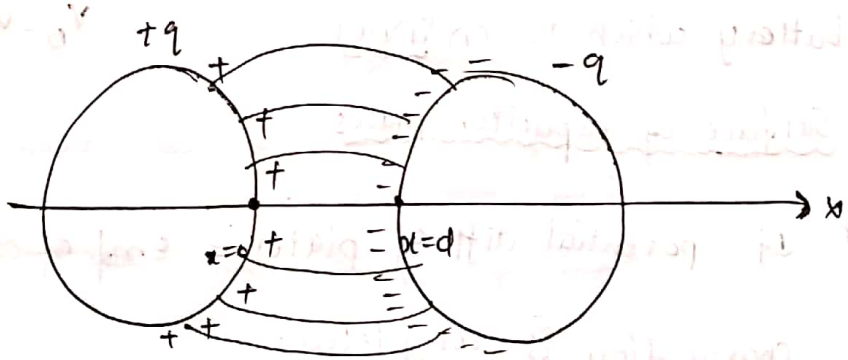
Recap :



$$V_A = \frac{kq}{R_1} - \frac{kq}{R_2}$$

$$V_B = \frac{kq}{R_2} - \frac{kq}{R_2}$$

Ex: Two spherical conductors carry equal and opposite charges. A non-uniform electric field is produced b/w the spheres as shown is given as $E = \frac{q}{\varepsilon_0 A}(1+Bx^2)$. Find capacity of capacitor so formed with these conductors.



Sol:

$$\int_{V_-}^{V_+} dV = -\int_{x=d}^{x=0} E \cdot d\bar{x}$$

$$V_+ - V_- = \int_{x=0}^{x=d} E \cdot d\bar{x}$$

$$= \frac{q}{\varepsilon_0 A}\left[x + \frac{B}{3}x^3\right]_0^d$$

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$$V_+ - V_- = \frac{q_0}{\varepsilon_0 A}\left(d + \frac{Bd^3}{3}\right)$$

$$C = \frac{q}{V_+ - V_-}$$

$$= \frac{q}{\frac{q}{\varepsilon_0 A}\left(d + \frac{Bd^3}{3}\right)}$$

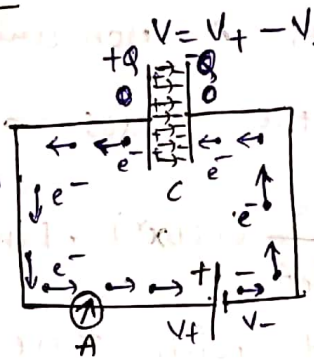
$$\boxed{C = \frac{\varepsilon_0 A}{d + \frac{Bd^3}{3}}}$$

Charging a capacitor :-

a) Initially plates are neutral :

i.e, $q = 0$ (Initially)

$\Rightarrow$ Potential diff b/w plates $= 0$ (Initially)



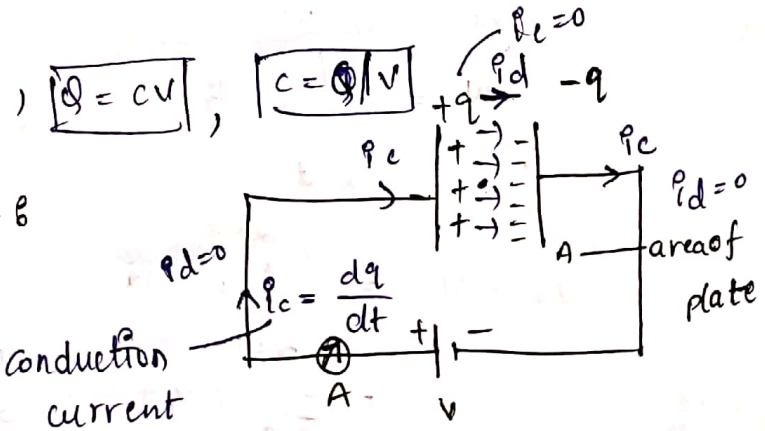
$V_b = V = V_+ - V_-$

** Charge only flow from battery which is on inner surface of capacitor plates.

** If potential diff of plates = Emf of cell then charge flow is completed.

we have, $\boxed{Q = CV}$, $\boxed{C = Q/V}$

During Charging :



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now, electric field b/w the plates

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}$$

$$\boxed{E = \frac{\sigma}{\varepsilon_0}}$$

$$\boxed{E = \frac{q}{\varepsilon_0 A}}$$

$$q = E \varepsilon_0 A$$

displacement current,

$$i_d = \frac{dq}{dt} = \frac{d}{dt}(\varepsilon_0 EA)$$

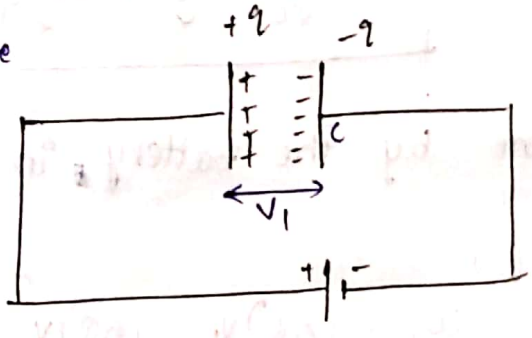
$$i_d = \varepsilon_0 A \frac{dE}{dt}$$

$$i_d = \varepsilon_0 \frac{d(EA)}{dt}$$

$$\boxed{i_d = \varepsilon_0 \frac{d\phi}{dt}} \quad *$$

* Energy stored in the capacitor :-

At any intermediate state, potential diff b/w the plates,



$$\boxed{V_1 = \frac{q}{C}} \quad -(1)$$

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* work done in adding small amount of charge dq ~~(by the battery)~~,

$$dw = V_1 dq$$

Total work done,

$$w = \int_{q=0}^{q=Q} dw$$

$$w = \int_{q=0}^{q=Q} V dq$$

$$= \frac{1}{C}\int_{q=0}^{q=Q} q \cdot dq$$

$$\boxed{w = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}qV}$$

* This work done is stored in the form of Potential energy blw the plates.

Energy stored ,

$$\boxed{U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}qV}$$

* work done by the battery in the process of charging ,

$$w_b = (\Delta Q)V_b = (\Delta Q)V = (Q-0)V$$

$$\boxed{w_b = QV = CV^2}$$

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* $W_b - W_{stored} = CV^2 - \frac{1}{2}CV^2 = \frac{1}{2}CV^2$.

↓

lost in the form of heat by the wires.

* heat produced in the wires,

$$H = \frac{1}{2}CV^2$$

* Imp points

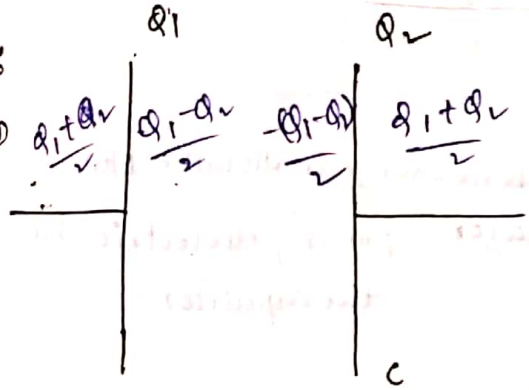
$W_b = CV$

50% | 50%

↓ heat generated

↓ Energy stored in capacitor.

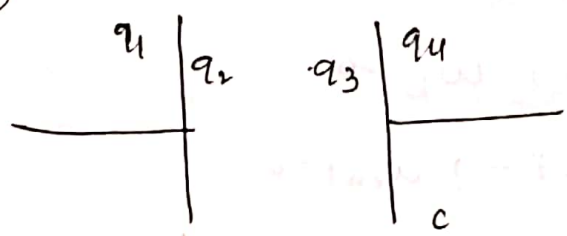
Ex 6



potential diff b/w plates of capacitor,

$$V = \frac{Q_1 - Q_2}{2C}$$

②



$q_1 = q_4$

$q_2 = -q_3$

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$$V = \frac{q_2}{C} = \frac{2q_2}{2C}$$

$$V = \frac{q_2 + q_2}{2C}$$

$$\boxed{V = \frac{q_2 - q_3}{2C}}$$

★★★★
Most Imp ⁶

(Law of ~~conservation of Energy~~)

Work Energy theorem ⁶-

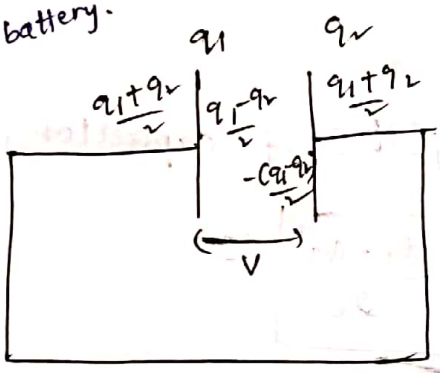
★ $\boxed{U_i + W_b + W_{ext} = U_f + H}$ *****

U_i → zero when initially no capacitor is just connected to battery.

W_b → zero when there is no battery

W_{ext} → zero when there is no change in distance b/w capacitor / placing dielectric b/w the capacitor.

Ex ⁶



Plates are connected with wires,

Find heat generated in the circuit.

Sol ⁶

no battery ⟶ $W_b = 0$

no external work ⟶ $W_{ext} = 0$

Initially there are charges (different) for plates

finally same charges so same potential ⟶ $U_f = 0$

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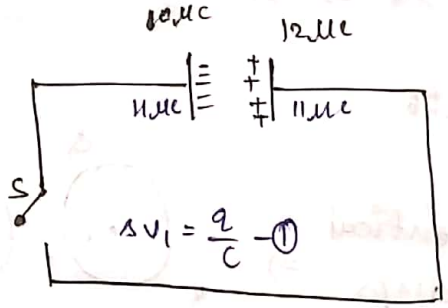
$U_i + 0 + 0 = 0 + H$

$$\boxed{H = U_i}$$

total energy is produced in the form of heat.

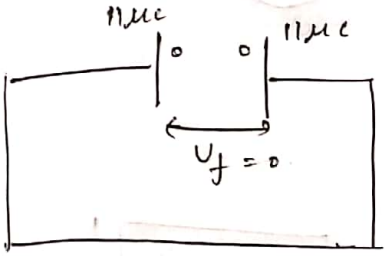
Ex 6

After switch is closed,



$$\Delta V_1 = \frac{q}{C} \quad -(1)$$

$$U_i = \frac{q^2}{2C}$$



$$\boxed{U_f = 0}$$

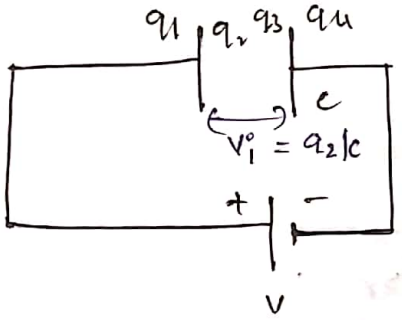
$U_i + \cancel{U_b} + \cancel{U_{ent}} = \cancel{U_f} + H$

$$\boxed{H = U_i} \Rightarrow \boxed{H = \frac{q^2}{2C}}$$

Ex 8

$W_{battery} = (\Delta Q) V$

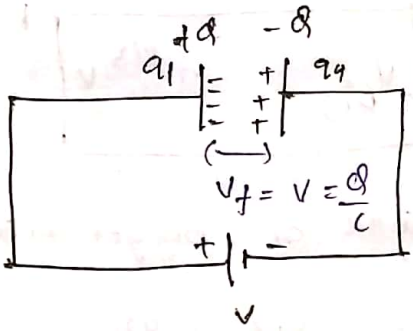
- V → emf of battery.
- ΔQ → charge flown through battery.



$$q_2 = -q_3$$

$$V_i = \frac{q_2}{C}$$

① $V_1 < V$ ⇒



$$U_i = \frac{q_1^2}{2C} = \frac{1}{2} C V_i^2$$

$$U_f = \frac{Q^2}{2C} = \frac{1}{2} C V^2$$

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& $W_b = (\Delta Q)V = (Q_f - Q_i)V = (Q - q/2)V$

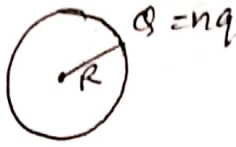
$W_{ext} = 0$

② $V_1 > V$ ⇒ battery is charging.

Imp points

n-identical drops ⇒

⇓ merged into single drop



$\boxed{Q = nq}$ *

* For each drop,

$V = \frac{q}{4\pi\varepsilon_0 r}$, & $C = 4\pi\varepsilon_0 r$

when drops merged,

$\frac{4}{3}\pi R^3 = n\left(\frac{4}{3}\pi r^3\right)$

$\boxed{R = n^{1/3} r}$ *

$V_{big} = \frac{Q}{4\pi\varepsilon_0 R}$

$= \frac{nq}{4\pi\varepsilon_0 n^{1/3} r}$

$\boxed{V_{big} = n^{2/3} V}$ *

& capacitance of bigger drop,

$C_{big} = 4\pi\varepsilon_0 R$

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$$C_{big} = n^{1/3} C$$

* Energy stored in bigger drop,

$$U_{big} = \frac{Q^2}{2C_{big}}$$

$$= \frac{(nq)^2}{2n^{1/3}C}$$

$$= \frac{n^{5/3} q^2}{2C}$$

$U_{each\ drop} = U$

$U = \frac{q^2}{2C}$

$$U_{big} = n^{5/3} U$$

* Surface charge density of each smaller drop,

$$\sigma = \frac{q}{4\pi r^2}.$$

& surface charge density of big drop,

$$\sigma_{big} = \frac{Q}{4\pi R^2}$$

$$= \frac{nq}{4\pi n^{2/3} r^2}$$

$$\sigma_{big} = n^{1/3} \sigma$$

Ex:-

Soap bubble radius (r) & thickness (t) is breaks & melts in the form of droplet.

V = potential

$$V = \frac{q}{4\pi\epsilon_0 r}$$

$$\Rightarrow C = 4\pi\epsilon_0 r$$

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droplet

$$V' = \frac{q}{4\pi\varepsilon_0 R} = \frac{q}{4\pi\varepsilon_0 (3t)^{1/3} r \cdot r^{-1/3}}$$

$$\boxed{V' = V\left(\frac{r}{3t}\right)^{1/3}}$$

Vol is const-

$$C' = 4\pi\varepsilon_0 R$$

$$= 4\pi\varepsilon_0 (3t)^{1/3} \cdot r \cdot r^{-1/3}$$

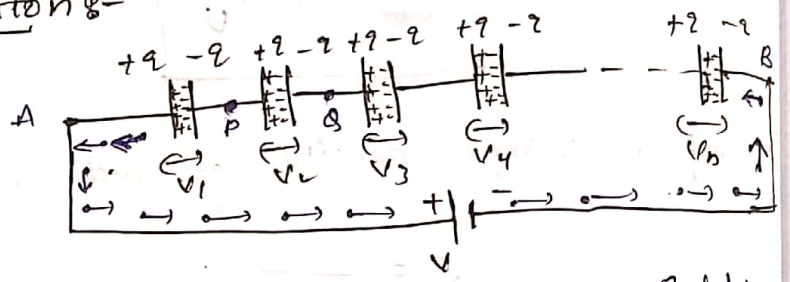
$$\boxed{C' = C\left(\frac{3t}{r}\right)^{1/3}}$$

$$\frac{4}{3}\pi r^3 t = \frac{4}{3}\pi R^3$$

$$R = (3r^2 t)^{1/3}$$

Combination of capacitors:

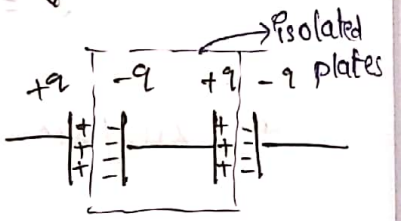
① Series combination:-



$$V_A - V_P = V_1 = \frac{q}{C_1}$$

$$V_P - V_Q = V_2 = \frac{q}{C_2}$$

⋮

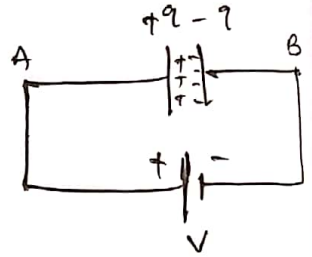


↳ here $q_{net} = 0$

$$V_A - V_B = V_A - V_P + V_P - V_Q + \cdots$$

$$\frac{q}{C_S} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3} + \cdots\cdots + \frac{q}{C_n}$$

$$\boxed{\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots\cdots + \frac{1}{C_n}} \quad *$$



$$\frac{1}{C_S} = \sum_{i=1}^{n} \frac{1}{C_i}$$

If all capacitors are identical,

$$C_1 = C_2 = C_3 = \cdots = C_n = C$$

$$\frac{1}{C_S} = \frac{n}{C} \Rightarrow \boxed{C_S = C/n} \quad *.$$

* For continuous capacitor elements in series, $\boxed{\frac{1}{C_e} = \int \frac{1}{dC}}$

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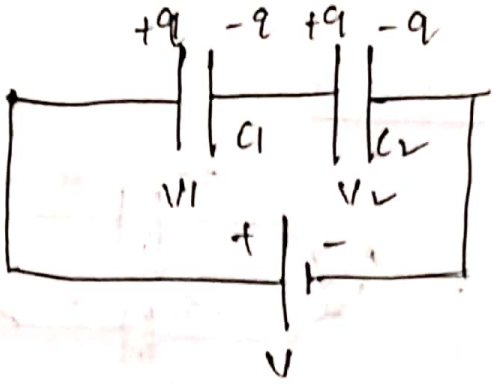

V_1/V_2 =

① $q = C_{eff} \cdot V$

$= C_1V_1 = C_2V_2$

$C_{eff} = \frac{C_1C_2}{C_1+C_2}$

② $\frac{V_1}{V_2} = \frac{C_2}{C_1}$ & $V_1+V_2 = V$

$$V_1 = \frac{VC_2}{C_1+C_2} \quad *$$

$$V_2 = \frac{VC_1}{C_1+C_2} \quad *$$

③ Energy stored in each capacitor,

$$U_1 = \frac{q^2}{2C_1} \qquad U_2 = \frac{q^2}{2C_2}$$

$$\Rightarrow U_{system} = U_1 + U_2 = \frac{q^2}{2}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)$$

$$U_{system} = \frac{q^2}{2C_{eff}} \quad *$$

$$U_{system} = \frac{1}{2} C_{eff} V^2$$

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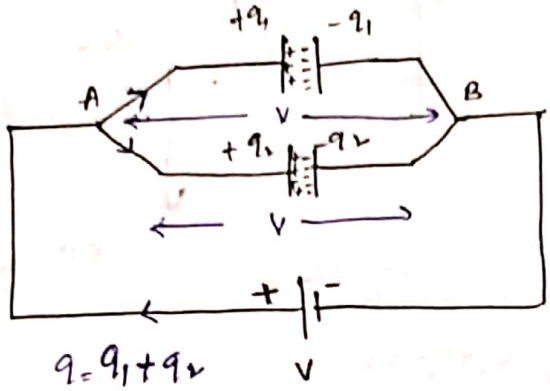
Parallel combination:



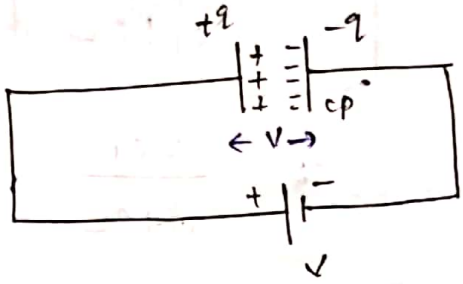
Charges:

$q_1 = C_1 V$

$q_2 = C_2 V$

& $q = q_1 + q_2$

$\Rightarrow \boxed{C_p = C_1 + C_2}$



$q = C_p V$

Points:

① For n-capacitors in parallel,

$$C_p = C_1 + C_2 + - - - C_n$$

If all are identical

then $C_1 = C_2 = C_3 = - - - C_n$

$$\boxed{C_p = nC}$$

② For continuous capacitor elements, in parallel,

$$C_p = \int dC$$

③ $q_1 = C_1 V$

$q_2 = C_2 V$

$\Rightarrow \frac{q_1}{q_2} = \frac{C_1}{C_2}$ & $q_1 + q_2 = q$

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$$q_1 = \frac{C_1 q}{C_1 + C_2}$$

& $$q_2 = \frac{C_2 q}{C_1 + C_2}$$

(4) Energy stored in capacitors,

$$U_1 = \frac{q_1^2}{2C_1} = \frac{1}{2} C_1 V^2$$

$$U_2 = \frac{q_2^2}{2C_2} = \frac{1}{2} C_2 V^2$$

$$\therefore \boxed{\frac{U_1}{U_2} = \frac{C_1}{C_2}}$$

(5) Total energy stored in circuits,

$$U = U_1 + U_2$$

$$= \frac{1}{2} C_1 V^2 + \frac{1}{2} C_2 V^2$$

$$= \frac{1}{2} (C_1 + C_2) V^2$$

$$\boxed{U = \frac{1}{2} C_p V^2}$$

(6) we have,

$$U_i + W_b + W_{ext} = U_f + H$$

& $$W_b = \Delta Q V$$

$$= qV$$

$$= (q_1 + q_2) V$$

$$\boxed{W_b = (C_1 + C_2) V^2}$$

$$\boxed{H = \frac{1}{2} (C_1 + C_2) V^2}$$ *

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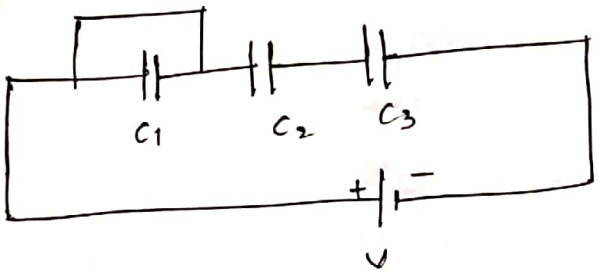
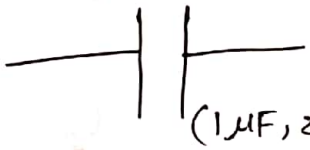
Exp



charge flow through $e_1 = 0$.

Ex: you are given a capacitor as shown.

To get (6μF, 12V) output, Find the no. of such capacitors connected in circuit



Sol:

no. of capacitors in a row,

$$n = \frac{V_{net}}{V} = \frac{12}{2} = 6$$

If potential is more than limiting voltage, air b/w plates ionise & charge leaks. (break down strength)

no. of rows,

$$m = \frac{C_{net}}{\frac{C}{n}} = \frac{6}{1/6} = 36.$$

$$m = \frac{n\,C_{net}}{C}$$

$$m = \frac{V_{net}}{V} * \frac{C_{net}}{C}$$

∴ Total no. of capacitors,

$$N = mn = 6 \times 36 = 216.$$

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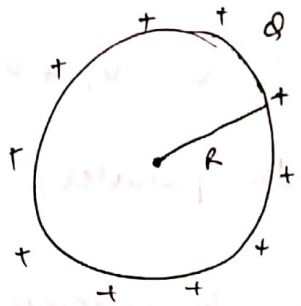
Ex:6 Find max charge present on the sphere of radius 1m placed in air. [$E = 3 \times 10^6$ v/m in air].



Solⁿ

$$E = \frac{KQ}{R^2}$$

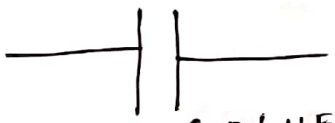
$$3 \times 10^6 = \frac{9 \times 10^9 \times Q}{(1)^2}$$

$$Q = \frac{1}{3} \times 10^{-3} \text{ C}$$

$$Q = 0.33 \times 10^{-3} \text{ C} = 0.33 \text{ mC}.$$

Ex! Two capacitors are as shown in figure. Find the max applied voltage across the combⁿ, if they are connected i) In series

ii) In parallel



$C_1 = 1\mu F$

$V_1 = 6kV$

↓

breakdown

$C_2 = 2\mu F$

$V_2 = 4kV$

↓

breakdown

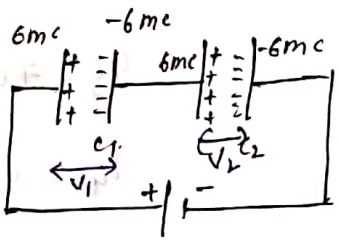
Solⁿ

① In series,

$$q_1 = C_1 V_1 = 1 \times 10^{-6} \times 6 \times 10^3$$

↓ max

$$= 6 \text{ mC}$$



& $q_2 = C_2 V_2 = 2 \times 10^{-6} \times 4 \times 10^3$

$$= 8 \text{ mC}$$

here,

$V = V_1 + V_2$

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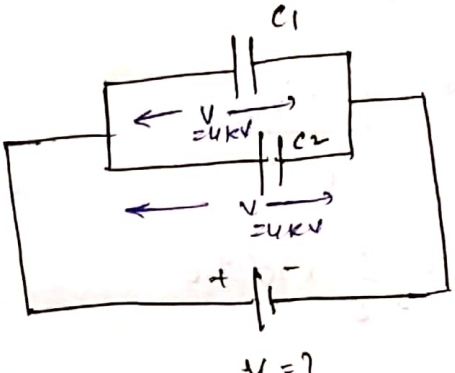
$$V_1 = \frac{q_1}{c_1} = \frac{6\times10^{-3}}{1\times10^{-6}} = 6\,kV$$

$$V_2 = \frac{q_2}{c_2} = \frac{6\times10^{-3}}{2\times10^{-6}} = 3\,kV$$

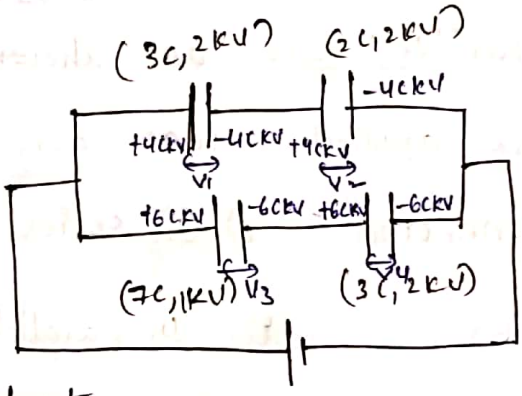
$$\therefore V = V_1 + V_2 = 6kV + 3kV = 9kV$$

(2) In paiallel,



min breakdown voltage is the applied voltage across the comb^n = 4kV

Ex6



what should be the max value of voltage applied across the comb^n without breakdown.

Soln

In First row,

$q_1 = 6ckV$

$q_2 = 4ckV$ $(q_2 < q_1)$

In 2nd row,

$q_3 = 7ckV$

$q_4 = 6ckV$ $(q_4 < q_3)$

$$V_1 = \frac{q_0}{c_1} = \frac{4C}{3C} = \frac{4}{3}$$

$$V_2 = \frac{q}{c_2} = \frac{4C}{2C} = 2$$

$$V' = V_1 + V_2 = \frac{4}{3} + 2 = \frac{10}{3}$$

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$$V_3 = \frac{q'}{C_3} = \frac{6}{7}$$

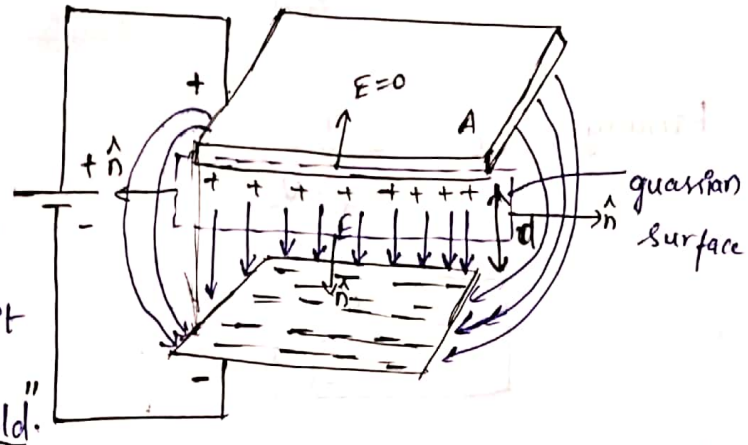
$$V_4 = \frac{q'}{C_4} = \frac{6}{3} = 2$$

$$\Big\} \quad V'' = V_3 + V_4 = \frac{6}{7} + 2 = \frac{20}{7}$$

max value of voltage $= V'' = \frac{20}{7}$ kV $(\because V'' < V')$

calculation of Capacitance of a parallel plate Capacitor

* E → uniform b/w the plates & non-uniform around the edges, it is called "fringing field".



* Here "fringing" can be neglected.

To find E:-

M-I:

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A}$$

M-II: guass law,

$$\phi = \frac{q}{\varepsilon_0}$$

$$EA = \frac{q}{\varepsilon_0}$$

$$\therefore \boxed{E = \frac{q}{\varepsilon_0 A}}$$

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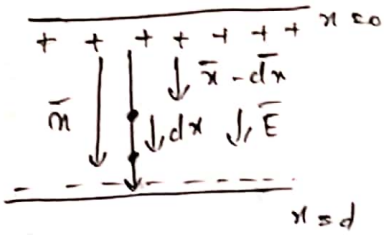
now, $\int_{V-}^{V+} dV = -\int_{x=d}^{x=0} \bar{E} \cdot d\bar{x}$



$$V_+ - V_- = \int_{x=0}^{x=d} \bar{E}\, d\bar{x}$$

$$= E d$$

$$V_+ - V_- = \frac{q d}{\varepsilon_0 A}$$

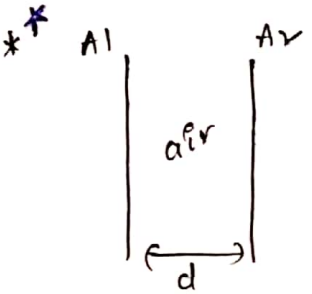
Finally, $C = \frac{q}{V_+ - V_-}$

$$\boxed{C = \frac{\varepsilon_0 A}{d}}$$

* If the gap b/w the plates is filled with dielectric K, then $C' = \frac{\varepsilon_0 K A}{d}$

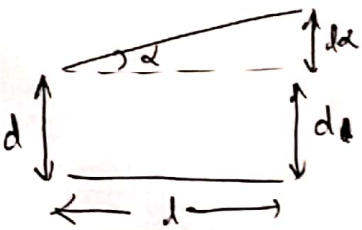
$$C' = \frac{\varepsilon_0 A}{d}(K)$$

$$\boxed{C' = KC}$$

** 

$A_2 < A_1$ then $\boxed{C = \frac{\varepsilon_0 A_2}{d}}$

~~Capacitance of a cap~~

*Exp 

seperation b/w plates is changing linearly. Then find C.

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$$C = \frac{\varepsilon_0 A}{d_{mean}} = \frac{\varepsilon_0 A}{d + \frac{d\alpha}{2}}$$

$$C = \varepsilon_0 A \left(d + \frac{d\alpha}{2}\right)^{-1}$$

$$C = \frac{\varepsilon_0 A}{d}\left(1 + \frac{d\alpha}{2d}\right)^{-1}$$

$$\boxed{C = \frac{\varepsilon_0 A}{d}\left(1 - \frac{d\alpha}{2d}\right)}$$

$$d_{mean} = \frac{d + d + d\alpha}{2} = \frac{2d + d\alpha}{2} = d + \frac{d\alpha}{2}$$

Capacitance of a spherical capacitor-

a) Concentric spherical conductors-

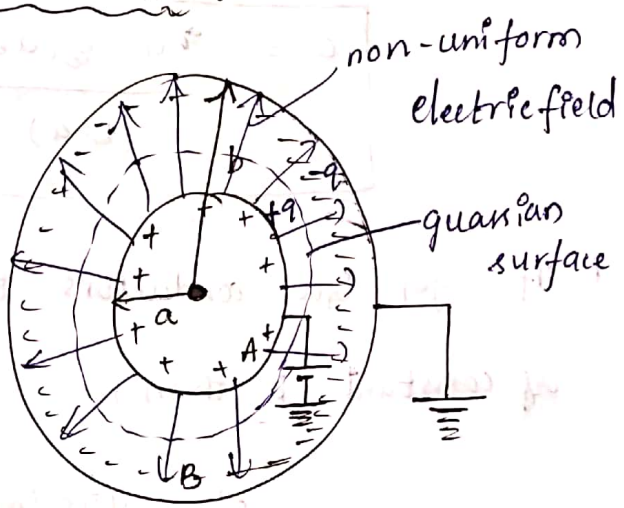
M-1: (absolute potential)

$$V_A = \frac{kq}{a} - \frac{kq}{b}$$

$$V_B = \frac{kq}{b} - \frac{kq}{b} = 0$$

$$V_A - V_B = \frac{kq}{a} - \frac{kq}{b}$$

$$= kq\left(\frac{1}{a} - \frac{1}{b}\right)$$



M-II:

$$\int \bar{E}\, d\bar{s} = \frac{q}{\varepsilon_0}$$

$$E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

$$E = \frac{q}{4\pi\varepsilon_0 r^2}$$

now,

$$\int_{V_+}^{V_+} dV = -\int_{r=b}^{r=a} E \cdot d\bar{r}$$

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$$V_+ - V_- = \int_{r=a}^{r=b} \frac{q}{4\pi\varepsilon_0 r^2} \cdot dr$$

$$= \frac{q}{4\pi\varepsilon_0} \left[-\frac{1}{r}\right]_a^b$$

$$V_+ - V_- = \frac{q}{4\pi\varepsilon_0} \left[\frac{1}{a} - \frac{1}{b}\right]$$

$$V_+ - V_- = \frac{q\,(b-a)}{4\pi\varepsilon_0 ab}$$

$$C = \frac{q}{V_+ - V_-}$$

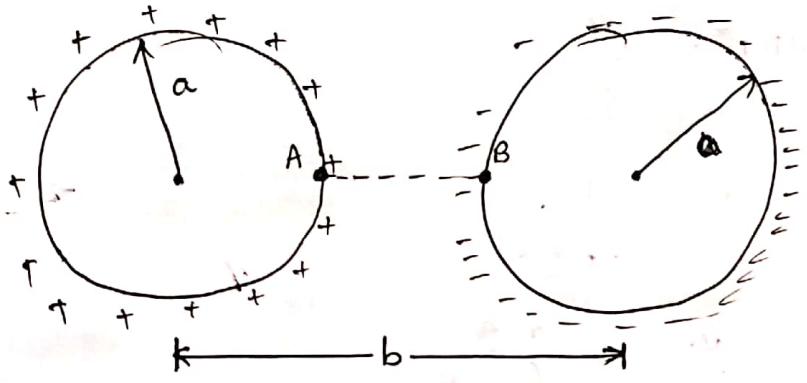
$$\boxed{C = \frac{q\,4\pi\varepsilon_0(ab)}{(b-a)}} \quad *$$

* If gap b/w conductors is filled with dielectric of constant k then,

$$C' = \frac{4\pi\varepsilon_0(ab)}{b-a}(K)$$

$$\boxed{C' = CK}$$

b) spherical conductors are seperated by certain distance



Find capacitance of the arrangement.

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MSE now,

$$V_A = \frac{kq}{a} - \frac{kq}{b-a}$$

$$V_B = -\frac{kq}{a} + \frac{kq}{b-a}$$

$$V_A - V_B = \frac{2kq}{a} - \frac{2kq}{b-a}$$

$$= 2kq\left(\frac{1}{a} - \frac{1}{b-a}\right)$$

$$= \frac{2kq}{a(b-a)}(b-a-a)$$

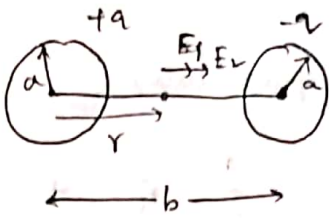
$$V_A - V_B = \frac{2kq(b-2a)}{a(b-a)}$$

Finally, $C = \frac{q}{V_A - V_B}$

$$C = \frac{a(b-a)}{2k(b-2a)}$$

$$\boxed{C = \frac{2\pi\varepsilon_0 a(b-a)}{(b-2a)}} \quad *$$

Method - II :



$$E = E_1 + E_2$$

$$= \frac{q}{4\pi\varepsilon_0 r^2} + \frac{-q}{4\pi\varepsilon_0 (b-r)^2}$$

$$\int_{V_+}^{V_-} dV = -\int_{r=b-a}^{r=a} \bar{E}\cdot d\bar{r}$$

* If gap b/w spheres is filled with dielectric of dielectric const ε_r then,

$$\boxed{C' = \frac{2\pi\varepsilon_0\varepsilon_r a(b-a)}{b-2a}}$$

* If $b >> a$

then $C' = \frac{2\pi\varepsilon_0\varepsilon_r a(b)}{b} \Rightarrow \boxed{C' = 2\pi\varepsilon_0\varepsilon_r a}$

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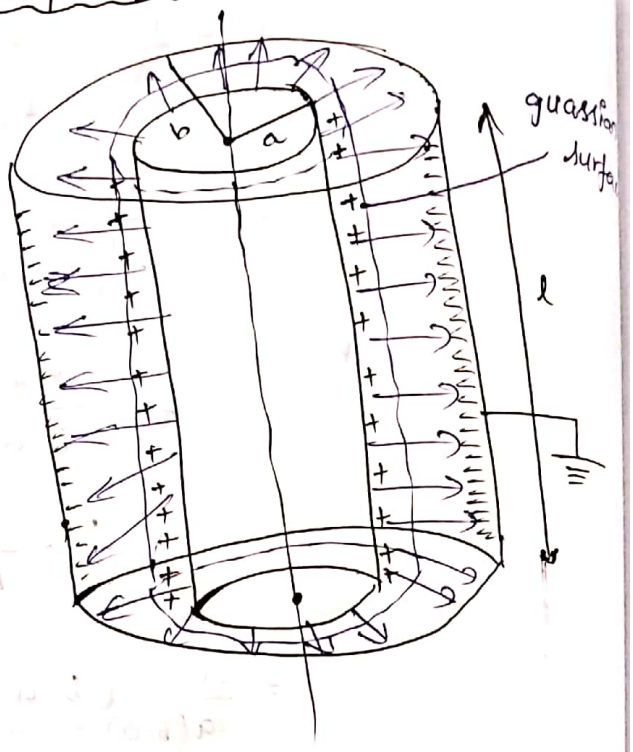
Capacity of a Cylindrical Capacitor :-



guass law,

$$\oint \bar{E} \cdot d\bar{s} = \frac{q}{\varepsilon_0}$$

$$E \times 2\pi r l = \frac{q}{\varepsilon_0}$$

$$E = \frac{q}{2\pi \varepsilon_0 r l}$$

now,

$$\int_{V-}^{V+} dV = -\int_{r=b}^{r=a} \bar{E} \cdot d\bar{r}$$

$$V_+ - V_- = \int_{r=a}^{r=b} \frac{q}{2\pi \varepsilon_0 l} \cdot dr$$

$$V_+ - V_- = \frac{q}{2\pi \varepsilon_0 l} \ln(b/a)$$

Finally,

$$\boxed{C = \frac{2\pi \varepsilon_0 l}{\ln(b/a)}} \quad *$$

* If the gap b/w the shells is filled with dielectric of 'k'.

$$\boxed{C' = \frac{2\pi \varepsilon_0 k l}{\ln(b/a)}}$$

Ex: Three capacitors of capacitances C_1 C_2 C_3 0.002, 0.004, 0.006 are connected in series. i) Can a potential of

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11000 V be applied to this battary?

(ii) what voltage will be recieved by each capacitor in the battery if puncture voltage of each capacitor is 4000V.

Sol^n

$$V_1 : V_2 : V_3 = \frac{1}{2} : \frac{1}{4} : \frac{1}{6}$$

$$= 6 : 3 : 2$$

$$11x = 11000$$

$$\boxed{x = 1000}$$

$$V_1 = 6000V, \quad V_2 = 3000V, \quad V_3 = 2000\,V$$

∴ C_1 will punture if 11000V is applied. ($\because$ 6000 > 4000).

$$q_1 = C_1V \qquad q_2 = C_2V \qquad q_3 = C_3V$$

$$= 0.002 \times 4000 \qquad = 16 \qquad = 24$$

$$= 8$$

$$V_1 = \frac{8}{0.002} = 4000$$

$$V_2 = \frac{8}{0004} = 2000$$

$$V_3 = \frac{8}{0.006} = \frac{8000}{6} = \frac{4000}{3}$$

$$V = V_1 + V_2 + V_3$$

$$= \left(6 + \frac{4}{3}\right) = \frac{18+4}{3} = \frac{22}{3}\,kV$$

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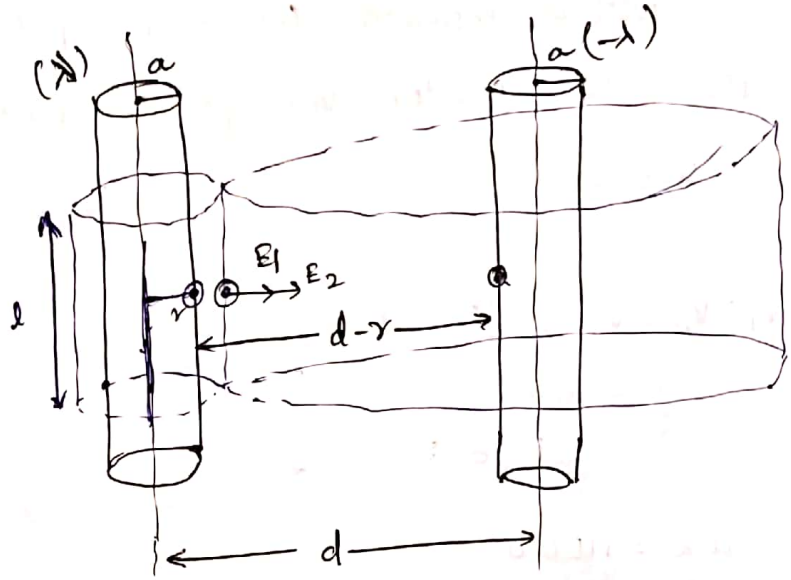
Capacitance of two parallel cylindrical conductors



$$E_1 \times 2\pi r l = \frac{q}{\varepsilon_0}$$

$$E_1 = \frac{q}{2\pi\varepsilon_0 r l}$$

& $$E_2 = \frac{q}{2\pi\varepsilon_0 (d-r) l}$$

net field,

$$E = E_1 + E_2$$

$$= \frac{q}{2\pi\varepsilon_0 l}\left[\frac{1}{r} + \frac{1}{d-r}\right]$$

now, $$\int_{V-}^{V+} dV = -\int_{r=d-a}^{r=a} \bar{E} \cdot d\bar{r}$$

$$V_+ - V_- = \int_{r=a}^{r=d-a} \frac{q}{2\pi\varepsilon_0 l}\left(\frac{1}{r} + \frac{1}{d-r}\right) dr$$

$$V_+ - V_- = \frac{q}{2\pi\varepsilon_0 l}\left[\int_{r=a}^{r=d-a} \frac{1}{r}.dr + \int_{r=a}^{r=d-a} \frac{1}{d-r}.dr\right]$$

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$$V_+ - V_- = \frac{q}{2\pi\varepsilon_0 l}\left[\ln\left(\frac{d-a}{a}\right) + \ln\left(\frac{a}{d-a}\right)\right]$$

$$= \frac{q}{2\pi\varepsilon_0 l} \times 2\ln\left(\frac{d-a}{a}\right)$$

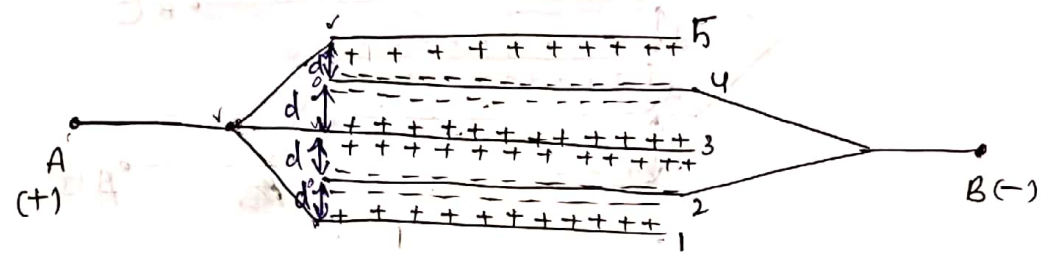
$$V_+ - V_- = \frac{q}{\pi\varepsilon_0 l}\ln\left(\frac{d-a}{a}\right)$$

finally, $C = \dfrac{q}{V_+ - V_-}$

$$\boxed{C = \frac{\pi\varepsilon_0 l}{\ln\left(\frac{d-a}{a}\right)}}$$

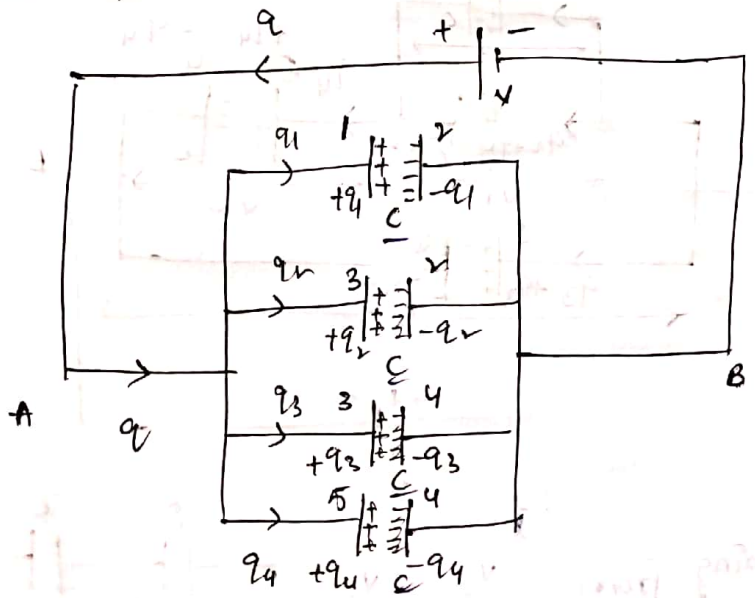
Capacitance of system of parallel plates

Ex 6



Find C_{AB}.

Sol 6



all capacitors have same capacitance bcoz they are at same distance.

$$C_{AB} = 4C = \frac{4\varepsilon_0 A}{d}.$$

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Plate ①, charge $= q_1 = CV = \frac{\varepsilon_0 AV}{d}$

Plate ②, charge $= -(q_2 + q_1) = -2CV = -\frac{2\varepsilon_0 AV}{d}$

Plate ③, charge $= q_2 + q_3 = \frac{2\varepsilon_0 AV}{d}$

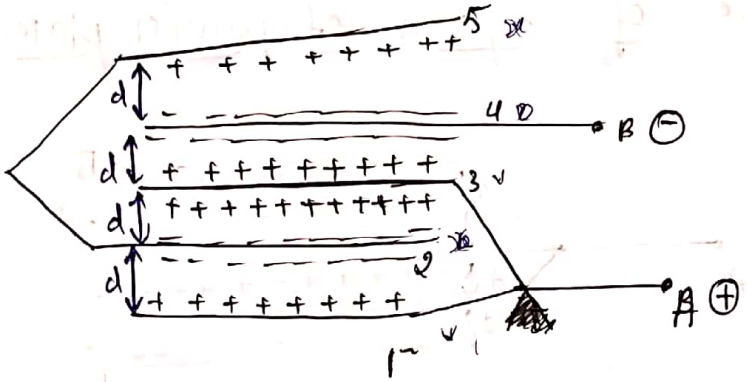
Plate ④, charge $= -(q_3 + q_4) = -\frac{2\varepsilon_0 AV}{d}$

* work done by battery $= C_{eff} V^2$

Energy stored in capacitor $= \frac{1}{2} C_{eff} V^2$

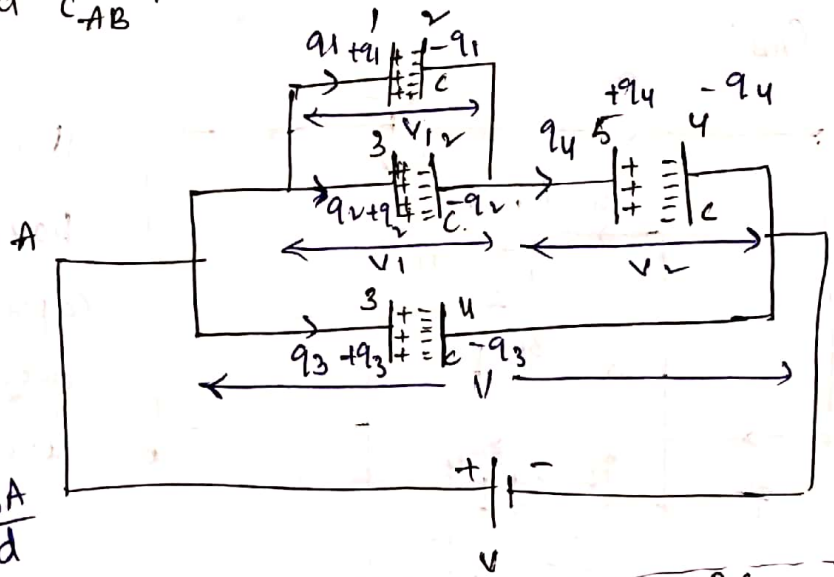
heat produced $= \frac{1}{2} C_{eff} V^2$ ..

Ex 6



Find C_{AB}.

Sol.



$C_{AB} = \frac{5C}{3}$

$C = \frac{\varepsilon_0 A}{d}$

charge flowing from battery,

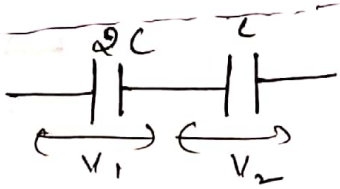
$q = C_{eff} V = \frac{5CV}{3}$

$V_2 = 2V_1$

$V_1 + V_2 = V$

$V_1 = V/3$

$V_2 = 2V/3$



$V_1 = \frac{q}{C_1}$, $V_2 = \frac{q}{C_2}$ $\Rightarrow \frac{V_1}{V_2} = \frac{C_2}{C_1} = \frac{1}{2}$

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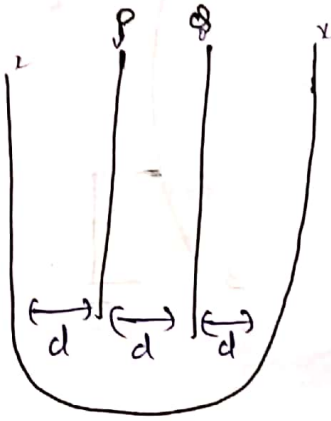
$q_1 = q_2 = CV_1 = \frac{CV}{3}$

$q_4 = CV_2 = \frac{2CV}{3}$

$q_3 = CV_1 = \frac{CV}{3}$

Exc 1)

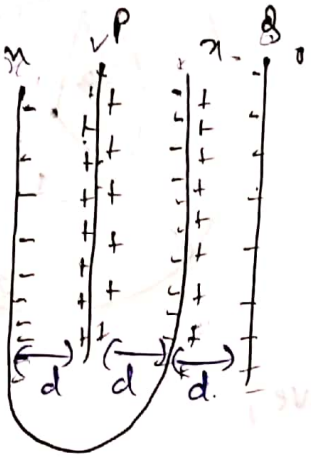


$C_{PQ} = ?$

Sols

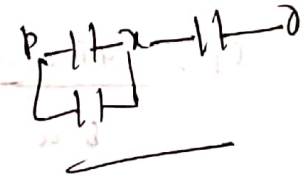
$3C$

2)



$C_{PQ} = ?$

$2C$



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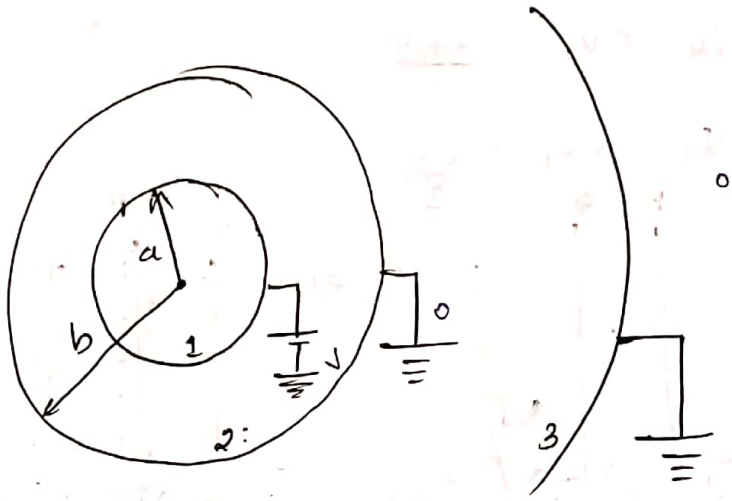
Combinations of spherical shells

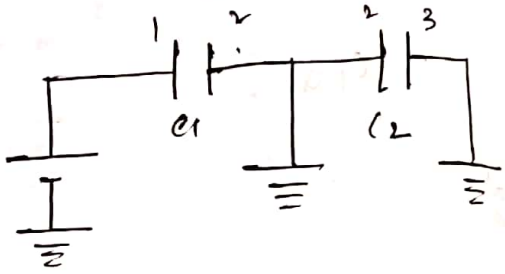
(1)



we have,

$$C = \frac{4\pi\varepsilon_0}{\frac{1}{a}-\frac{1}{b}}$$

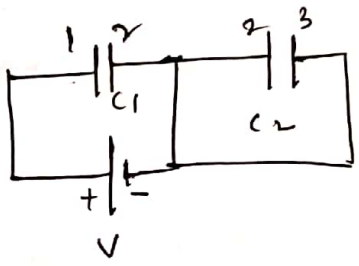
$\Rightarrow$



here, $C_1 = \dfrac{4\pi\varepsilon_0(ab)}{b-a}$

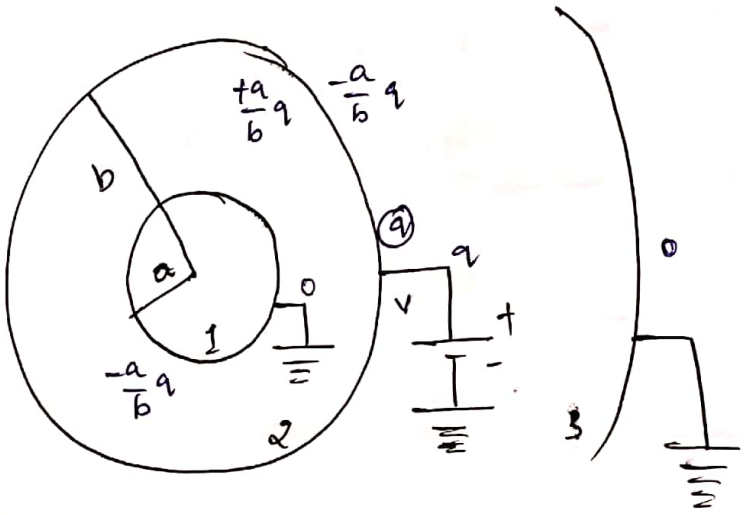
$$C_2 = \frac{4\pi\varepsilon_0}{\frac{1}{b}-\frac{1}{\infty}} = 4\pi\varepsilon_0 b$$

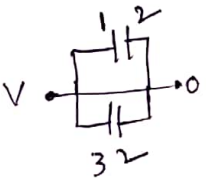
$=$



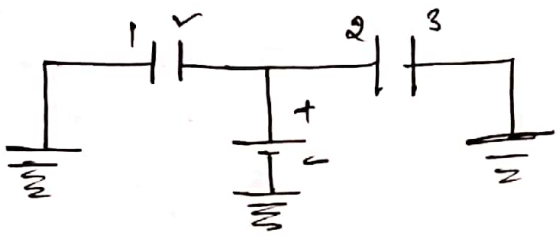
$$C_{eff} = C_1 = \frac{4\pi\varepsilon_0(ab)}{b-a}$$

(2)





$\Rightarrow$



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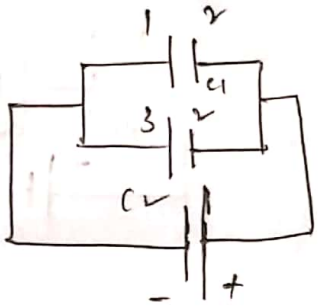
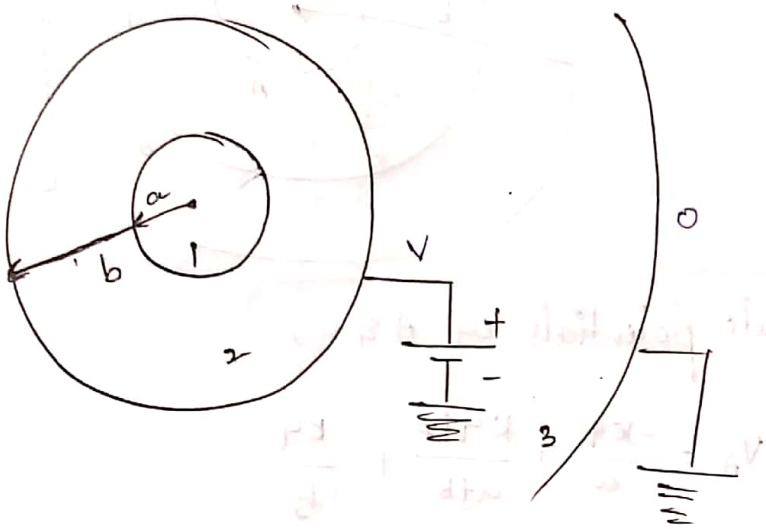


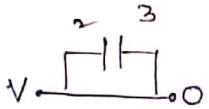
$$C_{eff} = C_1 + C_2 = \frac{4\pi\varepsilon_0 ab}{b-a} + 4\pi\varepsilon_0 b$$

$$= 4\pi\varepsilon_0 \left(\frac{ab + b^2 - ab}{b-a} \right)$$

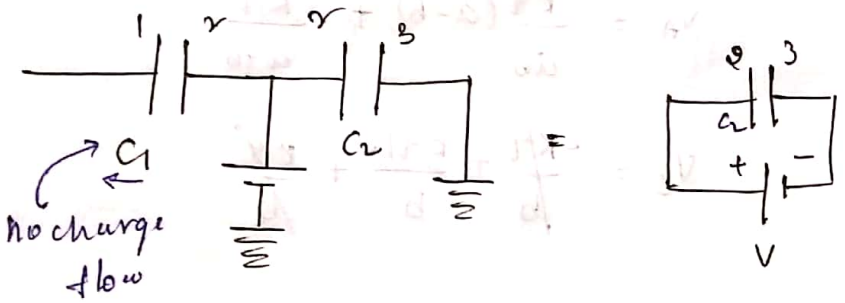
$$= \frac{4\pi\varepsilon_0 b^2}{b-a}.$$

(3)



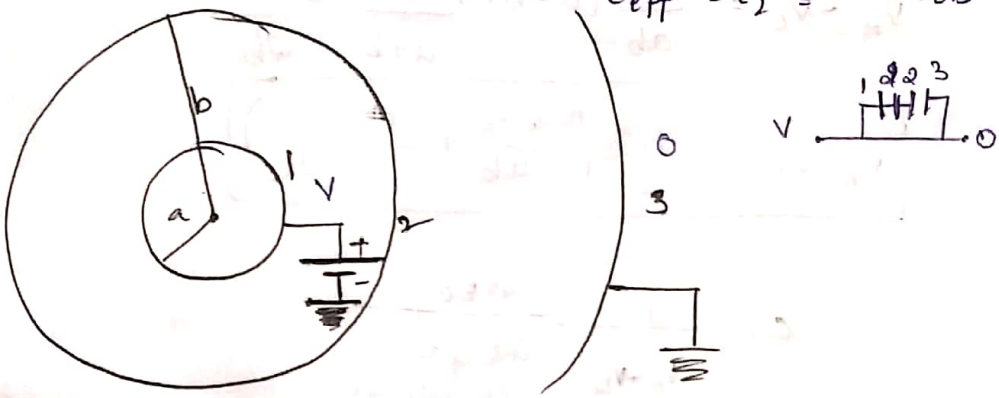


=)



$$C_{eff} = C_2 = 4\pi\varepsilon_0 b.$$

(4)



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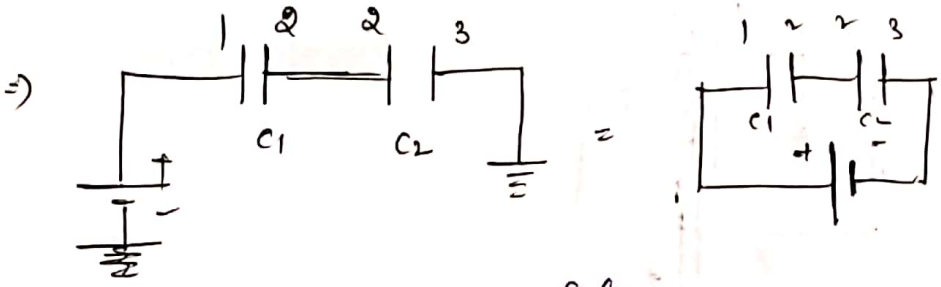
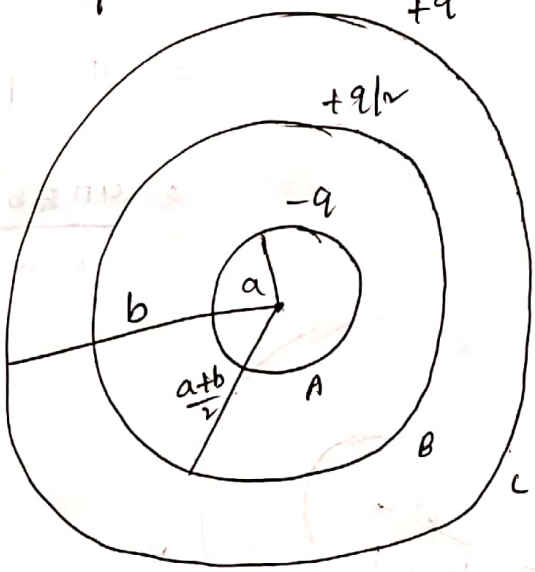


$$C_{eff} = \frac{C_1 C_2}{C_1 + C_2}$$

Ex6 Determine the capacitance of the system b/w A & C.



Solⁿ Absolute potentials of A & C,

$$V_A = \frac{-kq}{a} + \frac{kq/2}{\frac{a+b}{2}} + \frac{kq}{b}$$

$$V_A = \frac{kq}{ab}(a-b) + \frac{kq}{a+b}$$

$$V_C = \frac{-kq}{b} + \frac{kq/2}{b} + \frac{kq}{b}$$

$$V_A - V_C = \frac{kq}{ab}(a-b) + \frac{kq}{a+b} - \frac{kq}{2b}$$

$$\boxed{V_A - V_C = kq\left(\frac{a-b}{ab} + \frac{1}{a+b} - \frac{1}{2b}\right)}$$

$$C = \frac{q}{V_A - V_B} = \frac{4\pi\varepsilon_0}{\frac{a-b}{ab} + \frac{1}{a+b} - \frac{1}{2b}}$$

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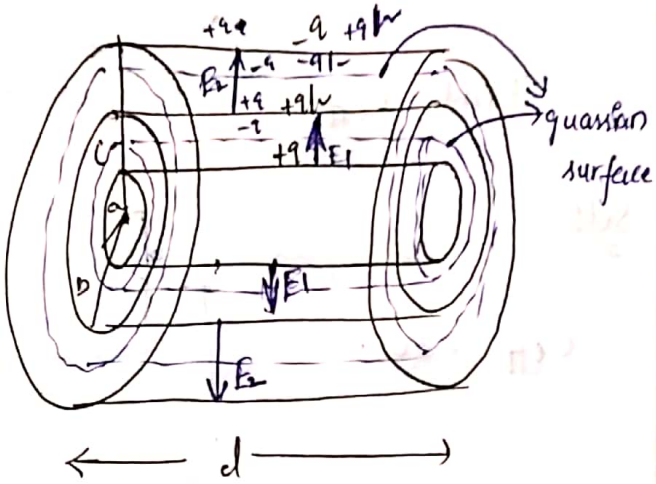
Ex! Three cylindrical shells of radii a, b, c & length l have charges, +q, +q/2, −q. Determine the capacitance of system b/w inner & outer cylinders.



now,

$$V_A - V_C$$

$$= V_A - V_B + V_B - V_C$$

$$= \int_{r=a}^{r=b} E_1 \cdot dr + \int_{r=b}^{r=c} E_2 \cdot dr$$

$E_1 \times 2\pi r l = \frac{q}{\varepsilon_0}$

$$\boxed{E_1 = \frac{q}{2\pi\varepsilon_0 r l}}$$

$E_2 \times 2\pi r l = \frac{3q}{2\varepsilon_0}$

$$\boxed{E_2 = \frac{3q}{4\pi\varepsilon_0 r l}}$$

now,

$$V_A - V_C = \frac{q}{2\pi\varepsilon_0 l} \int_{r=a}^{r=b} \frac{1}{r} \cdot dr + \frac{3q}{4\pi\varepsilon_0 l} \int_{r=b}^{r=c} \frac{1}{r} \cdot dr$$

$$V_A - V_C = \frac{q}{2\pi\varepsilon_0 l} \ln(b/a) + \frac{3q}{4\pi\varepsilon_0 l} \ln(c/b)$$

$$C = \frac{q}{V_A - V_C}$$

$$= \frac{2\pi\varepsilon_0 l}{\ln b - \ln a + \frac{3}{2}\ln c - \frac{3}{2}\ln b}$$

$$\boxed{C = \frac{2\pi\varepsilon_0 l}{\frac{3}{2}\ln c - \ln a - \frac{1}{2}\ln b}}$$

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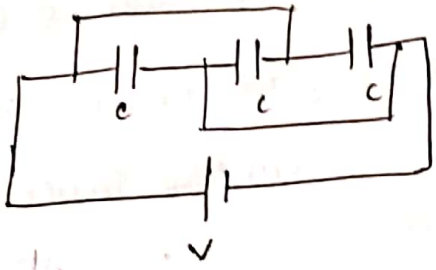
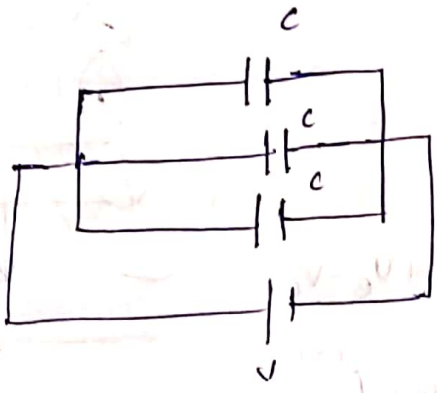
Ex 8



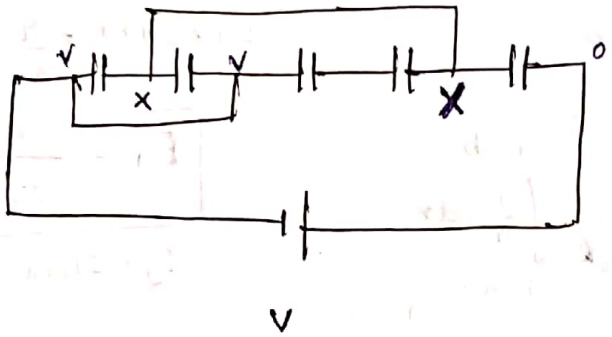
Find C_{eff}

Sol 8

$C_{eff} = 3C$.



Ex 8

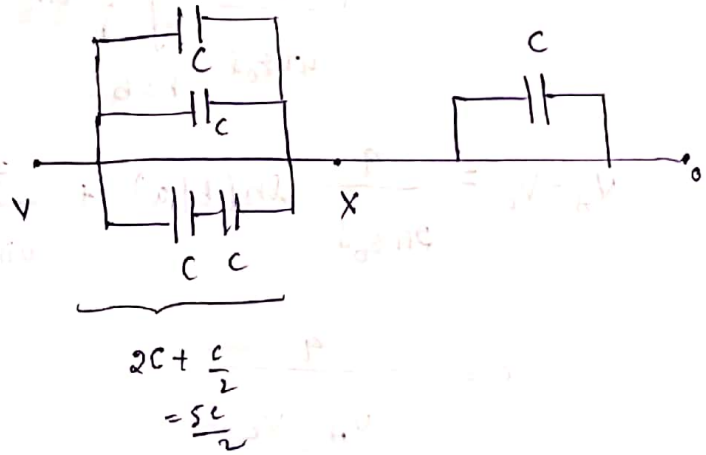


Find C_{eff}.

Sol 8

$$C_{eff} = \frac{\frac{5C}{2} \times C}{\frac{5C}{2} + C}$$

$$\boxed{C_{eff} = \frac{5C}{7}}$$



Ex 8 A capacitor of capacity 1 μF is charged to a potential of 100 V. The charging battery is then removed and connected to another capacitor of 2 μF. 1 plate of C_2 is earthed. The charges in C_1 & C_2 in steady state will be

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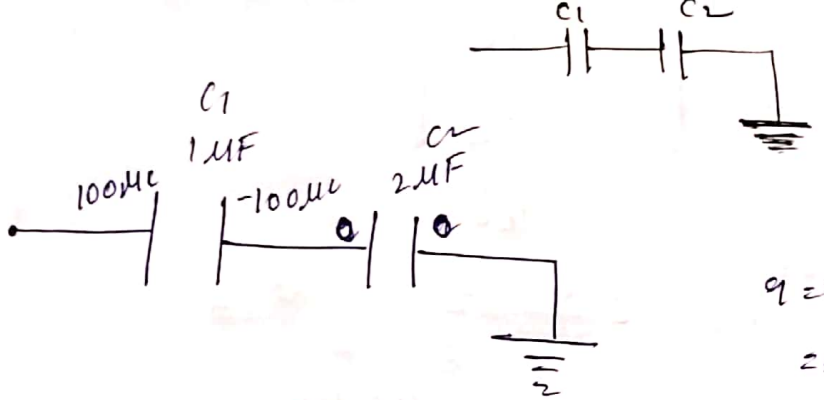
Sol:



$q = VC$

$= 100 \times 1$

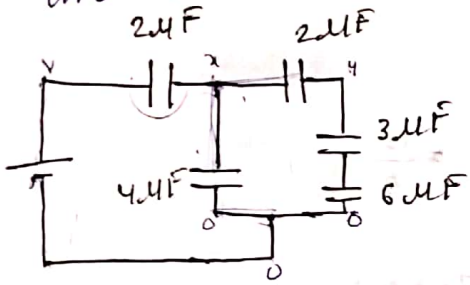
$= 100\mu C$

As C_2 capacitor is earthed,

V = potential of $C_2 = 0$.

As there is no connection of C_1 with battery or the given system is open circuit, there is no charge flow from C_1 to C_2 (or) from C_2 to C_1. ∴ Charges at steady state are 100μC & zero respectively.

Ex:



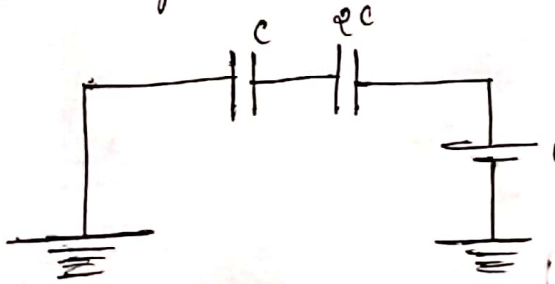
Find Emf of battery.

Sol:

Ex:

Find charge stored on C.

given $C = 6\mu F$



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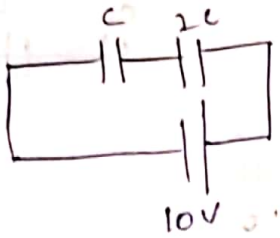
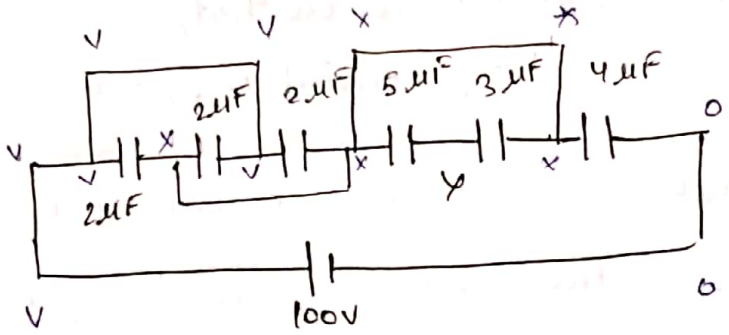
Sol:



$$C_{eff} = \frac{2C \times C}{3C} = \frac{2C}{3} = \frac{2 \times 6}{3} = 4$$

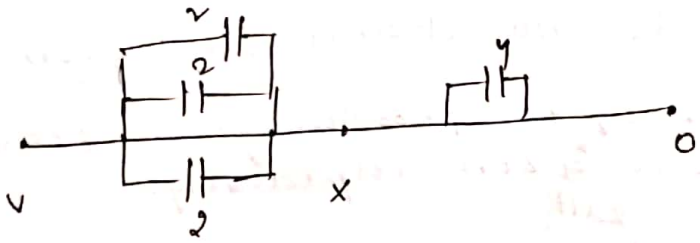
$$q = VC = 4 \times 10 = 40\mu C$$

Ex:



charge Stored on 5μF

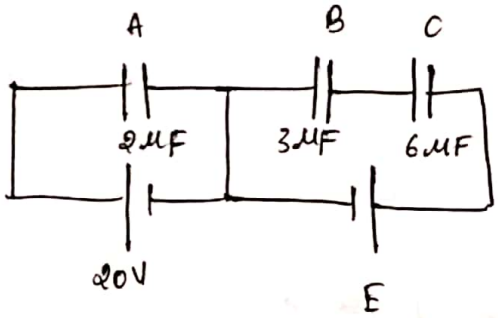
Sol:



$$C_{eff} = \frac{6 \times 4}{10} = \frac{24}{10} = \frac{12}{5}$$

Charge on 5μC = zero.

Ex: In the given circuit, if Emf is increased the steady state charge on A, B, C is ——.



$$q_A = VC_A$$

$$= 20 \times 2$$

$$\boxed{q_A = 40\mu C}$$

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q in second circuit = 2E ($\because C_{eff} = 2$)

So as E increases, common charge flow from B & C also increases.

And A in the 1st circuit is uneffected with E value.

Ex: A parallel plate capacitor is charged upto V_0 & then detached from charger. Now the distance b/w the plates is varied according to the law $d = d_0 + d_1 \sin\omega t$

i) The value of d_1 if potential diff across the capacitor changes according to the law $V = V_0 (1 + \frac{1}{2}\sin\omega t)$.

ii) the electric field b/w the plates varied according to the law

A) $E = \frac{V_0}{d_0}(1 + \sin\omega t)$

B) $E = \frac{V_0}{d_0}(1 + \frac{1}{2}\sin\omega t)$

C) will not vary. ✓

D) will vary arbitarly.

i) $C = \frac{q}{V}$

$$\frac{\varepsilon_0 A}{d} = \frac{q}{V}$$

$$\frac{\varepsilon_0 A}{d_0 + d_1 \sin\omega t} = \frac{q}{V_0(1 + \frac{1}{2}\sin\omega t)} \quad \text{—①}$$

at $t = 0$,

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$$\frac{\varepsilon_0 A}{d_0} = \frac{q}{V_0} \Rightarrow \boxed{q = \frac{\varepsilon_0 A V_0}{d_0}}$$

from ①,

$$\frac{\varepsilon_0 A}{d_0 + d_1 \sin\omega t} = \frac{\frac{\varepsilon_0 A V_0}{d_0}}{V_0\left(1 + \frac{1}{2}\sin\omega t\right)}$$

$$d_0 + d_1 \sin\omega t = d_0 + \frac{d_0}{2}\sin\omega t$$

$$d_1 \sin\omega t = \frac{d_0}{2}\sin\omega t$$

$$\boxed{d_1 = d_0/2}$$

(ii) $E = \frac{\sigma}{\varepsilon_0}$

$$= \frac{q}{\varepsilon_0 A}$$

$$= \frac{\varepsilon_0 A V_0}{d_0 \varepsilon_0 A}$$

$$\boxed{E = \frac{V_0}{d_0}}$$

(or)

$$E = \frac{\varepsilon_0 A V}{d \varepsilon_0 A}$$

$$= \frac{V}{d}$$

$$= \frac{V_0\left(1 + \frac{1}{2}\sin\omega t\right)}{d_0\left(1 + \frac{1}{2}\sin\omega t\right)} = \frac{V_0}{d_0}$$

∴ E will not vary.

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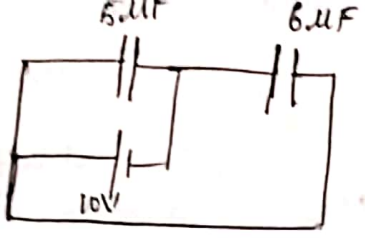
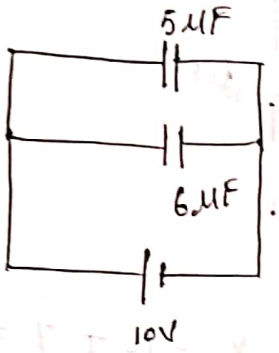
Ex6



The charge supplied by the battery in the circuit is _____.

Sol6



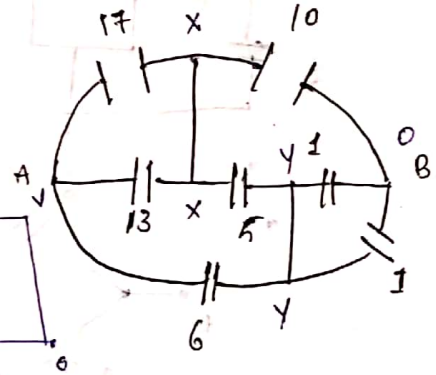
$C_{eff} = 6 + 5 = 11$

$q = C_{eff} V$

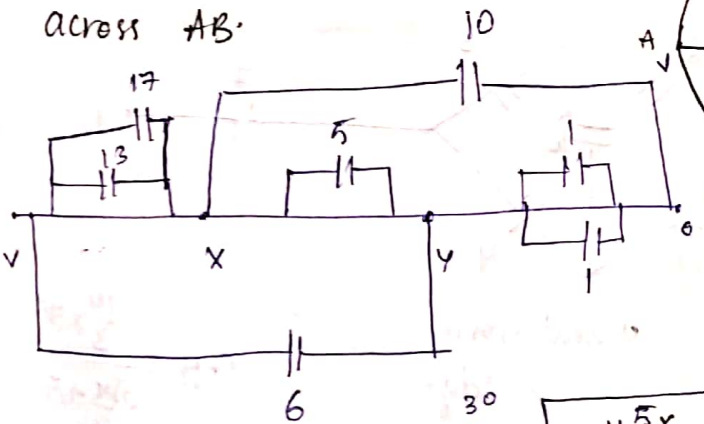
$= 11 \times 10$

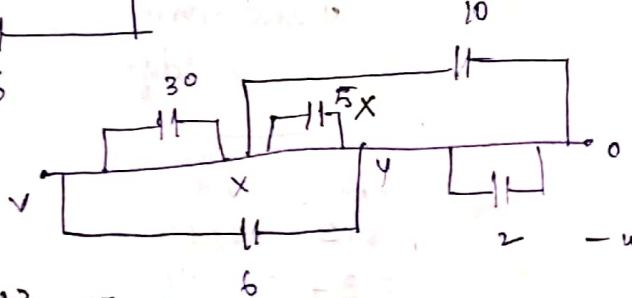
$q = 110 \mu C$

Ex: Find equivalent capacitance across AB.



Sol8





$\Rightarrow$

$C_{eff} = \frac{30 \times 10}{40} + \frac{12}{8} = \frac{15}{2} + \frac{3}{2} = \frac{18}{2} = 9.$

- wheat stone bridge.

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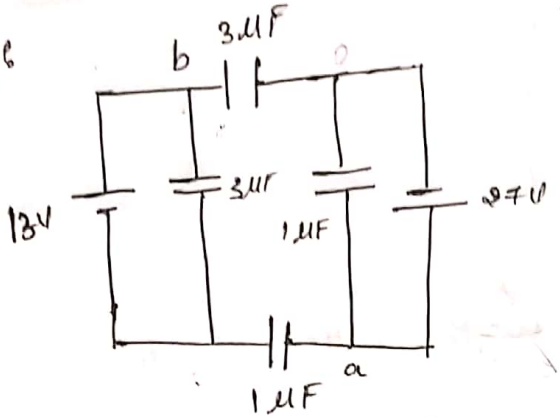
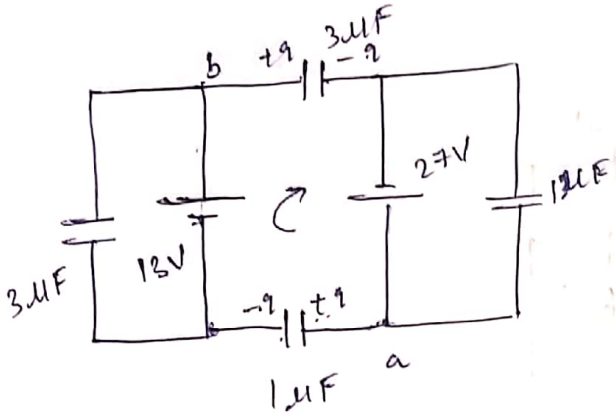
Exe



Potential difference b/w points a & b ?s

Solg



$13 - \frac{q}{3} + 27 - q = 0$

$\frac{4q}{3} = 40$

$q = 30$

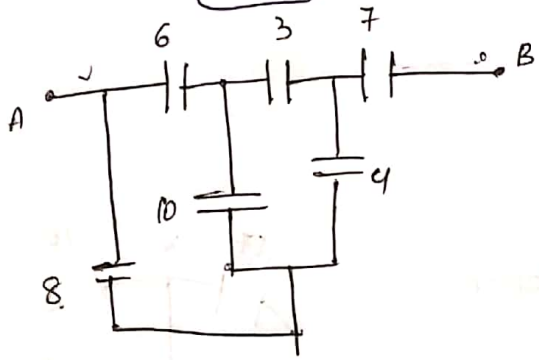
$V_b - 13 + q = V_a$

$V_b - V_a = 13 - q$

$V_a - V_b = q - 13$

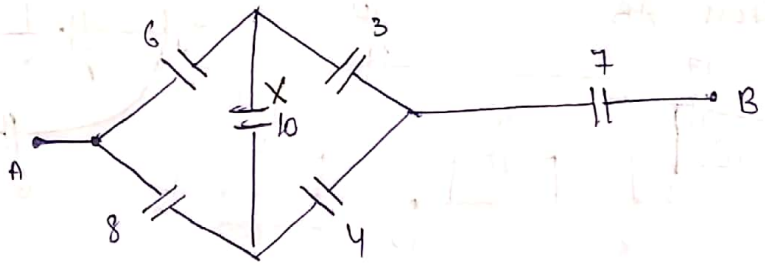
$V_a - V_b = 17$

Exe



The equivalent capacitance b/w A & B ?s

Solg



wheat stone bridge.

$\frac{6\times3}{6+3} + \frac{8\times4}{8+4}$

$= 2 + \frac{8}{3} = \frac{14}{3}$

$C_{eff} = \frac{\frac{14}{3}\times7}{\frac{21}{3}+7}$

$= \frac{14/3}{5/3}$

$= \frac{14}{5}$

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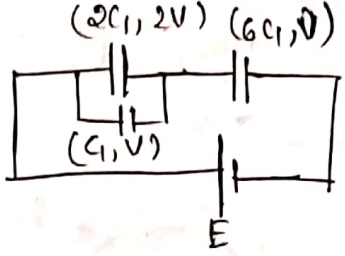
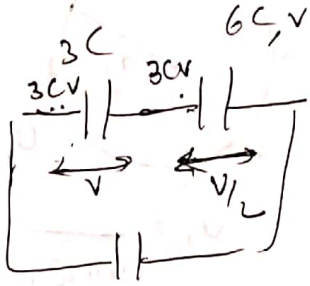


what is value of external emf such that no capacitor breaks.



$V = \frac{Q}{C}$

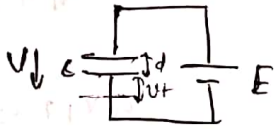
$= 0$

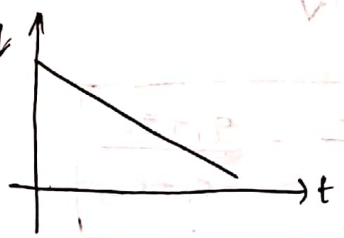
$V' = \frac{3CV}{6C}$

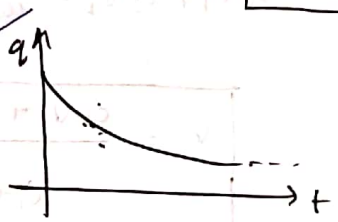
$V' = \frac{V}{2}$

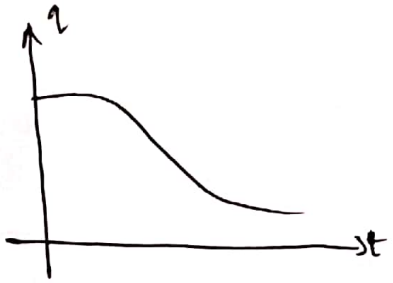
$V + \frac{V}{2} = \frac{3V}{2}$

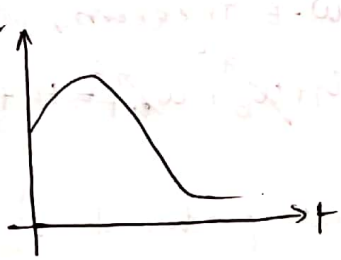
An ideal cell is connected across the capacitor as shown. Initial seperation b/w parallel plate is d. The lower plate is pulled down with ~~time~~ linear velocity v. Then the variation of charge on capacitor with time is given by



A) 

B) 

C) 

D) 

$Q = CV$

$Q = \frac{\epsilon_0 A}{d + vt} \times E$

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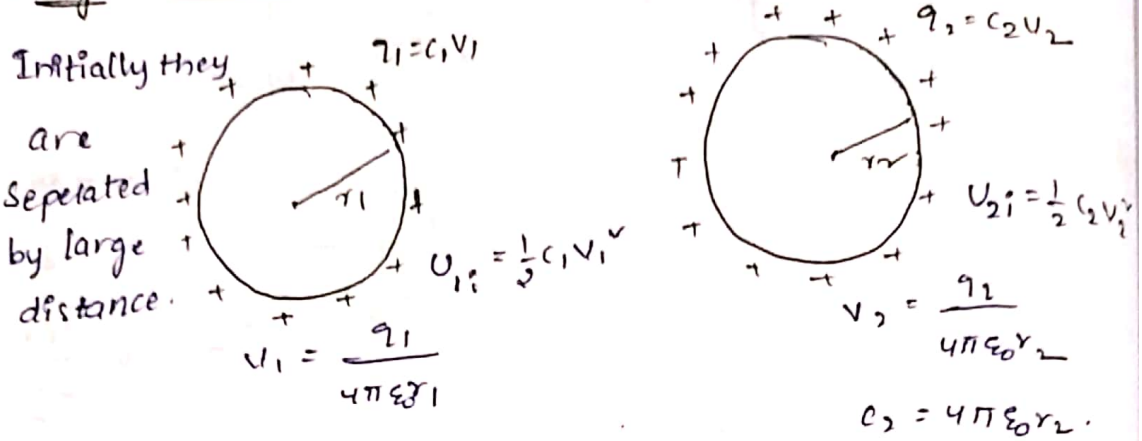
Charge distribution on metal conductors

Initially they are seperated by large distance.



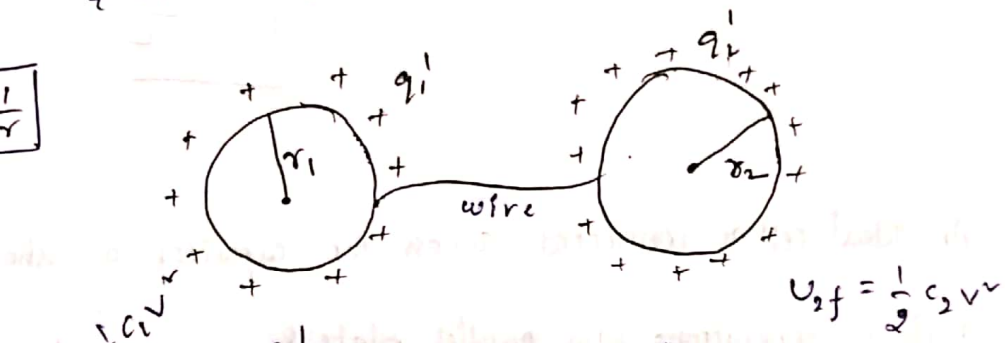
$$V_1 = \frac{q_1}{4\pi\varepsilon_0 r_1}$$

$$V_2 = \frac{q_2}{4\pi\varepsilon_0 r_2}$$

$$C_2 = 4\pi\varepsilon_0 r_2.$$

& $C_1 = 4\pi\varepsilon_0 r_1$

$\sigma \propto \frac{1}{r}$



$q_1' = C_1V$ $\quad$ $q_2' = C_2V$

* From law of conservation of charge,

$$q_1 + q_2 = q_1' + q_2'$$

$$C_1V_1 + C_2V_2 = (C_1+C_2)V$$

$$\boxed{V = \frac{C_1V_1 + C_2V_2}{C_1+C_2} = \frac{q_1+q_2}{C_1+C_2}}$$

* W-E Theorem,

$$U_i + \cancel{W_b}^{0} + \cancel{W_{ext}}^{0} = H + U_f$$

* heat produced $= U_i - U_f$

$$H = \frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2 - \frac{1}{2}(C_1+C_2)V^2$$

$$\boxed{H = \frac{1}{2}\left(\frac{C_1C_2}{C_1+C_2}\right)(V_1-V_2)^2}^{*}$$

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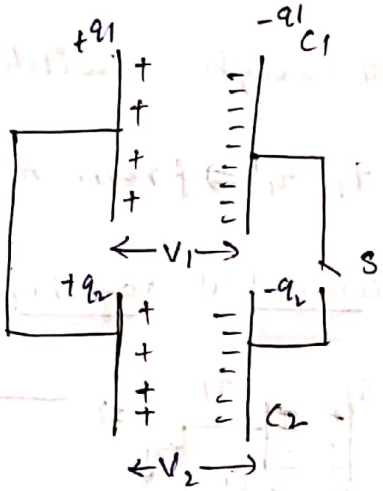
(or) $$H = \frac{1}{2}\frac{4\pi\varepsilon_0 r_1 r_2}{r_1 + r_2}(V_1 - V_2)^2$$

$\{H = ms\Delta\theta\}$

charge distribution in capacitors:

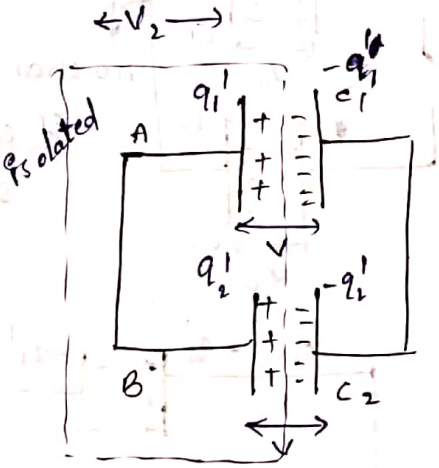
a) Like polarities are connected together:-



Before switch is closed –

$q_1 = C_1 V_1$

$q_2 = C_2 V_2$



After switch is closed –

Here,

Total charge is always const on isolated plates.

i.e, $q_1 + q_2 = q_1' + q_2'$

$$\Rightarrow V = \frac{q_1 + q_2}{C_1 + C_2} = \frac{\text{total charge}}{\text{total capacitance}}$$

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$ ← common potential.

* From W-E Theorem,

$$U_i + \cancel{W_b}^{0} + \cancel{W_{ent}}^{0} = U_f + H$$

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$$H = U_i - U_f$$

$$= \frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 - \frac{1}{2}(C_1+C_2)V^2$$

$$\boxed{H = \frac{1}{2} \frac{C_1 C_2}{C_1+C_2} (V_1-V_2)^2}$$

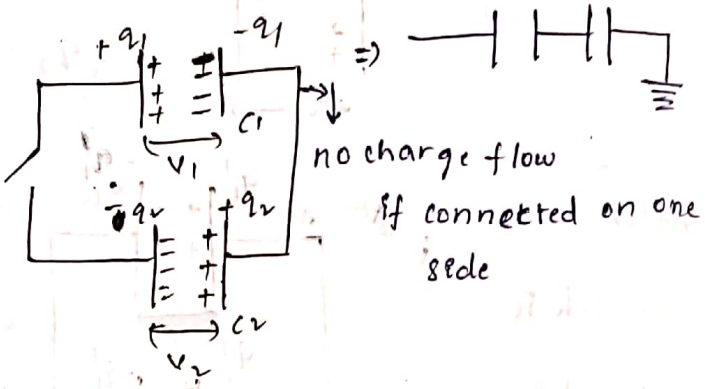
$\boxed{H = ms\Delta\theta}$

* charge flowing through switch,

$= q_1 - q_1' \Rightarrow$ from A to B

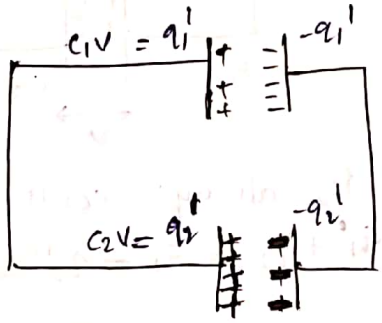
b) Unlike polarities are connected together -

Before switch is closed -



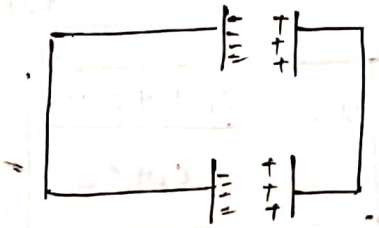
no charge flow if connected on one side

After switch is closed -

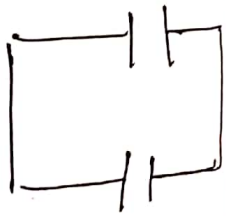


$C_1 V_1 > C_2 V_2$

(or)



(or)



$C_1 V_1 = C_2 V_2$

$C_1 V_1 < C_2 V_2$

From law of conservation of charge,

$$q_1 - q_2 = q_1' + q_2'$$

$$C_1 V_1 - C_2 V_2 = (C_1+C_2) V$$

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$$V = \frac{C_1V_1 - C_2V_2}{C_1 + C_2}$$

* Heat produced,

$$U_i + \cancel{W_b}^{0} + \cancel{W_{ent}}^{0} = U_f + H$$

$$H = U_i - U_f$$

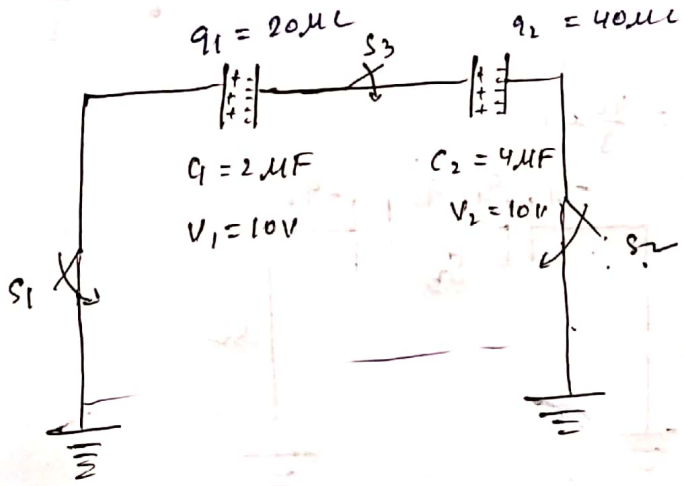
$$= \frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2 - \frac{1}{2}(C_1+C_2)V^2$$

$$H = \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_1+V_2)^2$$

* $H = ms\Delta\theta$

∴ Heat produced is more when unlike polarities are connected than like polarities are connected.

Ex6



* when only S_1 is closed, there is no charge flow through S_1.

* when only S_2 is closed, there is no charge flow through S_2.

* when only S_3 is closed, there is no charge flow through S_3.

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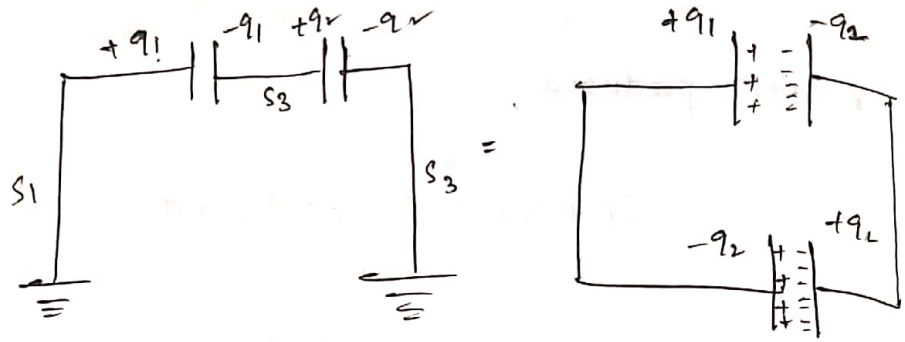
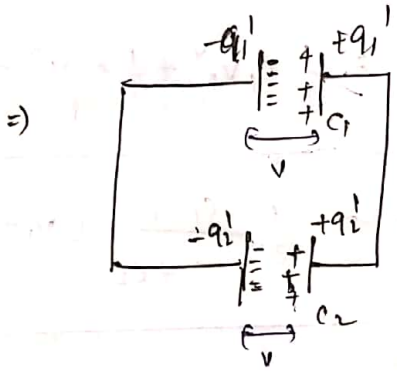
* when all switches are closed, there will be charge flow.



Just before closing switches.

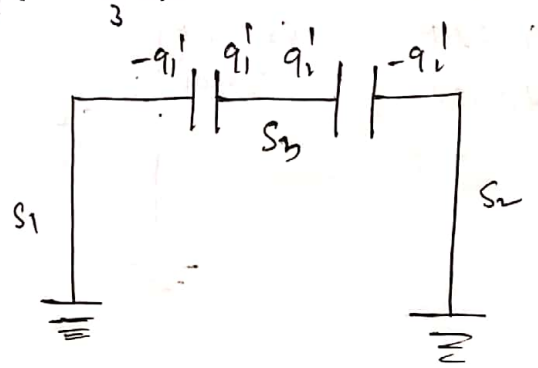


here $C_1V_1 < C_2V_2$.

$$V = \frac{C_2V_2 - C_1V_1}{C_1 + C_2}$$

$$q_1' = C_1V = \frac{20}{3}\mu C \qquad = \frac{20}{6} = \frac{10}{3}V$$

$$q_2' = C_2V = \frac{40}{3}\mu C$$



Charge flow through

1) S_1 i.e., $\Delta q_1 = q_1 - (-q_1')$

$= q_1 + q_1'$

$= 20 + \frac{20}{3}$

$= \frac{80}{3}$ (into earth)

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2) S_2 i.e, $\Delta q_2 = -q_2 - (-q_2')$

$= q_2' - q_2$

$= \frac{40}{3} - 40 = -\frac{80}{3}$ (into earth)

$= +\frac{80}{3}$ (from earth)

3) S_3 i.e, $\Delta q_3 = -q_1 - (+q_1')$

$= -q_1 - q_1'$

$= -\frac{80}{3}$ μc.

Heat produced,

M-I: $H = U_i - U_f$

$= \frac{1}{2} C_1 V_1^2 + \frac{1}{2} C_2 V_2^2 - \frac{1}{2}(C_1 + C_2) V^2$

$= \frac{1}{2} \times 2 \times 100 + \frac{1}{2} \times 4 \times 100 - \frac{1}{2}(6) \times \frac{100}{9}$

$= 300 - \frac{100}{3}$

$H = \frac{800}{3}$ J

M-II:

$H = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 + V_2)^2$

$= \frac{1}{2} \times \frac{2 \times 4}{6} \times (10 + 10)$

$= \frac{2}{3} \times 400 = \frac{800}{3}$

$H = \frac{800}{3}$ J

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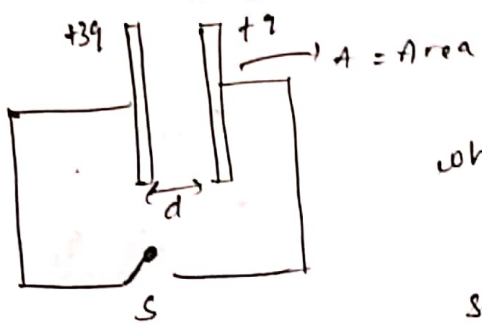
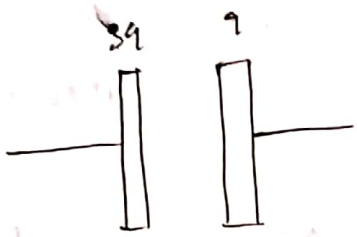
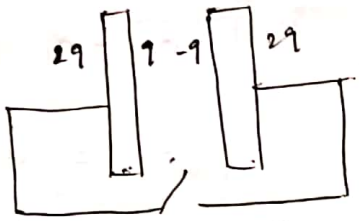
Ex:



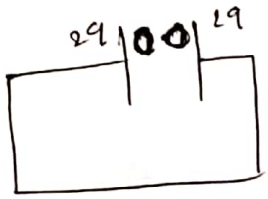
what is the heat produced when switch is closed.

Sol:





$\Rightarrow$



$U_f = 0.$

$$U_i = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2}\frac{q^2 d}{\varepsilon_0 A}$$

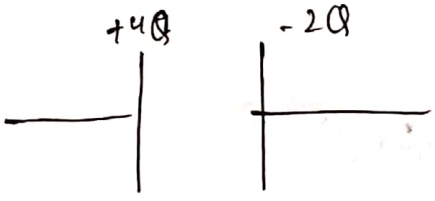
$U_f = 0.$

$$H = U_i - U_f$$

$$= \frac{1}{2}\frac{q^2 d}{\varepsilon_0 A} - 0$$

$$\boxed{H = \frac{q^2 d}{2\varepsilon_0 A}}$$

Ex:



The charge on the plates of capacitor are as shown. capacitor is now connected to an identical uncharged capacitor. Find the final potential difference across the capacitors.

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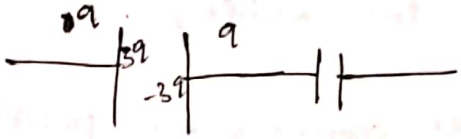
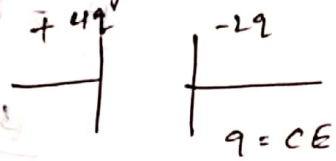
Sol6



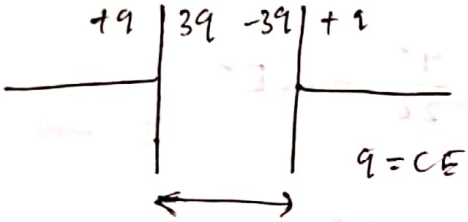
$$V = \frac{\text{total charge}}{C_1 + C_2}$$

$$\boxed{V = \frac{3q}{2C}}$$

Eg Two identical plates of capacitors are given charges as shown. Now the capacitor is connected to a battery of Emf E, with +ve plate of capacitor is connected to positive terminal of the battery. Then final charge on the capacitor plates is —.

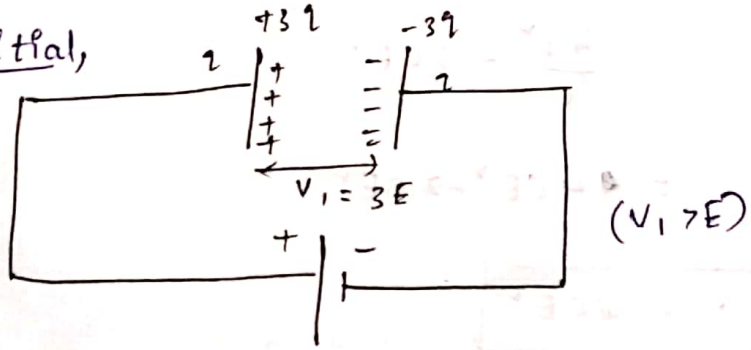


Sol:

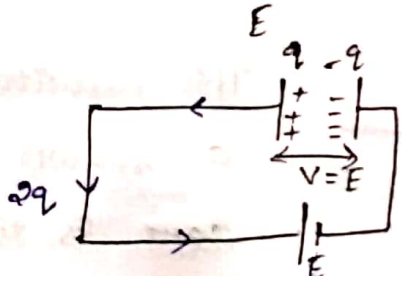


$$V_1 = \frac{3q}{C} = \frac{3CE}{C} = 3E$$

Initial,



Final,



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* work done on the battery.

* battery transfer charge from positive plate of capacitor to negative plate of capacitor through battery.

* work done by battery;

$$W_b = (\Delta Q)E$$

$$= -2qE$$

$$\boxed{W_b = -2CE^2}$$

* Heat produced,

$$H = U_i - U_f + W_b$$

$$= \frac{(3q)^2}{2C} - \frac{q^2}{2C} - 2CE^2$$

$$= \frac{9q^2}{2C} - \frac{q^2}{2C} - 2CE^2$$

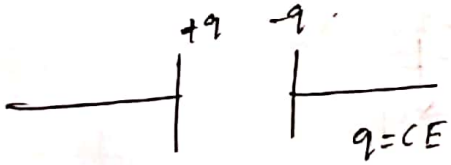
$$= \frac{8q^2}{2C} - 2CE^2$$

$$= \frac{4q^2}{C} - 2CE^2$$

$$= 4CE^2 - 2CE^2$$

$$\boxed{H = 2CE^2}$$

Ex 6



This capacitor connected as shown. Then what ~~work done~~ is the heat

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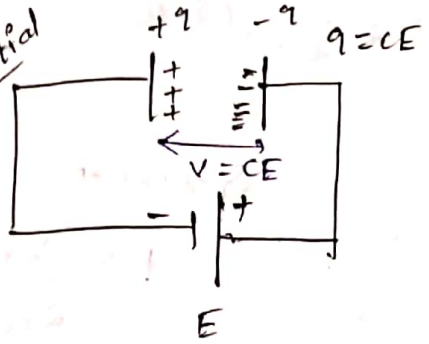
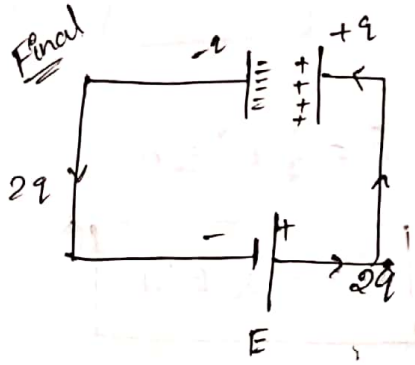
produced in the circuit.

Sol:

Initial



Final



$W_b = (2q)E$

$= (q-(-q))E$

$W_b = 2qE$

heat produced.

$H = U_i - U_f + W_b$

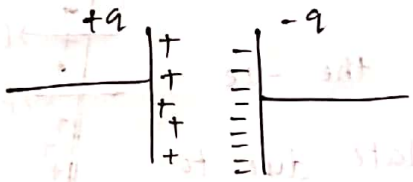
$= \frac{q^2}{2C} - \frac{q^2}{2C} + 2qE$

$$\boxed{H = 2CE^2}$$

Force of attraction b/w plates of capacitor:-

M-I:

* Energy stored in the capacitor,



$$U = \frac{q^2}{2C} = \frac{q^2 x}{2\varepsilon_0 A}$$

* Force b/w the plates of capacitor,

$$F = -\frac{dU}{dx}$$

$$= -\frac{d}{dx}\left(\frac{q^2 x}{2\varepsilon_0 A}\right)$$

$$\boxed{F = -\frac{q^2}{2\varepsilon_0 A}}$$

(-ve sign indicates force is attractive)

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$$\therefore \boxed{F = \frac{q^2}{2\varepsilon_0 A}} \quad * \qquad \Rightarrow F = \frac{1}{2}\frac{q^2}{\varepsilon_0^2 A^2}(\varepsilon_0 A)$$

$$F = \frac{1}{2} E^2 \varepsilon_0 A$$

$$\boxed{F = \frac{1}{2}\varepsilon_0 E^2 A} \quad *$$

$$F = \frac{q}{2} \times \frac{q}{\varepsilon_0 A}$$

$$\boxed{F = \frac{1}{2} E q} \quad *$$

Q → const

$$F = \frac{q^2}{2\varepsilon_0 A}.$$

V → const

$$F = \frac{1}{2}\varepsilon_0 \frac{V^2}{d^2} \cdot A.$$

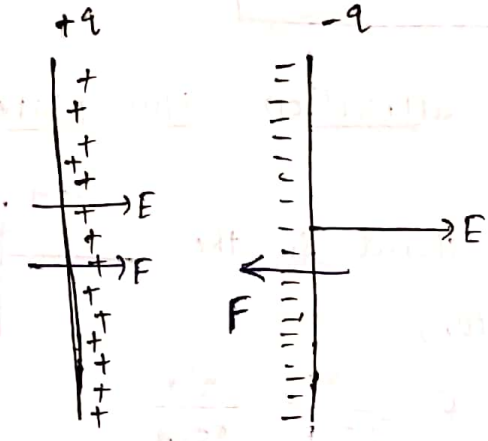
(or) $E = \frac{V}{d}$; d = distance b/w the plates.

$$\boxed{F = \frac{1}{2}\varepsilon_0 \frac{V^2}{d^2} A}$$

M-II ℓo

we have,

field at the -ve charged plate due to +ve charged plate is,



$$\boxed{E = \frac{\sigma}{2\varepsilon_0}}$$

∴ Force on the -ve charged plate,

$$F = Eq$$

$$= \frac{\sigma q}{2\varepsilon_0} = \frac{q^2}{2\varepsilon_0 A}$$

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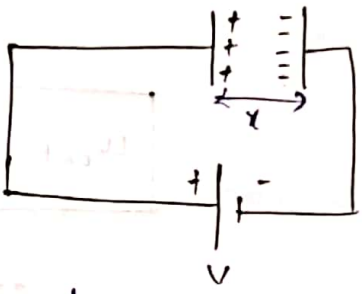
imp points

* work done by the External Agent :-

1) When battery is still continuing in the circuit :

Here,

separation b/w the plates is increased slowly from x_1 to x_2



* work is done by the external agent.,
* Battery is continuing / still present in the circuit,

potential difference $\Rightarrow V = \text{const.}$

$\therefore$ Force at any intermediate state,

Force b/w plates

$$F = \frac{1}{2} \varepsilon_0 \frac{V^2}{x^2} \cdot A = F_{ext}$$

* It is a variable force.

X $F = -\frac{dU}{dx}$

$$dU = -F\,dx$$

$$U_f - U_i = -\int_{x=x_1}^{x=x_2} \frac{1}{2} \varepsilon_0 \frac{V^2}{x^2} \cdot A \cdot dx$$

$$= \frac{1}{2} \varepsilon_0 V^2 A \left[-\frac{1}{x}\right]_{x_1}^{x_2}$$

$$= -\frac{1}{2} \varepsilon_0 V^2 A \left[\frac{1}{x_1} - \frac{1}{x_2}\right]$$

$$\boxed{\Delta U_f - U_i = -\frac{1}{2} \varepsilon_0 V^2 A \left(\frac{x_2 - x_1}{x_1 x_2}\right)}$$

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now,

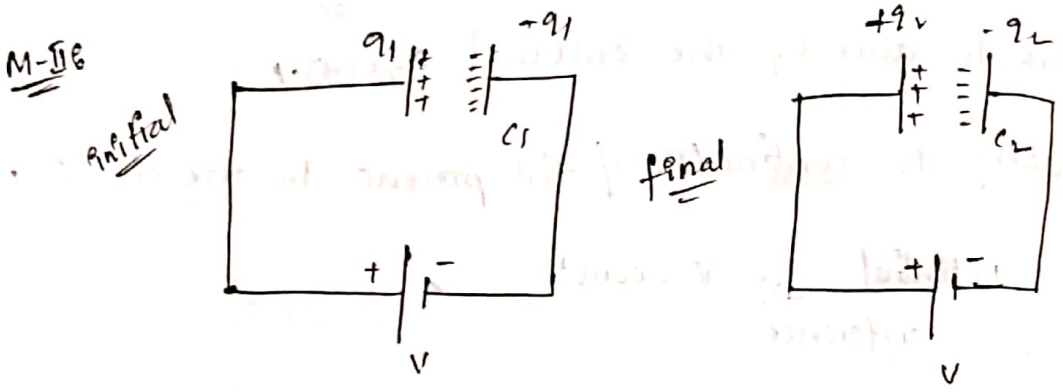
$$dW_{ext} = F_{ext} \cdot dx$$

$$W_{ext} = \int_{x_1}^{x_2} F_{ext} \cdot dx$$

$$= \frac{1}{2} \varepsilon_0 V^2 A \left[\frac{1}{x}\right]_{x_1}^{x_2}$$

$$\boxed{W_{ext.} = \frac{1}{2} \varepsilon_0 V^2 A \left(\frac{1}{x_1} - \frac{1}{x_2}\right)}$$

M-II 6



Initial,

$$C_1 = \frac{\varepsilon_0 A}{x_1}$$

$$V_1 = V$$

$$q_1 = C_1 V_1 = \frac{\varepsilon_0 A V}{x_1}$$

$$U_i = \frac{1}{2} C_1 V^2$$

$$= \frac{1}{2} \frac{\varepsilon_0 A}{x_1} V^2$$

Final,

$$C_2 = \frac{\varepsilon_0 A}{x_2}$$

$$V_2 = V$$

$$q_2 = C_2 V = \frac{\varepsilon_0 A V}{x_2}$$

$$U_f = \frac{1}{2} C_2 V^2$$

$$= \frac{1}{2} \frac{\varepsilon_0 A V^2}{x_2}$$

work done by battery,

$$W_b = (\Delta q) V$$

$$= (q_2 - q_1) V$$

$$= \left(\frac{\varepsilon_0 A}{x_2} - \frac{\varepsilon_0 A}{x_1}\right) V^2$$

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we have,

$$U_i + W_b + W_{ext} = U_f + H$$

Here plate are displaced very slowly, so heat produced in the circuit = 0.

$$\Rightarrow \boxed{H = 0}.$$

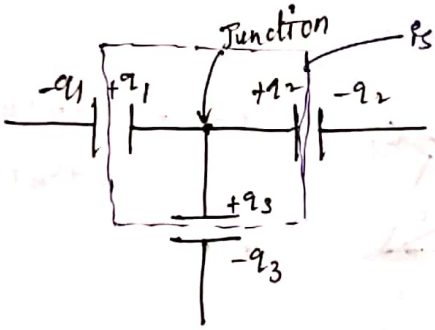
$$U_i + W_b + W_{ext} = U_f + H$$

$$\frac{1}{2}\frac{\varepsilon_0 A}{x_1} V^2 + (q_2 - q_1)V + W_{ext} = \frac{1}{2}\frac{\varepsilon_0 A}{x_2}(V^2)$$

$$\boxed{W_{ext} = \frac{1}{2}\varepsilon_0 A V^2\left(-\frac{1}{x_1} + \frac{1}{x_2}\right)}$$

Circuit Analysis :-

① Kirchoff's Law :-

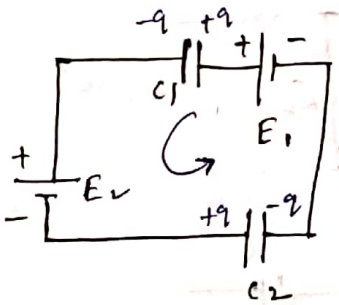


At junction,

$$\boxed{q_1 + q_2 + q_3 = 0}$$

② Loop Law :-

$$\Sigma V = 0 \Rightarrow \Sigma E + \Sigma \frac{q}{C} = 0$$



$$E_1 - \frac{q}{C_1} - E_2 - \frac{q}{C_2} = 0$$

$$E_1 - E_2 = q\left(\frac{1}{C_1} + \frac{1}{C_2}\right) \Rightarrow \boxed{E_1 - E_2 = \frac{q(C_1 + C_2)}{C_1 C_2}}$$

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Nodal Analysis :-

At any node, $\Sigma q = P$. $\boxed{q = CV}$

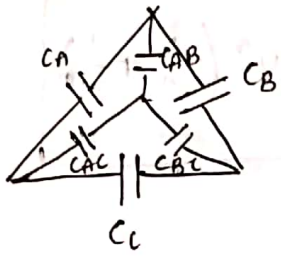
Delta Star :-

Resistors-

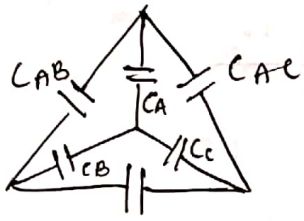
$$R_{AB} = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_{AB} = \frac{R_A R_B + R_B R_C + R_A R_C}{R_C}$$

Capacitors-



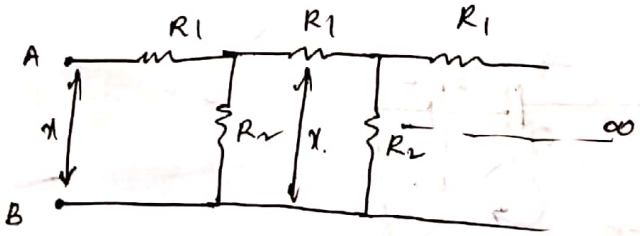
$$C_{AB} = \frac{C_A C_B + C_B C_C + C_A C_C}{C_C}$$



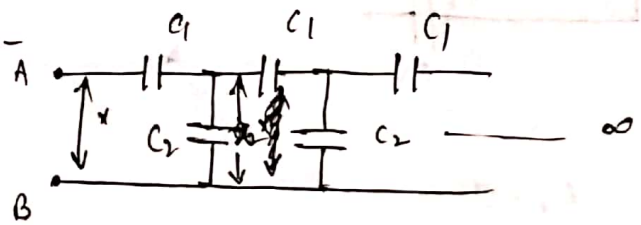
$$C_{AB} = \frac{C_A C_B}{C_A + C_B + C_C}$$

Infinite ladder problems :-

resistors-



capacitors-



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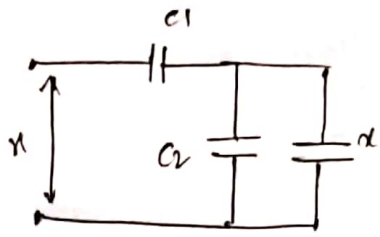


$$x = \frac{C_1(C_2+x)}{C_1+C_2+x}$$

solve for x.

* $C_{eff} = \frac{I}{V_A - V_B} = \frac{\text{Charge flowing through battery}}{\text{emf of battery}}$.

Ex: field b/w ● parallel plate capacitor is $E = 5\times10^6$ v/m & charge on plate is $0.5\,\mu c$ then Force acting on the plate is

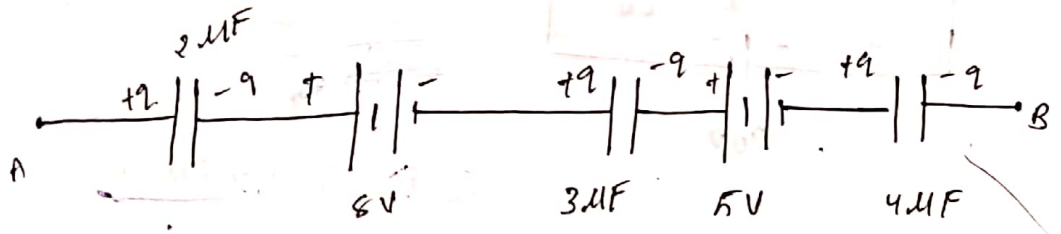
Sol:

$$F = \frac{1}{2}Eq$$

$$= \frac{1}{2}\times5\times10^6\times0.5\times10^{-6}$$

$$\boxed{F = 1.25\,N}$$

Ex:



If $V_A - V_B = 23$ V. Find potential difference across $3\mu F$ capacitor.

Sol:

$$V_A - \frac{q}{2} - 8 - \frac{q}{3} - 5 - \frac{q}{4} = V_B$$

$$V_A - V_B = q\left(\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right) + 13$$

$$10 = q\left(\frac{6+4+3}{12}\right) \Rightarrow \boxed{q = \frac{120}{13}\,\mu c}$$

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$$V = \frac{q}{C} = \frac{120/13}{3}$$

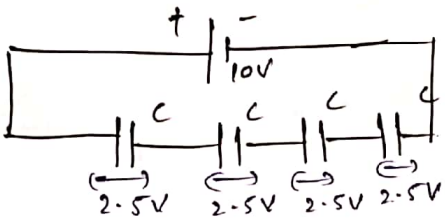
$$\boxed{V = 40/13\ V}$$

* Energy stored,

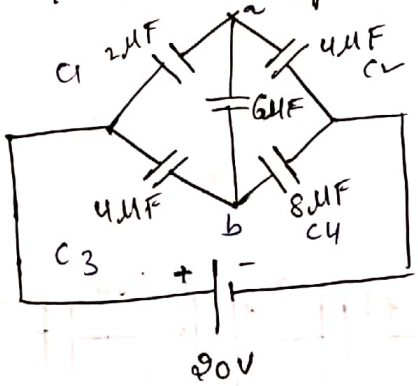
$$U = \frac{1}{2} C V^2$$

* work done by the battery is -ve.

Exe



wheat stone Bridge



$$\frac{C_1}{C_2} = \frac{1}{2}$$

$$\frac{C_3}{C_4} = \frac{1}{2}$$

$$\boxed{V_a = V_b}$$

C_5 is ineffective.

$$C_{eff} = \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4}$$

$$= \frac{2 \times 4}{6} + \frac{4 \times 8}{12}$$

$$= \frac{8 + 16}{6} = \frac{24}{6} = 4.$$

$$U = \frac{1}{2} C_{eff} V^2$$

$$H = \frac{1}{2} C_{eff} V^2$$

$$W_b = C_{eff} V^2$$

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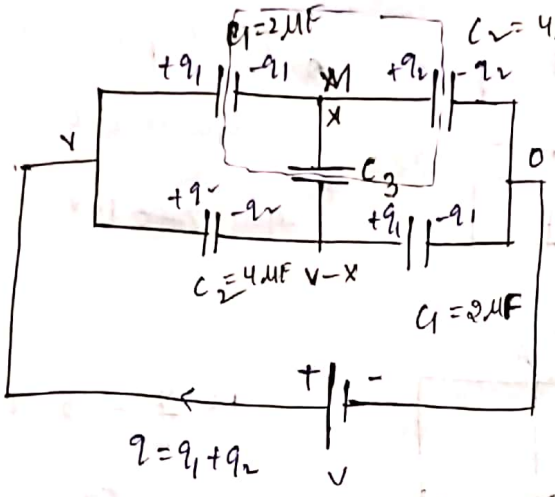
Ex 6



at node M, $\Sigma q = 0$

$(x-v)c_1 + (x-0)c_2 + (x-v+x)c_3 = 0$

$q_1 = (V-x)c_1$

$q_2 = (V-(V-x))c_2$

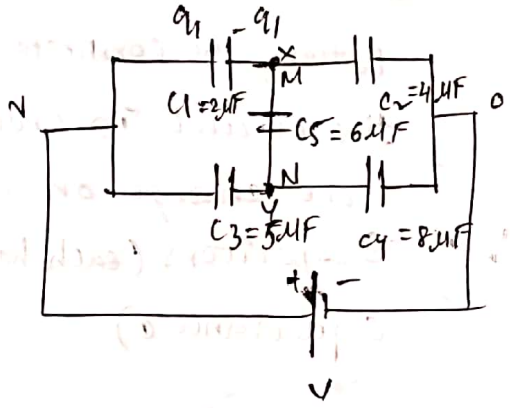
Finally, $c_{eff} = \frac{q_1+q}{V}$

$-q_1 + q_2 + x = 0$

$x = q_1 - q_2$

$$c_{eff} = \frac{q}{V} = \frac{q_1+q_2}{V}$$

Ex 8



here, at node M,

$(x-v)c_1 + (x-0)c_2 + (x-y)c_5 = 0$ — ①

at node N,

$(y-v)c_3 + (y-x)c_5 + (y-0)c_4 = 0$ — ②

Solve ①, ② for x, y.

now, $q_1 = (V-x)c_1$

$q_2 = (V-y)c_3$

$$c_{eff} = \frac{q_1+q_2}{V}$$

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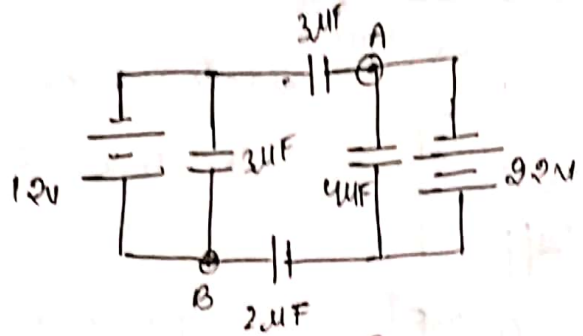
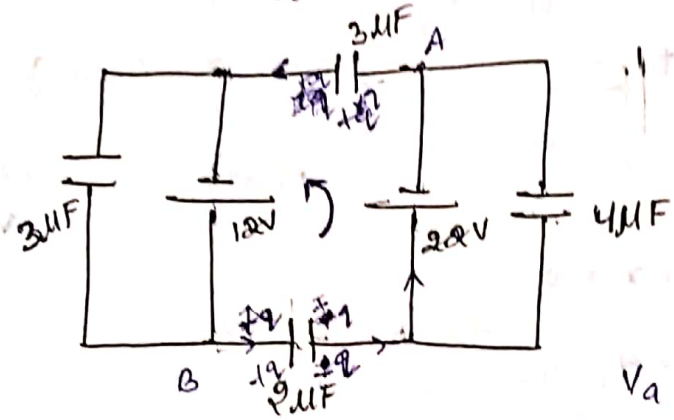
Ex 6



Find potential difference b/w A & B.

Ans: ~~7440~~ 16V

Sol 6



$$12 - \frac{q}{2} - 22 - \frac{q}{3} = 0$$

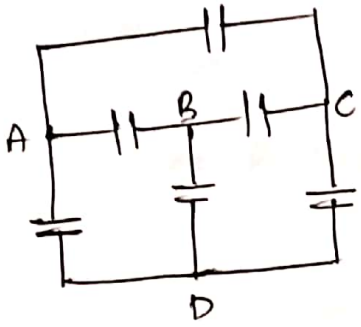
$$\frac{5q}{6} = -10$$

$$q = +12\mu C$$

$$V_a + 22 - \frac{q}{2} = V_b$$

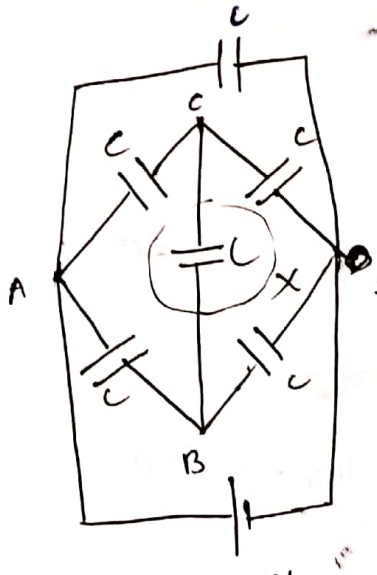
$$V_a - V_b = -22 + 6 = -16\,V$$

Ex 8



Between which pts, a battery is connected in order to store charge on all the 6 capacitors (each having capacitance C)

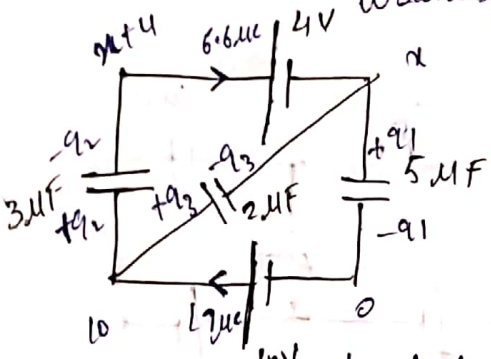
Sol 8



⇒ similarly at all pts we get wheatstone bridge & one of 6 capacitors is un effected by anything.

at any two points.

So battery is not connected between A & B (or) B & A.



w done by battery = −26.4 μJ

Total w done

= 190 − 26.4

= 163.6 μJ.

w done by battery = 190 μJ

At node, $\Sigma q = 0$

$$(x-0)(5) + (x-10)(2) + (x+4-10)(3) = 0$$

$$5x + 2x - 10 + 3x - 18 = 0$$

$$10x = 38$$

$$\boxed{x = 3.8\ V}$$

$$q_1 = 3.8 \times 5 = 19\,\mu c$$

$$q_2 = (10 - 7.8) \times 3 = 2.2 \times 3 = 6.6\,\mu c$$

$$q_3 = (10 - 3.8) \times 2 = 6.2 \times 2 = 12.4\,\mu c$$

$$\text{Heat} = \frac{163.6}{2} = 81.8\,\mu J$$

Stored energy = 81.8 μJ

$$U_1 = \frac{1}{2} \times 19 \times 19 \times \frac{1}{5} = \frac{361}{10} = 36.1\,\mu J$$

$$U_2 = \frac{1}{2} \times 6.6 \times 6.6 \times \frac{1}{5} = 7.26\,\mu J$$

$$U_3 = \frac{1}{2} \times 12.4 \times 12.4 \times \frac{1}{2} = 38.44\,\mu J$$

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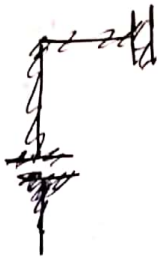
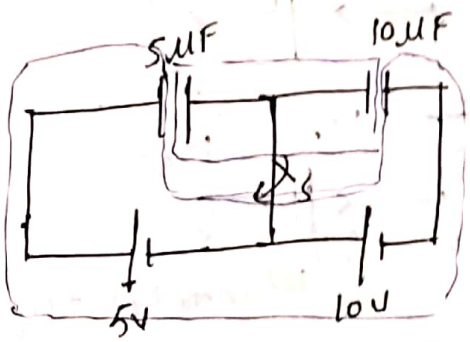
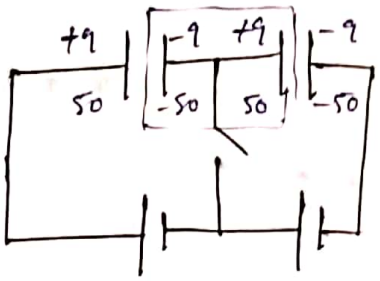
Switch Problems:





In these problems work done by external agent is zero.

* Before closing the switch,



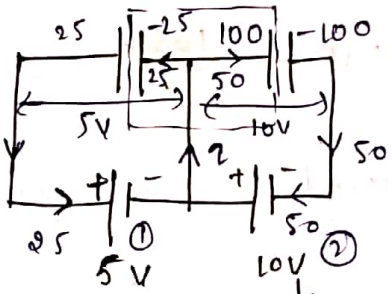
$$10 + 5 - \frac{q}{5} - \frac{q}{10} = 0$$

$$15 = q\left(\frac{2+1}{10}\right)$$

$$q = \frac{150}{3}$$

$$\boxed{q = 50\mu C}$$

* After closing the switch,



$$q = 100 - 25 = 75\mu C$$

↳ charge

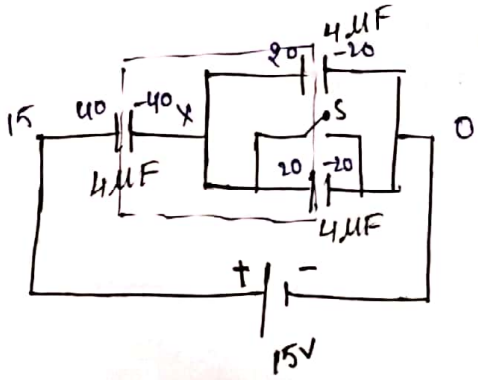
Charging W_b = -ve

Discharging W_b = +ve

* $W_{b_1} = 25(5) = 125$

$W_{b_2} = -10 \times 50 = +500$

Ex 6



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Solb at node, $\Sigma q = 0$

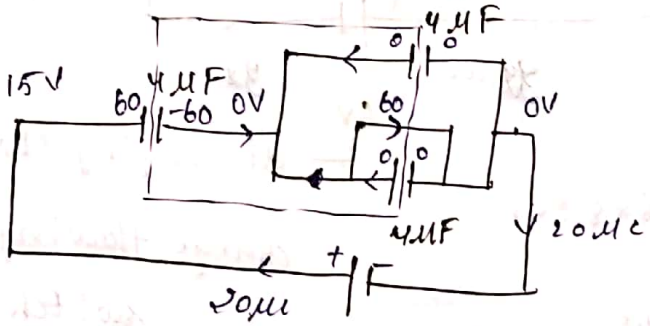
$$(x-15)4 + (x-0)4 + (x-0)4 = 0$$

$$12x = 60$$

$$\boxed{x = 5\,V}$$

charge through switch = 0

* After switch is closed ⇒



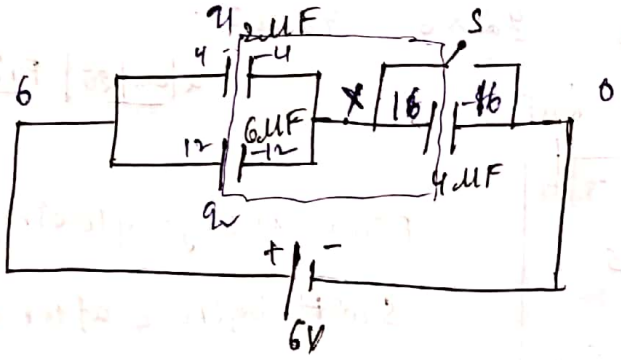
15 V → discharging W_b = +ve

charge through switch = 60 μC

Charge flown through battery = 20

$$W_b = 20 \times 15 = 300\,\mu J.$$

Ex6



$V = \frac{q}{C}$

$q = VC$

Solb

at node, $\Sigma q = 0$

$$(x-6)(2) + (x-6)(6) + (x-0)(4) = 0$$

$$2x - 12 + 6x - 36 + 4x = 0$$

$$12x = 48$$

$$\boxed{x = 4}$$

$$q = \frac{(6-4)(8)}{} = 16.$$

$q_1 : q_2 = 2 : 6 = 1 : 3$

$q_1 = \frac{16}{4} = 4 \quad q_2 = 12$

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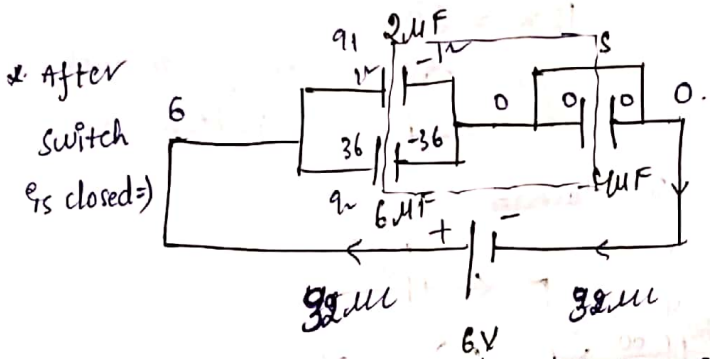
Before closing switch,

charge flowing through switch

= 0 μC.



* After switch is closed ⇒

discharging w_b = +ve

$q = 6 \times 8 = 48$

charge flowing through switch = 48μC.

$q_1 = \frac{48}{4} = 12$

$q_2 = 3 \times \frac{48}{4} = 36$

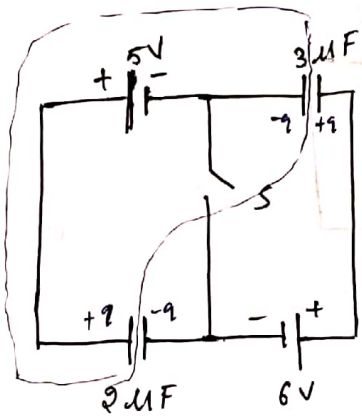
charge flowing through battery = 48 − 16

= 32 μC

$w_b = 32 \times 6 = 192 \mu J$.

2|05|25| Friday

Ex 6



Find charge flowing through switch before & after closing.

Sol 6

before closing the switch,

$$-\frac{q}{2} + 5 - \frac{q}{3} + 6 = 0$$

$$+q\left(\frac{1}{3} + \frac{1}{2}\right) = 11$$

$$\boxed{q = \frac{66}{5}} = 13.2$$

Charge flow through switch = 0.

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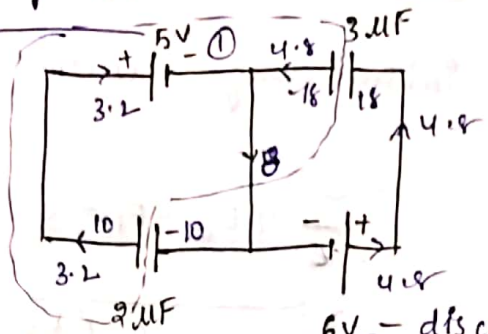
* After closing the switch.



charging, w_b = -ve

6V ② – discharge, w_b = +ve

charge flow through battery = 8 μc

$$w_{b_1} = 3.2 \times 5 = -16\ \mu J$$

$$w_{b_2} = 4.8 \times 6 = 28.8\ \mu J$$

$$w_b = 28.8 - 16 = 12.8\ \mu J \Rightarrow H = \frac{12.8}{2} = 6.4\ \mu J$$

* $U_i + w_b = U_f + H$

$$\frac{1}{2} \times \frac{3}{5} \times 11 \times 11 + 12.8 = \frac{1}{2} \times 2 \times 5^2 + \frac{1}{2} \times 3 \times 6^2 + H$$

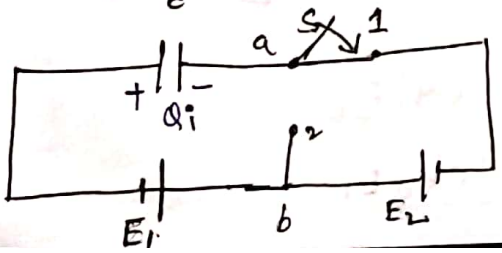
$$\frac{3 \times 121}{5} + 12.8 = 25 + 18 \times 3 + H$$

$$0.6 \times 121 + 12.8 = 25 + 54 + H$$

$$72.6 + 12.8 - 79 = H$$

$$\boxed{H = 6.4}$$

Ex 8 what amount of heat will be generated in the circuit after the switch S is shifted from position 1 to position 2?



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Sol:

* when switch is in position 1.

Q_i = Initial charge with polarities as shown

$$-\frac{Q_i}{C} + E_2 - E_1 = 0$$

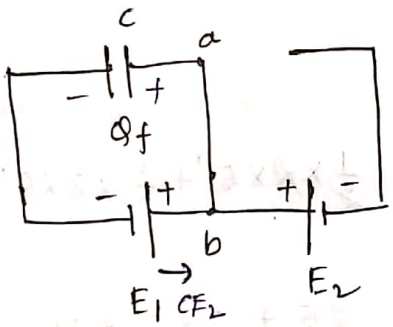
$$\boxed{Q_i = C(E_2 - E_1)}$$

If $E_2 > E_1$ then $Q_i = +ve$

If $E_2 < E_1$ then $Q_i = -ve$

If $E_2 = E_1$ then $Q_i = 0$

* when switch is shifted to position 2.



now, $Q_f = CE_1$

charge flowing through switch,

$$\Delta Q = Q_i - Q_f$$

$$= -C(E_2 - E_1) - CE_1$$

$$= -CE_2 + CE_1 - CE_1$$

$$= -CE_2$$

* $W_b = (\Delta Q) E_1$

$= CE_2 E_1$

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$$U_i = \frac{1}{2} C(E_2 - E_1)^2 = \frac{1}{2} C E_1^2 + \frac{1}{2} C E_2^2 - C E_1 E_2$$

$$U_f = \frac{1}{2} C E_2^2$$

$$W_b = C E_1 E_2$$

$$W_{ext} = 0$$

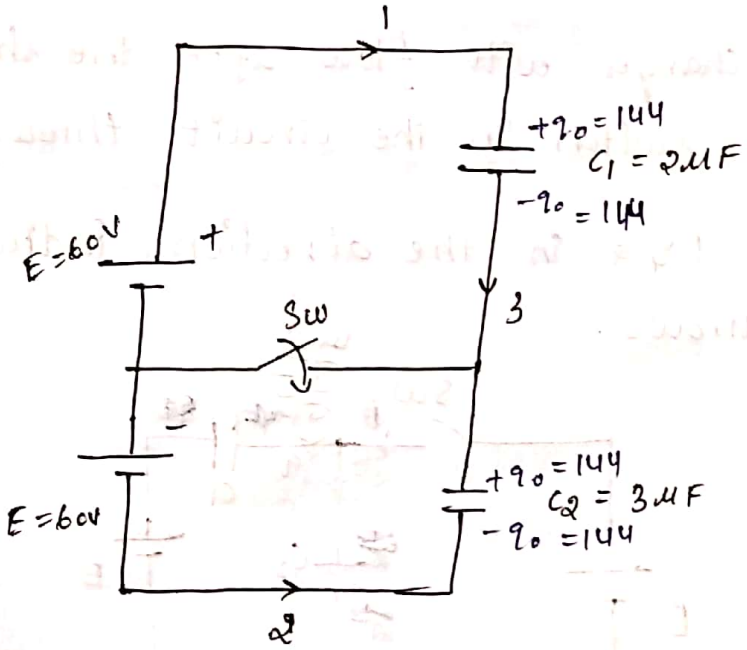
$$H = ?$$

$$H = U_i - U_f + W_b$$

$$= \frac{1}{2} C E_1^2 + \frac{1}{2} C E_2^2 - C E_1 E_2 - \frac{1}{2} C E_1^2 + C E_1 E_2$$

$$\boxed{H = \frac{1}{2} C E_2^2}$$

Ex.6 Find the charges which will flow after the ~~charging~~ shortening of the switch Sw through the section 1, 2 & 3 in the directions indicated in arrow.



Sol.

$V_1 : V_2 = 3 : 2$

$$V_1 = \frac{3}{5} \times 120 = 3 \times 24$$

$$V_2 = \frac{2}{5} \times 120 = 2 \times 24$$

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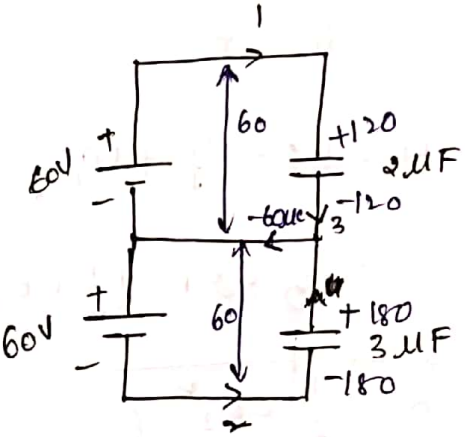
Before closing the switch:

$$-\frac{q_0}{2} - \frac{q_0}{3} + 120 = 0$$

$$\boxed{q_0 = 144\,\mu C}$$

when switch is closed,



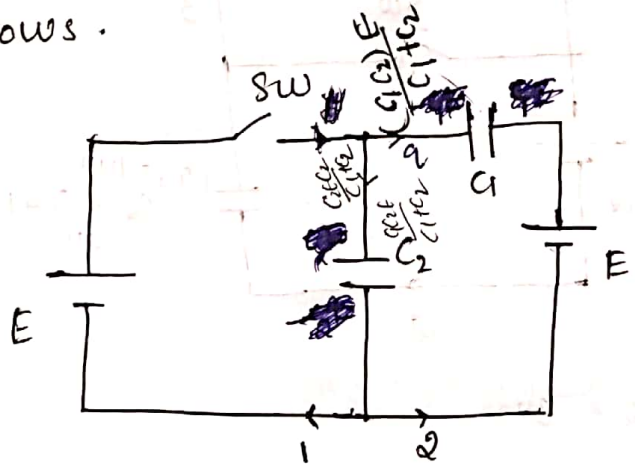
charge through 1
$= -24\,\mu C$

Charge through 2
$= -36\,\mu C$

Charge through 3
$= -24\,\mu C$

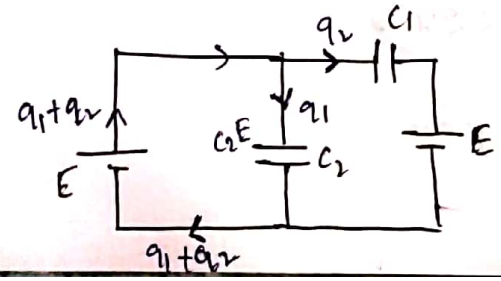
Charge through switch $= -60$

Ex 6 what charges will flow after the shortening of the switch in the circuit through the sections 1 & 2 in the directions indicated by the arrows.



when switch is opened $q = \frac{C_1 C_2 (E}{C_1 + C_2}$

Sol 6



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$$E = \frac{q_1}{C_2}$$

$$\boxed{q_1 = C_2 E}$$

$$-\frac{q_1}{C_2} + E + \frac{q_2}{C_1} = 0$$

$$\boxed{q_2 = 0}$$

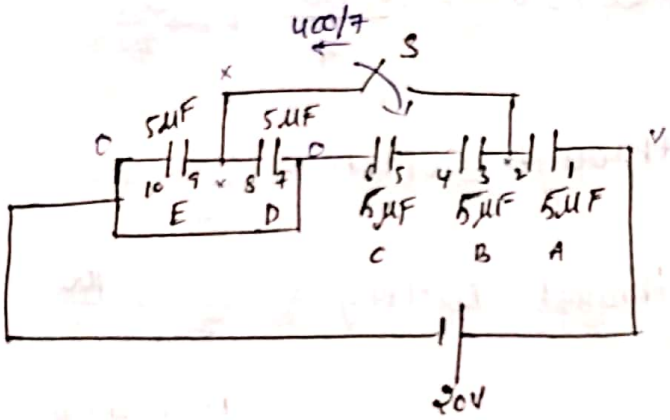
∴ charge through 1 .

$$= C_2 E$$

charge through 2

$$= -\frac{C_1 C_2 E}{C_1 + C_2}$$

Ex6



Find ① charge flowing through switch.

② Heat produced in the circuit.

Sol6

before closing switch,

D, E are short circuited.

Charge through A, B, C $= 20 \times \frac{5}{3}$

$= \frac{100}{3}$.

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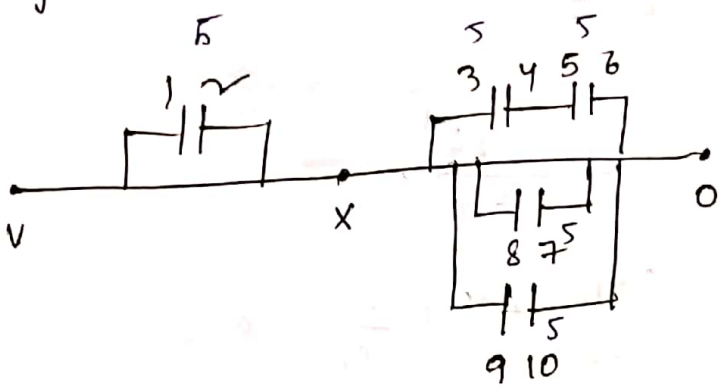
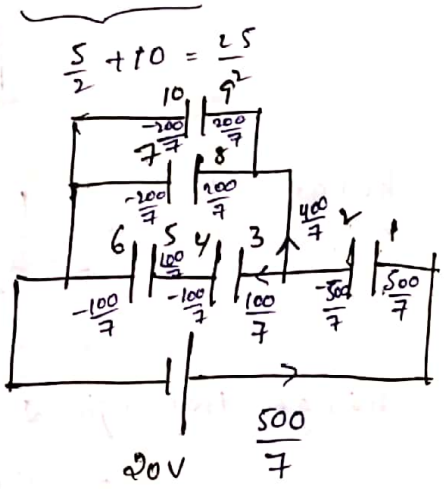
After closing switch,



$$C_{eff} = \frac{\frac{25}{2} \times 5}{\frac{25}{2} + 10}$$

$$\boxed{C_{eff} = \frac{25}{7}\,\mu F}$$

$$q = \frac{25}{7} \times 20 = \frac{500}{7}\,\mu c$$



$q_{through\ switch}$ charge through switch $= \frac{400}{7}\,\mu c$

Charge through battery $= \frac{500}{7} - \frac{100}{3}$

$$= \frac{1500 - 700}{21}$$

$$= \frac{800}{21}$$

battery is discharging.

W_b = +ve

$$W_b = \frac{800}{21} \times 20 = \frac{16000}{21}$$

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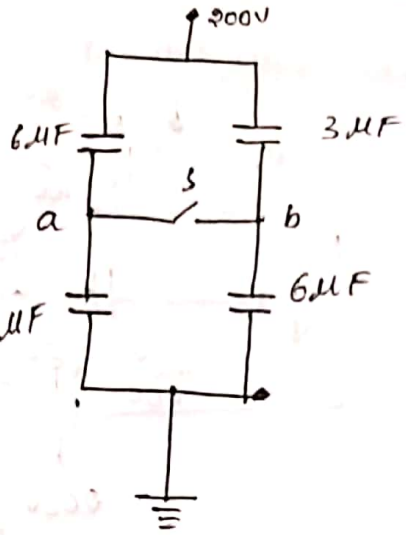
Ex:

a) Find V_{ab} when switch is opened. $\frac{200}{3}$

b) Find absolute potentials at a & b $\frac{400}{3}$ $\frac{200}{3}$

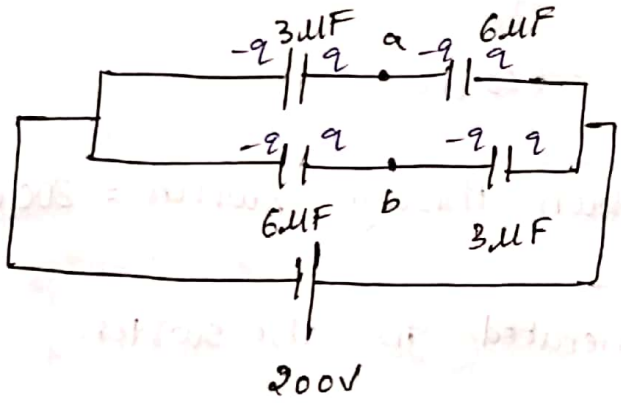
c) Find charge flowing through switch. 300

d) Heat generated in the switch.



Soln

* before switch is closed,



$C_{eff} = 4$ $\Rightarrow$ $q = \frac{4}{6} \times 200 = 800\,\mu C$.

i) $V_a - \frac{400}{3} + \frac{400}{6} = V_b$

$$V_a - V_b = \frac{400}{3} - \frac{200}{3}$$

$$\boxed{V_a - V_b = \frac{200}{3}}$$

ii) $V_a = \frac{400}{3}$, $V_b = \frac{200}{3}$.

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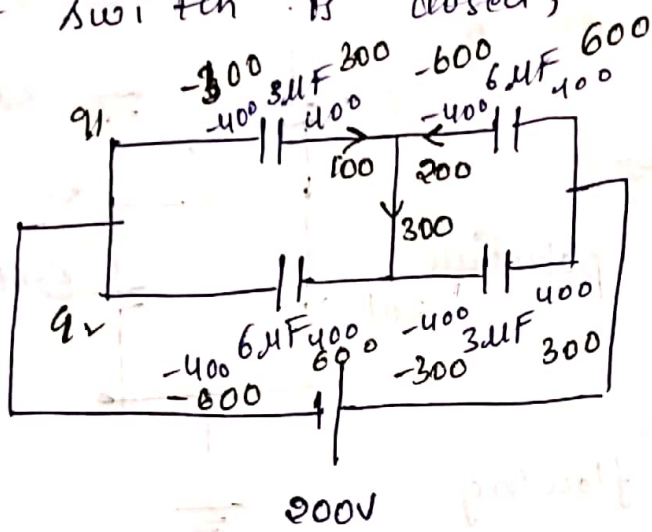
* After switch is closed,



$C_{eff} = \frac{q}{2} \Rightarrow q = 900\mu C$

$q_1 : q_2 = 1 : 2$

$q_1 = \frac{900}{3} = 300\mu C$

$q_2 = 600\mu C$

3) ∴ charge flocon through switch = 300μC.

4) Heat generated in the switch,

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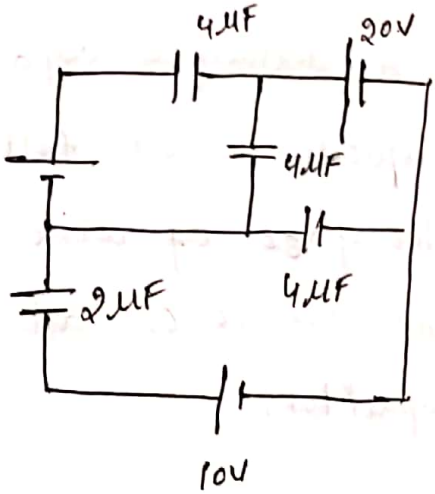
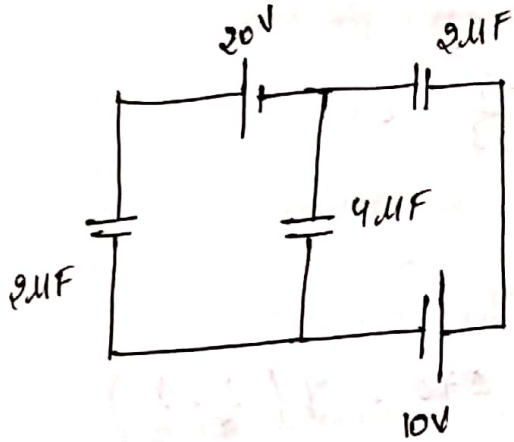
Ex 6

Find charge & energy stored in all capacitors.



Ex 6



Find charge & energy stored in all capacitors.

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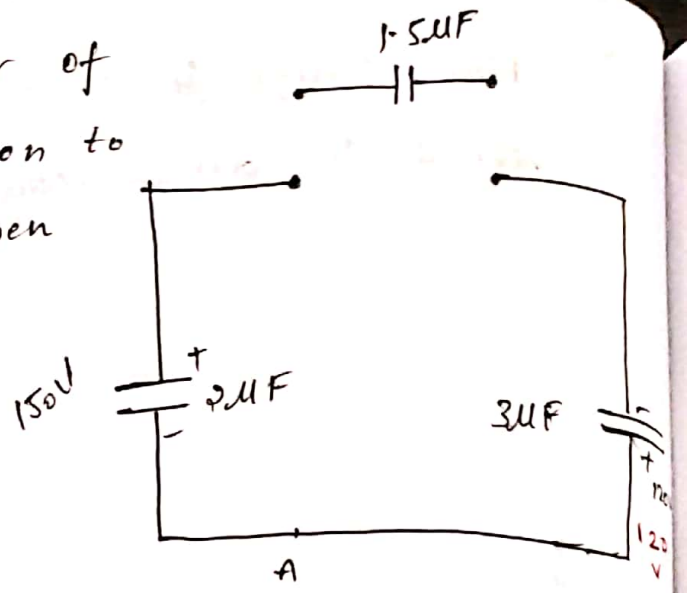
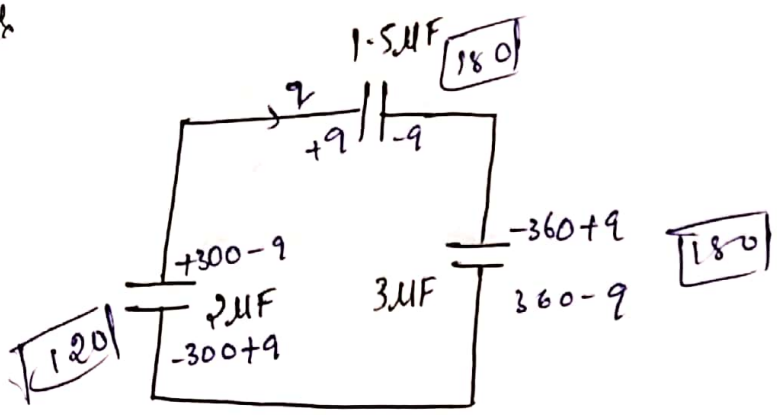
Ex: A discharged capacitor of capacity 1·5μF falls on to the free of wire then find charge on all capacitors.



Sol:



$$\frac{(300-q)}{2} - \frac{q}{1.5} + \frac{360-q}{3} = 0$$

$$150 - \frac{q}{2} - \frac{2q}{3} + 120 - \frac{q}{3} = 0$$

$$270 = +\frac{q}{2} + \frac{q}{2}$$

$$270 = q$$

$$q = 180$$

Ent: Find charge on the capacitors C_1, C_2, C_3 are q_1, q_2, q_3 respectively. Then

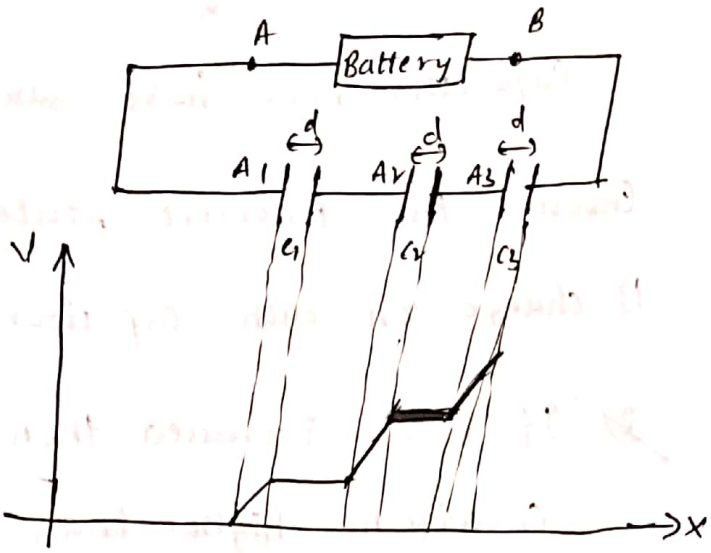
(1)

1) $q_1 > q_2 > q_3$

2) $q_1 < q_2 > q_3$

3) $q_1 = q_2 = q_3$

4) $q_1 > q_2 < q_3$



(2) If area of the capacitor plates are A_1, A_2 & A_3 respectively. then,

1) $A_1 < A_2 > A_3$

2) $A_1 > A_3 > A_2$

3) $A_1 = A_2 = A_3$

4) $A_1 > A_2 > A_3$

(3) If the magnitudes of electric field between their plates are E_1, E_2 & E_3 respectively. then

1) $E_1 > E_2 > E_3$

2) $E_1 < E_2 > E_3$

3) $E_1 > E_2 > E_3$

4) $E_1 < E_2 < E_3$

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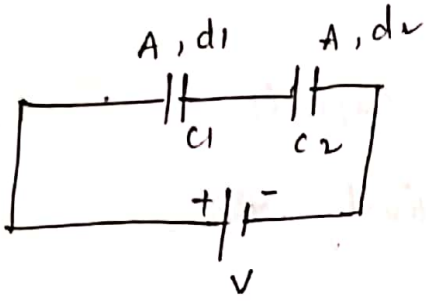
Ex6



$(C_1 < C_2)$

Capacitors here have same plate area.

Choose the incorrect statement.

1) charge on each capacitor is same.

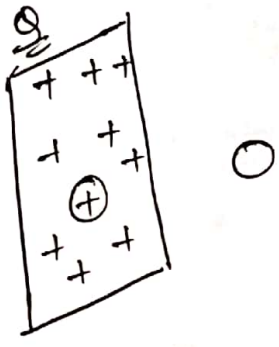
2) If V is increased then increase in energy of C_1 will be higher than that of C_2.

3) potential drop across C_1 is more than C_2.

4) electric field in C_1 is higher than that of C_2.

Ex8 A conductor is charged from electroporous by repeated contacts with the plate which after each contact is recharged with a quantity q of electricity from the electroporous if q is the charge of the conductor. after 1st repetetion/ operation. Calculate the Ultimate charge.

Sol8



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* In contact,

$$V_{cond} = V_{plate}$$

$$\frac{q_1}{C_1} = \frac{Q - q_1}{C_2} \quad - (1)$$

$$\Rightarrow \frac{C_1}{C_2} = \frac{q_1}{Q - q_1}$$

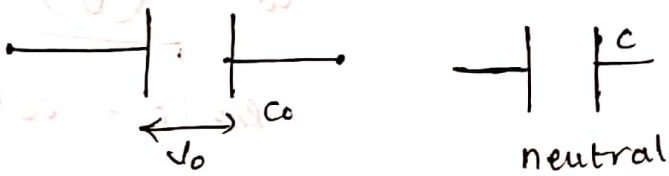
* If q_{max} is the ultimate charge,

$$\frac{q_{max}}{C_1} = \frac{Q}{C_2}$$

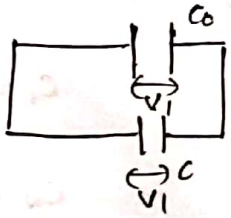
$$q_{max} = \frac{Q C_1}{C_2} \quad - (2)$$

from (1) & (2) $\Rightarrow$ $\boxed{q_{max} = \frac{q_1 Q}{Q - q_1}}$

Ex:6



Kept in contact & charge is distributed till they attain same potential. The capacitor is then charged & discharged repeatedly. After n^{th} contact if potential of C_0 becomes V then value of C is $C_0\left(\left(\frac{V_0}{V}\right)^{1/n} - 1\right)$



$$V_1 = \frac{C_0 V_0}{C_0 + C}$$

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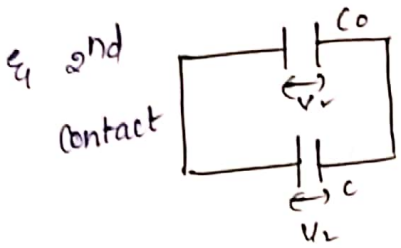


After 2nd contact

$$V_2 = \frac{C_0 V_1}{C_0 + C}$$

$$= \frac{C_0}{C_0 + C}\left(\frac{C_0 V_0}{C_0 + C}\right)$$

$$V_2 = \left(\frac{C_0}{C_0 + C}\right)^2 V_0$$

After nth contact,

$$V_n = \left(\frac{C_0}{C_0 + C}\right)^n V_0$$

(or)

$$V = \left(\frac{C_0}{C_0 + C}\right)^n V_0$$

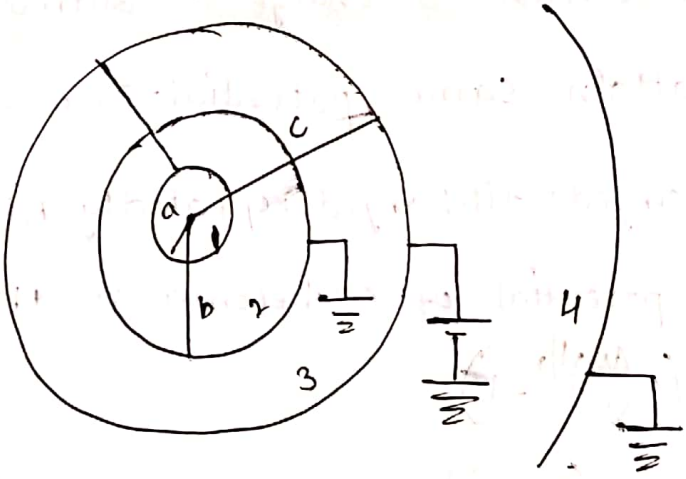
$$\left(\frac{V}{V_0}\right)^{1/n} = \frac{C_0}{C_0 + C}$$

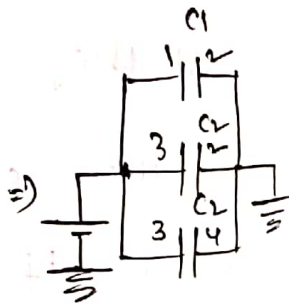
$$\frac{C_0 + C}{C_0} = \left(\frac{V_0}{V}\right)^{1/n}$$

$$1 + \frac{C}{C_0} = \left(\frac{V_0}{V}\right)^{1/n}$$

$$\therefore \boxed{C = C_0\left(\left(\frac{V_0}{V}\right)^{1/n} - 1\right)}$$

Ex 6





$C_{eff} = C_1 + C_2 + C_3$

$$C_1 = \frac{4\pi\varepsilon_0 ab}{b-a}$$

$$C_2 = \frac{4\pi\varepsilon_0 bc}{c-b}$$

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$$C_3 = 4\pi\varepsilon_0 c$$

$$C_{eff} = \frac{4\pi\varepsilon_0 ab}{b-a} + \frac{4\pi\varepsilon_0 bc}{c-b} + 4\pi\varepsilon_0 c .$$

* capacity = 0 $\rightarrow$ open circuit

capacity = ∞ $\rightarrow$ short circuit

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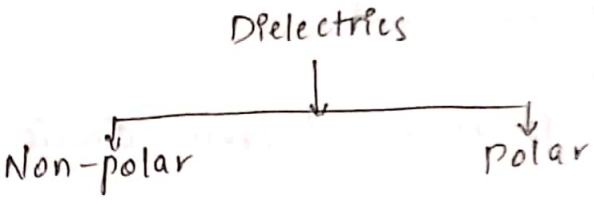
5/05/25 | Monday

* Dielectrics:-

* These are the materials having very few number of free charge carriers, hence called insulators.

* No Dielectric is a perfect insulator.

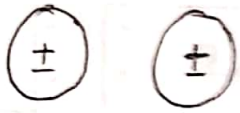


Non-polar dielectrics

* In the absence of electric field, displacement of each molecule = 0.

* centre of positive charge & centre of negative charge coincides.

* These molecules are symmetric.



* Ex: CO_2, H_2, N_2 etc..

* Every particle will be polarised when placed in electric field.

$P = q \times d = 0$

$P_i = 0$ — dipolemoment.

Polar dielectrics

* In the absence of external field, dipolemoment $\neq 0$.

* These molecules are in ellipsoidal shape.

* But $P_{net} = 0$

* Ex: H_2O.

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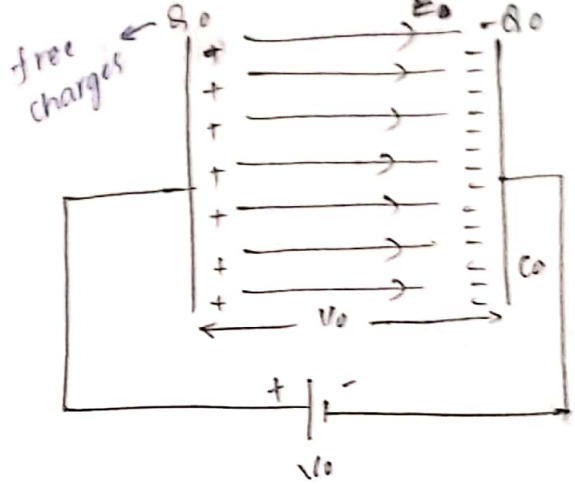
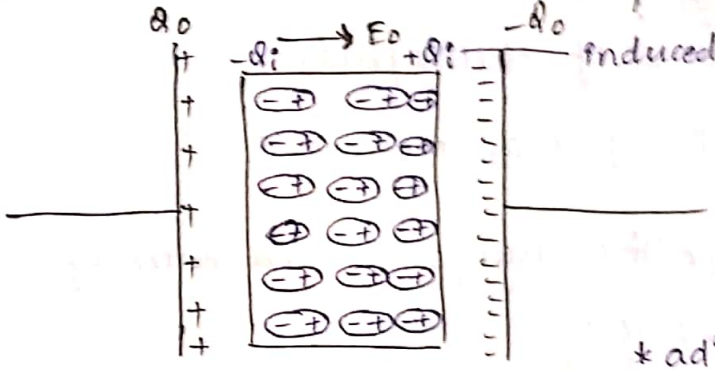
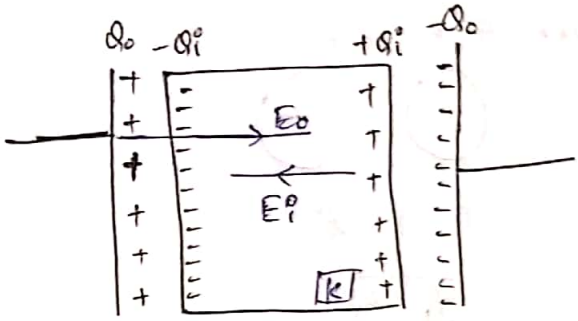


$$C_0 = \frac{Q_0}{V_0}$$

$$E_0 = \frac{V_0}{d}$$

let the battery is removed from circuit and dielectric is inserted between the plates





* adjacent positive & negative charges has strong forces and their effect will be zero.
* charge lies only on surface of plate of dielectric.

net field inside the dielectric,

$$\bar{E} = \bar{E}_0 + \bar{E}_i'$$

$$\bar{E} = E_0 - E_i' \Rightarrow E < E_0 \text{ & } E = \frac{E_0}{K} \quad *$$

we have, $K = \frac{E_0}{E}$

& $V = \frac{V_0}{K}$

we have, $E_0 = \frac{V_0}{d}$

$$E = \frac{V}{d}$$

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$$\frac{E_0}{K} = \frac{V}{d}$$

$$\frac{V_0}{dK} = \frac{V}{d}$$

$$\boxed{V = V_0/K}$$

* Capacitance,

$$C_0 = \frac{Q_0}{V_0}$$

$$C = \frac{Q}{V} = \frac{Q_0}{V_0/K}$$

$$C = K\frac{Q_0}{V_0}$$

$$\boxed{C = KC_0}$$

Again, $E = E_0 - E_i$

$$\frac{E_0}{K} = E_0 - E_i$$

$$\boxed{E_i = E_0\left(1 - \frac{1}{K}\right)}$$ — enclusive induced electricfield expression.

$$\frac{\sigma_i}{\varepsilon_0} = \frac{\sigma_0}{\varepsilon_0}\left(1 - \frac{1}{K}\right)$$

$$\boxed{\sigma_i = \sigma_0\left(1 - \frac{1}{K}\right)}$$ - Induced surface charge density.

K = dielectric const

K = 1 — for air

K = ∞ - for metal

$$\boxed{q_i = q_0\left(1 - \frac{1}{K}\right)}$$

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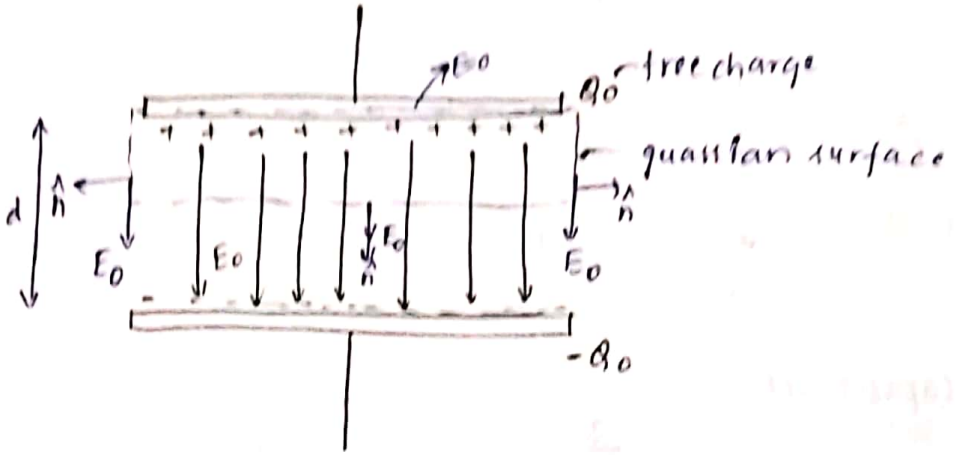
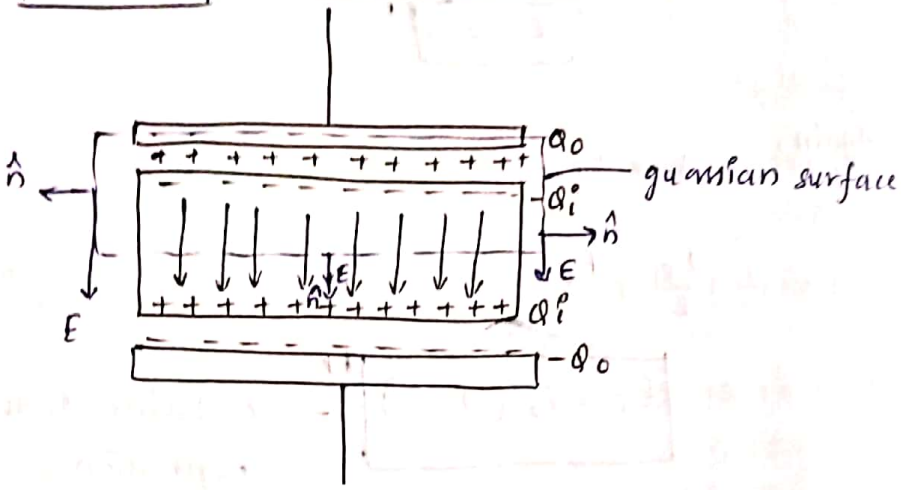
* Guass Law in dielectric -



$$\phi_E = E_0 A$$

$$\frac{Q_0}{\varepsilon_0} = E_0 A$$

$$\boxed{E_0 = \frac{Q_0}{\varepsilon_0 A}} \Rightarrow \boxed{\oint \vec{E} \cdot d\vec{s} = \frac{Q_0}{\varepsilon_0}}$$



$$\phi = E A$$

$$E A = \frac{Q_0 - Q_i'}{\varepsilon_0}$$

$$E A = \frac{Q_0 - Q_0\left(1 - \frac{1}{k}\right)}{\varepsilon_0}$$

$$E A = \frac{Q_0}{\varepsilon_0 K}$$

$$\boxed{E = \frac{Q_0}{\varepsilon_0 K A}}$$

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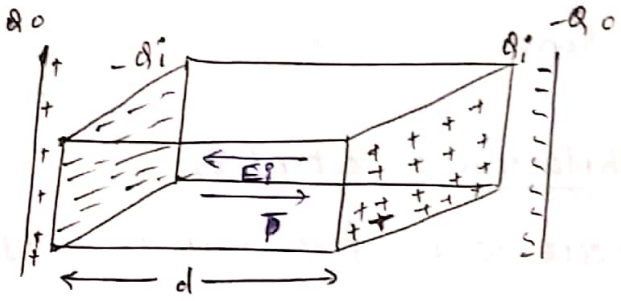
net field $EA = \frac{q_0}{\varepsilon_0 k}$ free charge

$$\oint \bar{E} \cdot d\bar{s} = \frac{q_0}{\varepsilon_0 k} \quad *$$

$$k \oint \bar{E} \cdot d\bar{s} = \frac{q_0}{\varepsilon_0}$$

Electric polarisation ($\bar{P}$) :-

* It is the net dipolement of the dielectric per unit volume.



we have,

$$\bar{P}_{net} = q_i \times \bar{d}$$

$$\bar{P} = \frac{\bar{P}_{net}}{vol}$$

$$\bar{P} = \frac{q_i d}{A d}$$

$$P = \frac{q_i'}{A}$$

$$P_i = \sigma_i'$$

* Electric polarisation is also defined as induced surface charge density of the dielectric.

clearly, $E_i = \frac{q_i'}{\varepsilon_0 A}$

& $P_i = \frac{q_i'}{A}$

$$E_i = \frac{P_i}{\varepsilon_0}$$

In vector form, $\bar{E}_i = -\frac{\bar{P}_i}{\varepsilon_0}$

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* As net electric field increased, all dipoles gradually comes in the direction of field and net dipole increases, and electric polarisation also increases.

we have,

$$P \propto E \Rightarrow \boxed{P = \varepsilon_0 \chi E}$$

χ = susceptability.

* for χ very large, then small E produce large polarisation.

Electric displacement vector ($\bar{D}$):-

* It connects with free charges only.
* It doesnot depends on presence of dielectric b/w the plates of capacitor.
* The free charge per unit area is called electric displacement.
* It's direction is from +ve charge to -ve charge.

i.e, $$D = \frac{q_0}{A}$$

$$DA = q_0$$

$$\boxed{\oint_S \bar{D} \cdot d\bar{s} = q_0}$$

we have, $$E_0 = \frac{q_0}{\varepsilon_0 A}$$

$$D = \frac{q_0}{A_0}$$

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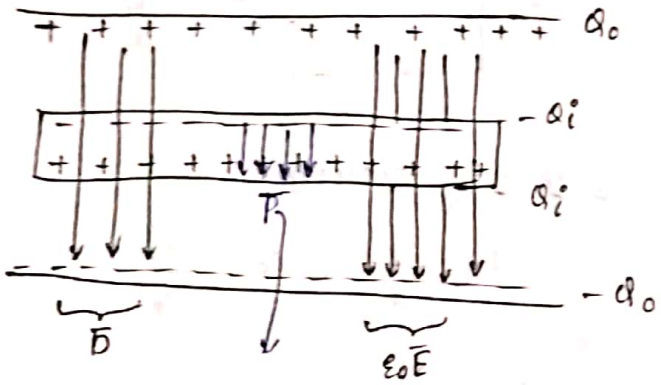
$$E_0 = \frac{D}{\varepsilon_0}$$

$$\boxed{D = \varepsilon_0 E_0}$$ *

Relation b/w $\bar{D}$, $\bar{E}$ & $\bar{P}$ is-



we have, polarisation

$$\bar{E} = \bar{E}_0 + \bar{E}_i$$

$$\bar{E} = \bar{E}_0 - \frac{\bar{P}}{\varepsilon_0}$$

$$\varepsilon_0\bar{E} = \varepsilon_0\bar{E}_0 - \bar{P}$$

$$\varepsilon_0\bar{E} = \bar{D} - \bar{P}$$

$$\boxed{\bar{D} = \varepsilon_0\bar{E} + \bar{P}}$$ *

Relation b/w susceptability & dielectric const (K) is

we have,

$$\varepsilon_0 E_0 = \varepsilon_0 E + \chi E$$

$$E_0 = E + \frac{\chi}{\varepsilon_0} E$$

$$kE = E\left(1 + \frac{\chi}{\varepsilon_0}\right)$$

$$\boxed{k = 1 + \frac{\chi}{\varepsilon_0}}$$ * & $k = \frac{\varepsilon}{\varepsilon_0}$

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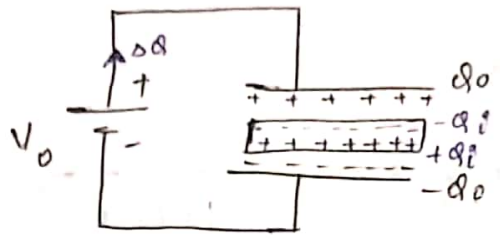
Effect of dielectric on capacitors:-

1) when the battery is continuing in the circuit:-



* without dielectric,

$$E_0 = \frac{V_0}{d}, \quad C_0 = \frac{Q_0}{V_0}$$

$$U_i = \frac{1}{2} C_0 V_0^2$$

* Now, dielectric is slowly inserted, [K = 0]

∴ with dielectric,

$$V = V_0$$

$$E = \frac{V}{d} = \frac{V_0}{d} = E_0$$

$$\boxed{C = KC_0}$$

$$\frac{Q}{V_0} = K\frac{Q_0}{V_0}$$

$$\boxed{Q = KQ_0}$$

Charge flowing through battery,

$$\Delta Q = Q - Q_0$$

$$= KQ_0 - Q_0$$

$$\Delta Q = (K-1)Q_0$$

Work done by battery,

$$W_b = (\Delta Q) V_0$$

$$= (K-1) Q_0 V_0$$

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$$W_b = (K-1) C_0 V_0^2$$

& $U_f = \frac{1}{2} K C_0 V_0^2$

we have,

$$U_i + W_b + W_{ext} = U_f + \cancel{H}^0$$

$$W_{ext} = U_f - U_i - W_b$$

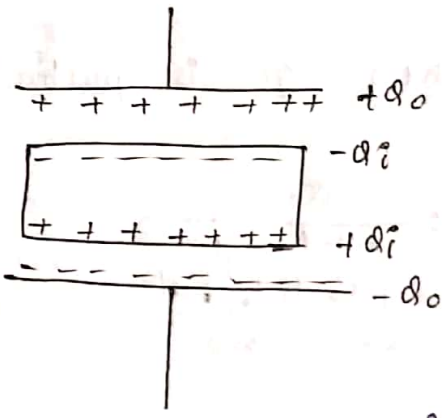
$$= \frac{1}{2} K C_0 V_0^2 - \frac{1}{2} C_0 V_0^2 - (K-1) C_0 V_0^2$$

$$= \frac{1}{2} (K-1) C_0 V_0^2 - (K-1) C_0 V_0^2$$

$$W_{ext} = -\frac{1}{2} (K-1) C_0 V_0^2$$

2) when battery is removed from circuit:-

After charging the capacitor, battery is disconnected from circuit & dielectric is slowly inserted b/w the plates.


+ + + +

$$W_{ext} = U_f - U_i$$

$$= \frac{Q_0^2}{2KC_0} - \frac{Q_0^2}{2C_0}$$

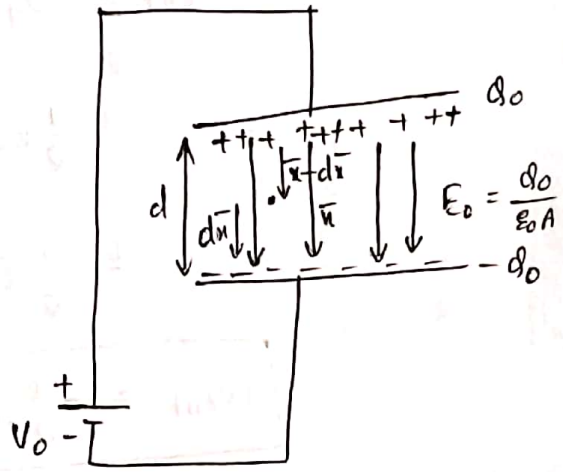
$$\boxed{W_{ext} = \frac{Q_0^2}{2C_0}\left(\frac{1}{K} - 1\right)} \quad *$$

3) when gap is partially filled with dielectric-

*

$$C = \frac{Q_0}{V_+ - V_-}$$

$$= \frac{Q_0}{-\int_{x=d}^{x=0} \bar{E}_0 \cdot dx}$$

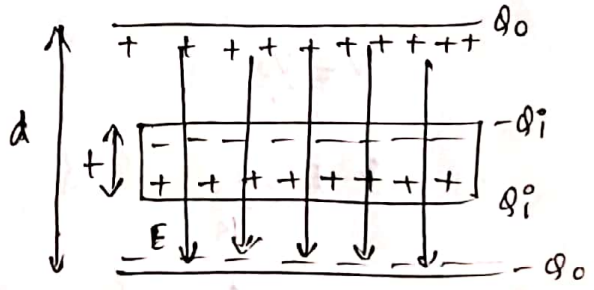


$$= \frac{Q_0}{\int_{x=0}^{x=d} E_0 \, dx}$$

$$= \frac{Q_0}{\frac{Q_0 d}{\varepsilon_0 A}} = \frac{\varepsilon_0 A}{d}.$$

* when gap is partially filled with dielectric,

$$C = \frac{Q_0}{\int_{x=0}^{x=d-t} E_0 \cdot dx + \int_{x=0}^{t} E \cdot dx}$$



$$C = \frac{Q_0}{\frac{Q_0}{\varepsilon_0 A}(d-t) + \frac{Q_0 t}{\varepsilon_0 K A}}$$

$$\boxed{C = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}} \quad *$$

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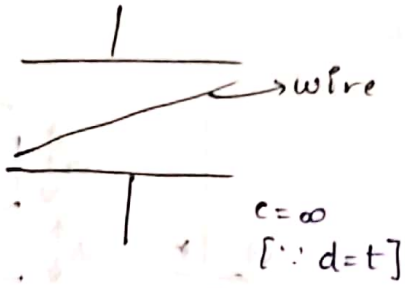
Points

1) If metal fills the gap,



$k = \infty$

$$C = \frac{\varepsilon_0 A}{d-t}$$

2) To restore the original capacity,

Let x be the further increase in seperation b/w the plates.

* C without dielectric = C with dielectric.

$$\frac{\varepsilon_0 A}{d} = \frac{\varepsilon_0 A}{d - t\left(1-\frac{1}{k}\right)+x}$$

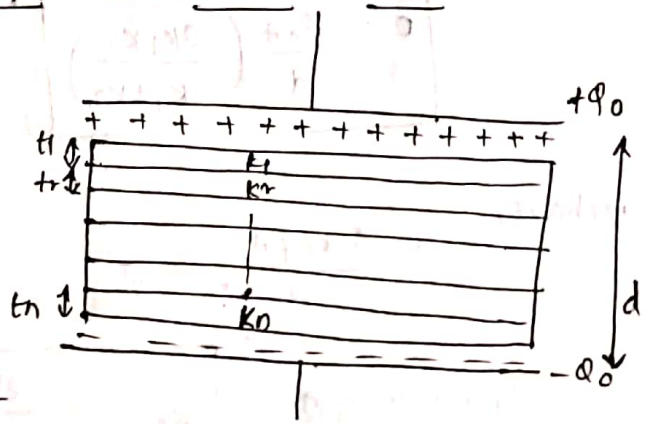
$$d = d - t\left(1-\frac{1}{k}\right)+x$$

$$x = t\left(1-1/k\right) \quad *$$

$$k = \frac{t}{t-x} \quad *$$

Gap fills with different dielectric slabs :-

$$C = \frac{q_0}{\int_{x=0}^{t_1} E_1\,dx + \int_{x=0}^{t_2} E_2\,dx + \cdots + \int_{x=0}^{t=t_n} E_n\,dx}$$



$$= \frac{q_0}{\frac{q_0 t_1}{\varepsilon_0 K_1 A} + \frac{q_0 t_2}{\varepsilon_0 K_2 A} + \cdots + \frac{q_0 t_n}{\varepsilon_0 K_n A}}$$

$$\Rightarrow C = \frac{\varepsilon_0 A}{\frac{t_1}{K_1} + \frac{t_2}{K_2} + \cdots + \frac{t_n}{K_n}}$$

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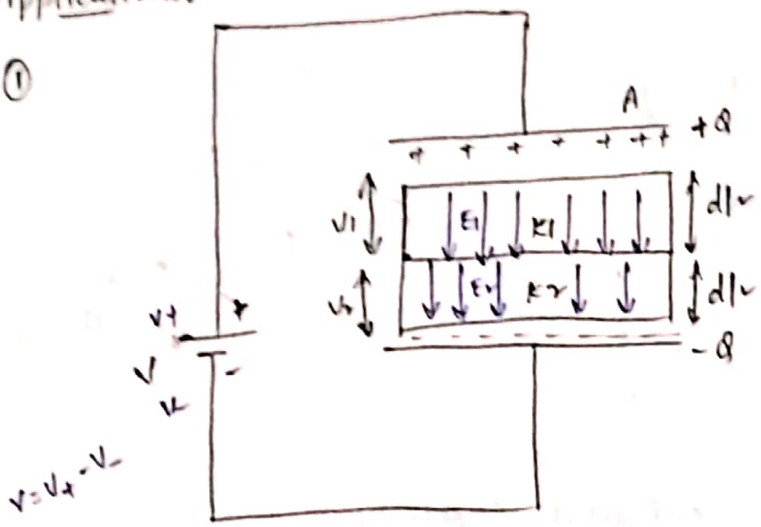
6/05/25 | Tuesday

Applications

①



$V = V_+ - V_-$

Ans-1:

$V = V_1 + V_2$

where, $V_1 = E_1\left(\frac{d}{2}\right) = \frac{Q}{\varepsilon_0 K_1 A}\left(\frac{d}{2}\right)$

$V_2 = E_2\left(\frac{d}{2}\right) = \frac{Q}{\varepsilon_0 K_2 A}\left(\frac{d}{2}\right)$

$$V = \frac{Qd}{2\varepsilon_0 A}\left(\frac{1}{K_1} + \frac{1}{K_2}\right)$$

$$\boxed{V = \frac{Qd}{2\varepsilon_0 A}\left(\frac{K_1 + K_2}{K_1 K_2}\right)}$$

$$C = \frac{Q}{V} = \frac{2\varepsilon_0 A}{d}\left(\frac{K_1 K_2}{K_1 + K_2}\right)$$

$$\boxed{C = \frac{\varepsilon_0 A}{d}\left(\frac{2K_1 K_2}{K_1 + K_2}\right)} \quad -①$$

we have,

$$C = \frac{\varepsilon_0 K_{eff} A}{d} \quad -②$$

from ① & ②

$$\boxed{K_{eff} = \frac{2K_1 K_2}{K_1 + K_2}}$$

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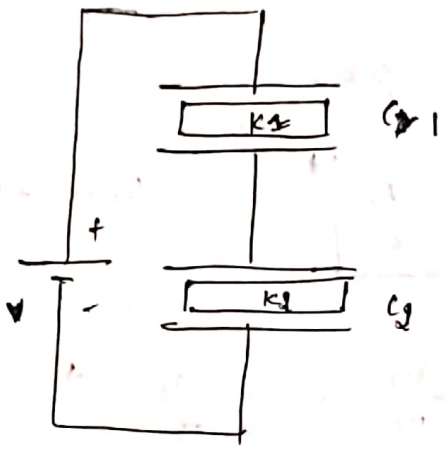
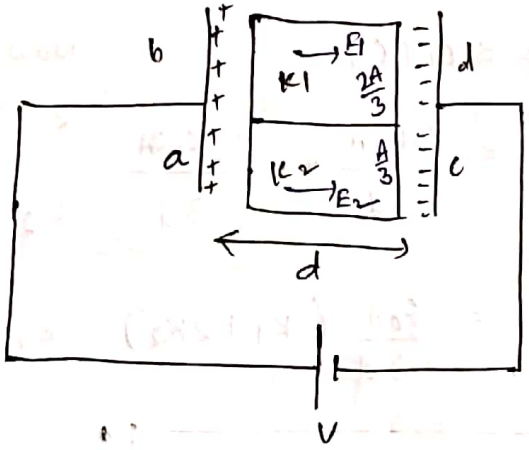
M-II:



$$C_1 = \frac{\varepsilon_0 K_1 A}{d/2} \qquad C_2 = \frac{\varepsilon_0 K_2 A}{d/2}$$

$$\boxed{C_{eff} = \frac{C_1 C_2}{C_1 + C_2}}$$

(2)



since conducting surface is always equipotential surface.

$$V_a = V_b \text{ \& } V_c = V_d$$

$$V_a - V_c = V_b - V_d$$

$$\frac{V_a - V_c}{d} = \frac{V_b - V_d}{d}$$

$$\boxed{E_1 = E_2}$$

$$\frac{E_{01}}{K_1} = \frac{E_{02}}{K_2}$$

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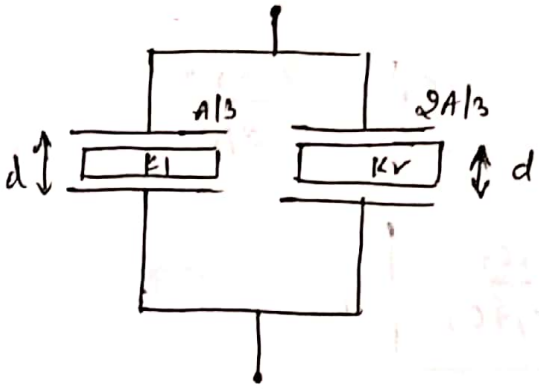
$$\frac{\sigma_1}{\varepsilon_0 K_1} = \frac{\sigma_2}{\varepsilon_0 K_2}$$

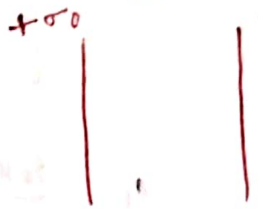
$$\boxed{\frac{\sigma_1}{K_1} = \frac{\sigma_2}{K_2}} \quad *$$

σ_1, σ_2 → free surface charge densities.

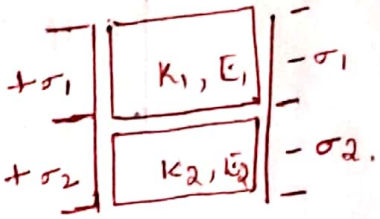
where $\sigma_1 = \frac{Q_{01}}{A_1}$, $\sigma_2 = \frac{Q_{02}}{A_2}$.



Before insertion :-



After insertion :-



we have,

$$C_{eff} = C_1 + C_2$$

$$= \frac{\varepsilon_0 K_1 A}{3d} + \frac{\varepsilon_0 K_2 2A}{3d}$$

$$= \frac{\varepsilon_0 A}{3d}(K_1 + 2K_2)$$

$$\boxed{C_{eff} = \frac{\varepsilon_0 A}{d}\left(\frac{K_1 + 2K_2}{3}\right)} \quad *$$

we have,

$$C_{eff} = \frac{\varepsilon_0 K_{eff} A}{d}$$

$$\boxed{K_{eff} = \frac{K_1 + 2K_2}{3}} \quad *$$

Now

$E_1 \neq \frac{E_0}{K_1} \neq \frac{\sigma_0}{\varepsilon_0 K_1}$

$E_2' \neq \frac{E_0}{K_2} \neq \frac{\sigma_0}{\varepsilon_0 K_2}$

$E_1 = \frac{E_1'}{K_1} = \frac{\sigma_1}{\varepsilon_0 K_1}$

$E_2 = \frac{E_2'}{K_2} = \frac{\sigma_2}{\varepsilon_0 K_2}$

* charge density on the plates will change in such a way that

$$\boxed{E_1 = E_2} \quad ***$$

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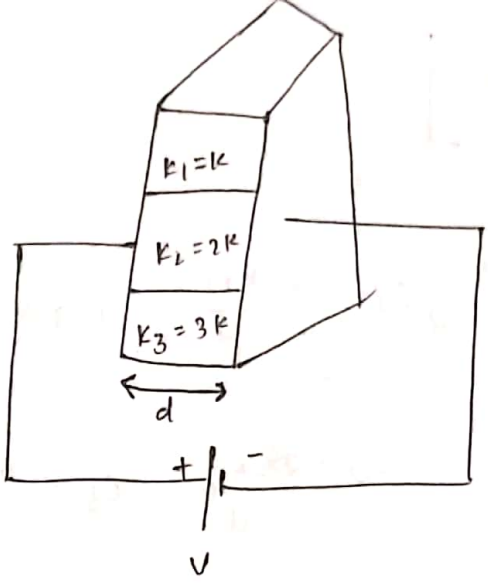
Ex 8



1) Find ratio of max & min surface charge density.

2) Find the induced (bound) charge density in the middle part of the capacitor.

3) If all dielectric occupy equal volume, find their effective capacitance.

Sol 8

1) we have,

$$E = \frac{V}{d} = \text{same in all capacitors.}$$

$$\frac{\sigma}{\varepsilon_0 K} = \frac{V}{d}$$

$$\boxed{\sigma = \frac{\varepsilon_0 K V}{d}} \rightarrow \text{free surface charge density}$$

$$\sigma_{max} = \frac{\varepsilon_0 3K V}{d} = 3\sigma$$

$$\sigma_{min} = \frac{\varepsilon_0 K V}{d} = \sigma$$

$$\boxed{\sigma_{max} : \sigma_{min} = 3:1}.$$

2) $\sigma_p = \sigma_0 \left(1 - \frac{1}{K}\right)$

for middle part, $\sigma_{i_2} = \frac{\varepsilon_0 (2K) V}{d} \left[1 - \frac{1}{2K}\right)$

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$$\sigma_{f_2} = \frac{\varepsilon_0 V (2k-1)}{d}$$

3) $C_1 = \frac{\varepsilon_0 K_1 A_1}{d}$, $C_2 = \frac{\varepsilon_0 K_2 A_2}{d}$, $C_3 = \frac{\varepsilon_0 K_3 A_3}{d}$

$C_1 = \frac{\varepsilon_0 KA}{3d}$ $\quad C_2 = \frac{2\varepsilon_0 KA}{3d}$ $\quad C_3 = \frac{3\varepsilon_0 KA}{3d}$

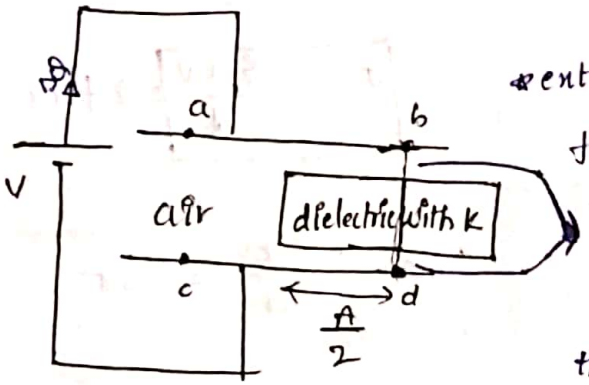
$C_{eff} = C_1 + C_2 + C_3$

$= \frac{6\varepsilon_0 KA}{3d}$

$$C_{eff} = \frac{2\varepsilon_0 KA}{d}$$

$$K_{eff} = 2K$$ *

**** If capacitor plates are partially filled with dielectric.



without dielectric,

$\sigma_0 = \frac{q_0}{A}$

$q_0 = \sigma_0 A$

↓

total charge on capacitor.

* extra charge flowing from battery is stored in these gaps in order to maintain same field in the overall region

with dielectric,

$V_a = V_b$ & $V_c = V_d$

$V_a - V_c = V_b - V_d$

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$E_{air} = E_{med}$

$\frac{\sigma_{air}}{\epsilon_0} = \frac{\sigma_{med}}{K\epsilon_0}$ → free surface charge densities.

$$\boxed{\sigma_{med} = K\sigma_{air}} \quad *$$

* Total charge on the capacitor plate,

$$Q = \sigma_{air}\left(\frac{A}{2}\right) + \sigma_{med}\left(\frac{A}{2}\right)$$

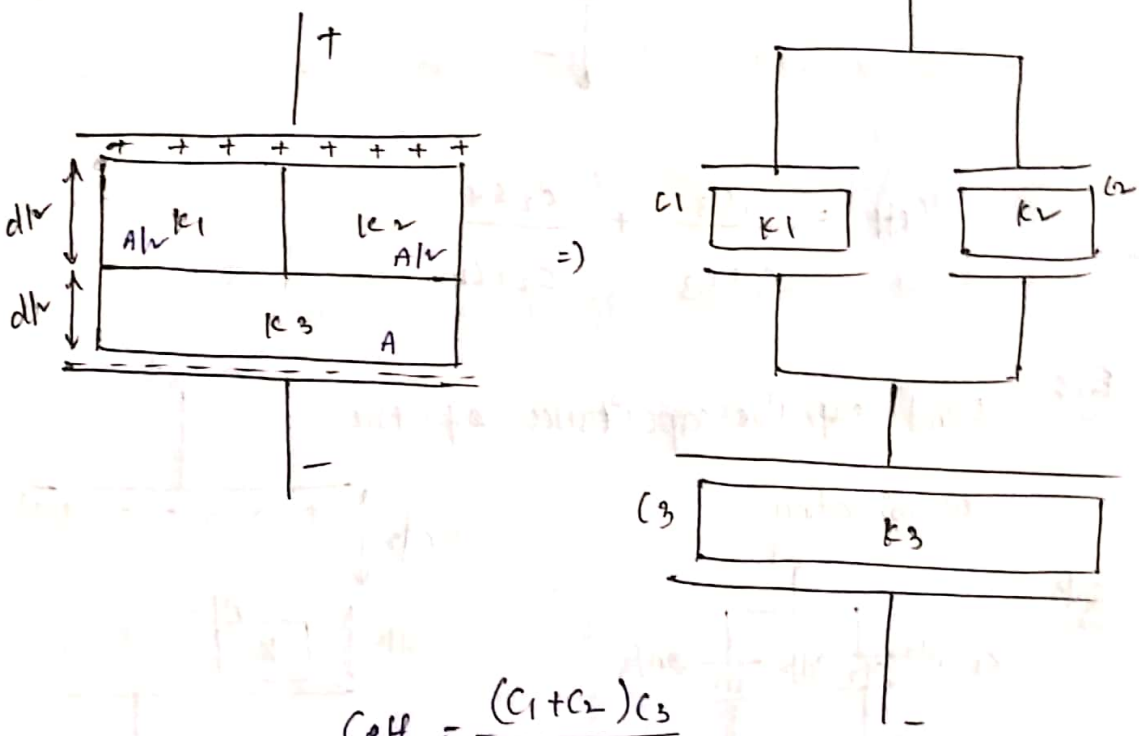
$$= (\sigma_{air} + \sigma_{med})\frac{A}{2}$$

$$\boxed{Q = \frac{(1+K)\sigma_{air} A}{2}} \quad (K>1)$$

i.e, $\boxed{Q > Q_0}$

$\Delta Q = Q - Q_0$. (from battery)

(3)



$$C_{eff} = \frac{(C_1 + C_2)C_3}{C_1 + C_2 + C_3}$$

This is wrong bcoz potential difference across C_1 & C_2 is not same.

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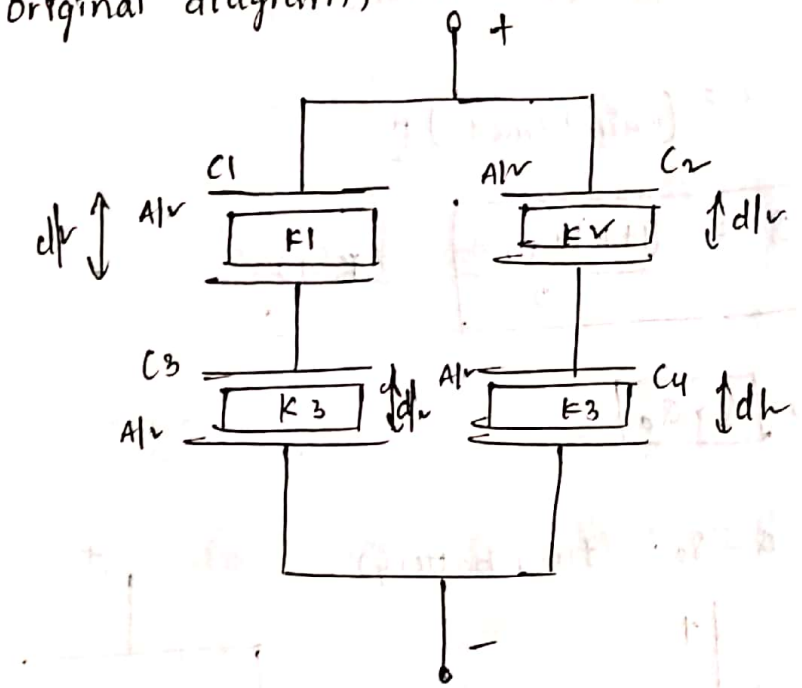
Actually,

$$V_1 = E_1\left(\frac{d}{2}\right) = \frac{Q}{\varepsilon_0 K_1\left(\frac{A}{2}\right)}\left(\frac{d}{2}\right) = \frac{Q_0 d}{\varepsilon_0 K_1 A}$$

& $$V_1' = E_2\left(\frac{d}{2}\right) = \frac{Q_0}{\varepsilon_0 K_2\left(\frac{A}{2}\right)}\left(\frac{d}{2}\right) = \frac{Q_0 d}{\varepsilon_0 K_2 A}$$

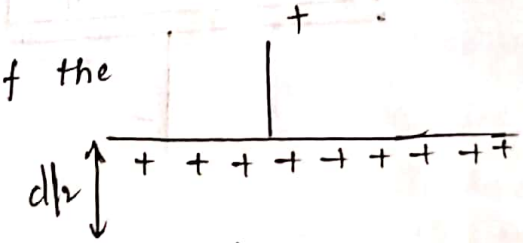
clearly $\boxed{V_1 \neq V_1'}$

original diagram,

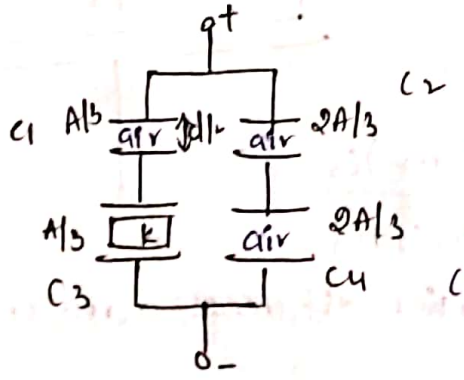


$$C_{eff} = \frac{C_1 C_3}{C_1 + C_3} + \frac{C_2 C_4}{C_2 + C_4}$$

Ex 6 Find effective capacitance of the combination.



Sol 6



$$C_{eff} = \frac{C_1 C_3}{C_1 + C_3} + \frac{C_2 C_4}{C_2 + C_4}$$

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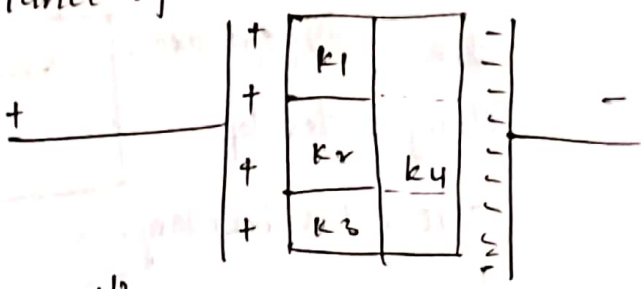
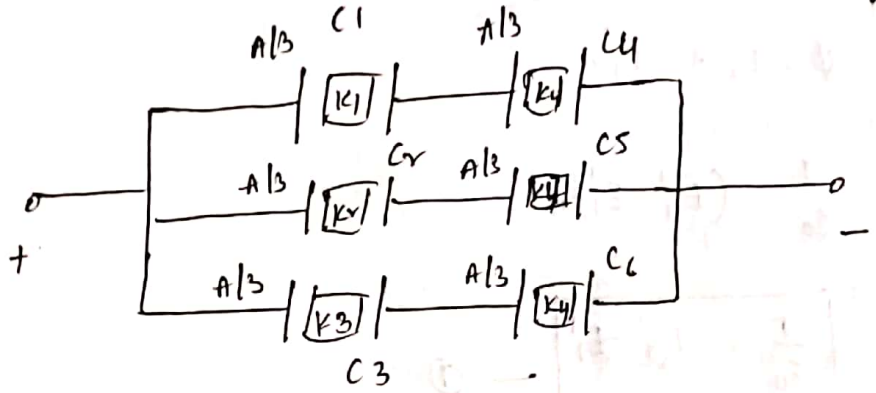
Ex6 Find effective capacitance of comb^n.

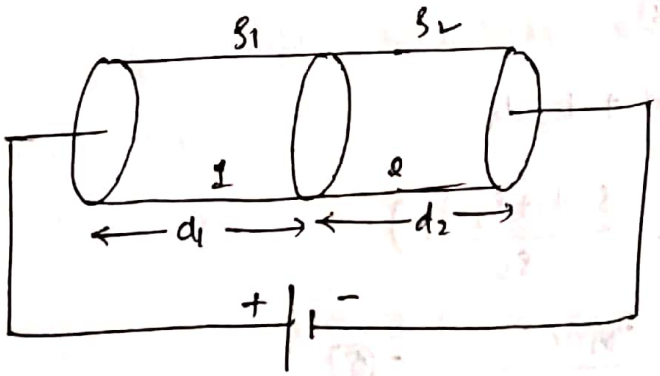


Sol6



$$C_{eff} = \frac{C_1 C_4}{C_1 + C_4} + \frac{C_2 C_5}{C_2 + C_5} + \frac{C_3 C_6}{C_3 + C_6}$$

Ex6 The gap b/w the paralllel plate capacitor is filled with dielectric layers 1 & 2 with thickness d_1 & d_2, permitivity ε_{D_1} & ε_{D_2} and resistivities ρ_1 & ρ_2. A DC voltage V is applied to the capacitor with electric field directed from layer 1 to layer 2. Find σ. the surface charge density of extraneous charges develops at the boundary b/w the dielectric layers.



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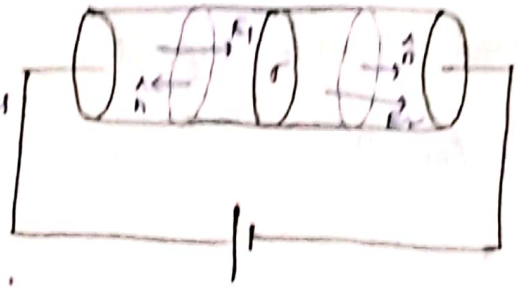
Sol:- without dielectric,



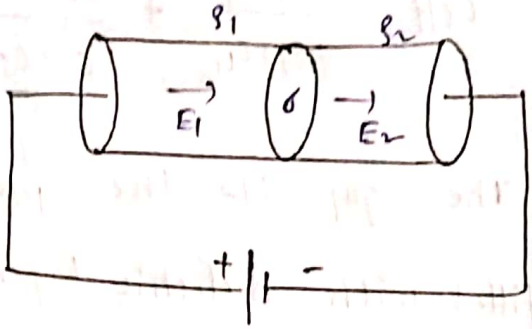
only extraneous charges develop across the boundary.

$$\phi = E_2' A - E_1' A$$

$$\frac{q}{\varepsilon_0} = (E_2' - E_1')A$$

$$\boxed{\frac{\sigma}{\varepsilon_0} = E_2' - E_1'} \quad \text{— (1)}$$

with dielectric,



(1) $q_1 = q_2$

$$T_1 = T_2$$

$$\frac{E_1}{\rho_1} = \frac{E_2}{\rho_2} \quad \text{— (2)}$$

(2) $V = E_1 d_1 + E_2 d_2$ — (3)

(3) Guass law in dielectric,

$$E_2' - E_1' = \frac{\sigma}{\varepsilon_0}$$

$$\varepsilon_2 E_2 - \varepsilon_1 E_1 = \frac{\sigma}{\varepsilon_0} \quad \text{— (4)}$$

Put (2) in (3),

(3) $\Rightarrow V = E_2 \frac{\rho_1}{\rho_2} d_1 + E_2 d_2$

$$V = E_2 \left(\frac{\rho_1 d_1 + \rho_2 d_2}{\rho_2} \right)$$

$$E_2 = \frac{\rho_2 V}{\rho_1 d_1 + \rho_2 d_2} \quad \text{— (5)}$$

V # As resistivities are different, fields are different so some extaneous charges is developed.

$$\boxed{\oint_S \vec{E} \cdot d\vec{s} = \frac{q_0}{\varepsilon_0}}$$

we have,

$$E_2 = \frac{E_2'}{\varepsilon_2}$$

$$E_1 = \frac{E_1'}{\varepsilon_1}$$

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put (5) & (2) in (4)

$$\Rightarrow \frac{\varepsilon_2 \rho_2 V}{\rho_1 d_1 + \rho_2 d_2} - \varepsilon_1 \frac{\rho_1}{\rho_2}\left(\frac{\rho_2 V}{\rho_1 d_1 + \rho_2 d_2}\right) = \frac{\sigma}{\varepsilon_0}$$

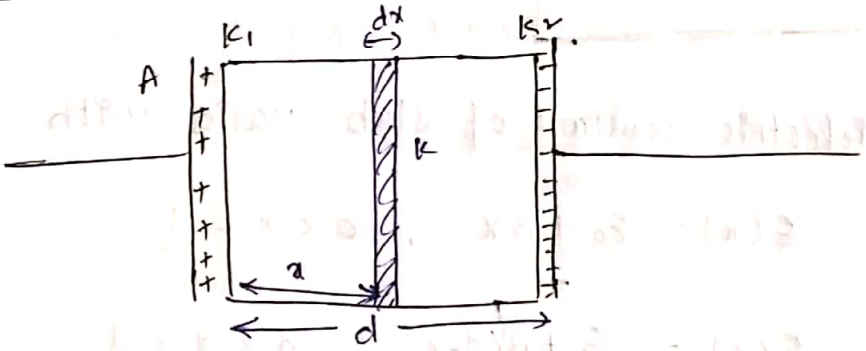
$$\frac{V(\rho_2 \varepsilon_2 - \rho_1 \varepsilon_1)}{\rho_1 d_1 + \rho_2 d_2} = \frac{\sigma}{\varepsilon_0}$$

$$\boxed{\sigma = \frac{V \varepsilon_0 (\rho_2 \varepsilon_2 - \rho_1 \varepsilon_1)}{\rho_1 d_1 + \rho_2 d_2}}$$

surface charge density of developed extraneous charge.

* $\sigma = 0$ when $\rho_2 \varepsilon_2 = \rho_1 \varepsilon_1$.

Ex: Variable dielectric constant :-



Q. dielectric const of slab increases linearly from K_1 to K_2. Find effective dielectric constant.

Sol: elementary capacitor,

$$dC = \frac{\varepsilon_0 K A}{dx} \quad \text{where, } K = K_1 + \left(\frac{K_2 - K_1}{d}\right)x$$

$$dC = \frac{\varepsilon_0 A}{dx}\left[K_1 + \frac{(K_2 - K_1)x}{d}\right]$$

All such elementary capacitors are in series,

$$\therefore \frac{1}{C} = \int \frac{1}{dC}$$

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$$\frac{1}{C} = \int_{x=0}^{x=d} \frac{dx}{\varepsilon_0 A \left(K_1 + \left(\frac{K_2 - K_1}{d}\right)x\right)}$$

$$\frac{1}{C} = \frac{1}{\varepsilon_0 A} \cdot \frac{\left[\ln\left(K_1 + \left(\frac{K_2 - K_1}{d}\right)x\right)\right]_{x=0}^{x=d}}{\frac{K_2 - K_1}{d}}$$

$$\frac{1}{C} = \frac{d}{\varepsilon_0 A (K_2 - K_1)} \left[\ln\left(K_1 + K_2 - K_1\right) - \ln(K_1)\right]$$

$$\boxed{C_{eff} = \frac{\varepsilon_0 A}{d} \frac{(K_2 - K_1)}{\ln(K_2/K_1)}}$$

$$\boxed{K_{eff} = \frac{K_2 - K_1}{\ln(K_2/K_1)}}$$

Ex: dielectric constant of slab varies with distance x

Is $\varepsilon(x) = \varepsilon_0 + \beta x \quad , \quad 0 < x < \frac{d}{2}$

$\varepsilon(x) = \varepsilon_0 + \beta(d - x) \quad , \quad \frac{d}{2} < x < d$

Find 1) eff capacitance

2) eff dielectric const

3) for what value of β capacitance becomes twice the capacitance without dielectric.

use $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

$\approx x$

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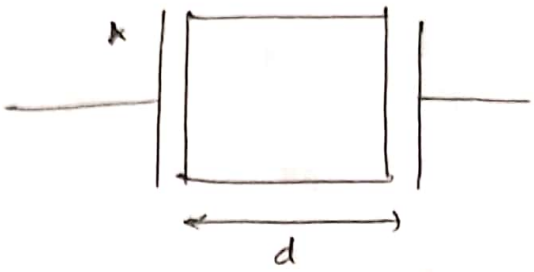
Sol 6



$$dC = \frac{\varepsilon_0 K A}{dx}$$

$$= \frac{\varepsilon_0 A (K)}{dx}$$

$$\frac{1}{C} = \int \frac{1}{dC} = \int \frac{dx}{\varepsilon_0 A K}$$

$$\frac{1}{C} = \frac{1}{\varepsilon_0 A} \int_{x=0}^{x=d/2} \frac{dx}{\varepsilon_0 + \beta x} + \frac{1}{\varepsilon_0 A} \int_{x=d/2}^{x=d} \frac{dx}{\varepsilon_0 + \beta d - \beta x}$$

$$\frac{1}{C} = \frac{1}{\varepsilon_0 A} \left[\frac{\ln(\varepsilon_0 + \beta x)}{\beta} \right]_0^{d/2} + \frac{1}{\varepsilon_0 A} \left[\frac{\ln(\varepsilon_0 + \beta d - \beta x)}{-\beta} \right]_{d/2}^{d}$$

$$\frac{1}{C} = \frac{1}{\varepsilon_0 A \beta} \left[\ln(\varepsilon_0 + \beta d/2) - \ln(\varepsilon_0) \right]$$

$$+ \frac{1}{\varepsilon_0 A \beta} \left[\ln(\varepsilon_0) - \ln(\varepsilon_0 + \frac{\beta d}{2}) \right]$$

$$\frac{1}{C} = \frac{1}{\varepsilon_0 A \beta} \ln\left(\frac{\varepsilon_0 + \beta d/2}{\varepsilon_0} \right) + \frac{1}{\varepsilon_0 A \beta} \ln\left(\frac{\varepsilon_0 + \frac{\beta d}{2}}{\varepsilon_0} \right)$$

$$\boxed{C_{eff} = \frac{\varepsilon_0 A \beta}{2 \ln\left(\frac{\varepsilon_0 + \beta d/2}{\varepsilon_0} \right)}}$$

$$C_{eff} = \frac{\varepsilon_0 A \beta}{2 \ln\left[1 + \frac{\beta d}{2\varepsilon_0}\right]} = \frac{\varepsilon_0 A \beta}{2\left(\frac{\beta d}{2\varepsilon_0}\right)}$$

$$\boxed{C_{eff} = \frac{\varepsilon_0 A}{d} (\varepsilon_0)}$$

$$\Rightarrow \boxed{K_{eff} = \varepsilon_0}$$

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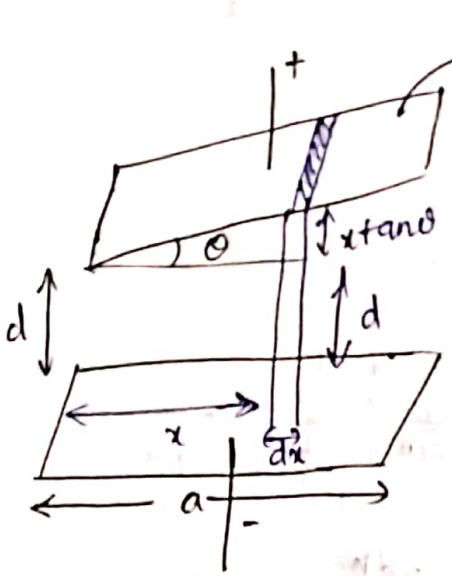
Ex:



$$\tan\theta = \frac{d}{a}$$

$$dc = \frac{\varepsilon_0 a\, dx}{d + x\tan\theta}$$; elementary capacitor,

∴ all such capacitors are in parallel.

$$C = \int_{x=0}^{x=a} dc$$

$$= \int_{x=0}^{x=a} \frac{\varepsilon_0 a\, dx}{d + x\tan\theta}$$

$$= \varepsilon_0 a \int_{x=0}^{x=a} \frac{dx}{d + x\tan\theta}$$

$$= \frac{\varepsilon_0 a}{\tan\theta}\left[\ln(d + x\tan\theta)\right]_0^a$$

$$= \frac{\varepsilon_0 a}{\tan\theta}\left[\ln(d + a\tan\theta) - \ln d\right]$$

$$= \frac{\varepsilon_0 a}{\tan\theta}\left[\ln\left(1 + \frac{a}{d}\tan\theta\right)\right]$$

$$\boxed{C = \frac{\varepsilon_0 a^2}{d}\ln 2}$$

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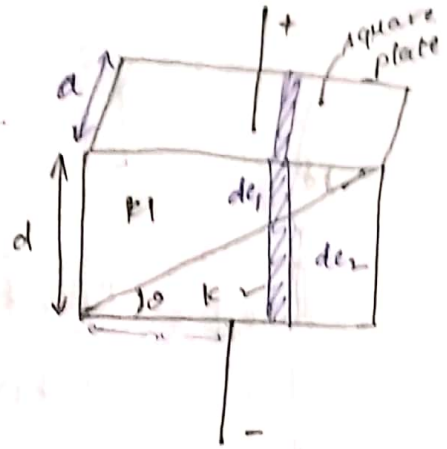
Ex 6 Find effective capacitance & effective dielectric const.



Sol:

now,

$$dc_1 = \frac{\varepsilon_0 k_2 a\,dx}{x\tan\theta}$$

$$-\tan\theta = \frac{d}{a}$$

$$dc_1 = \frac{\varepsilon_0 k_1 (a\,dx)}{d - x\tan\theta}$$

here dc_1 & dc_2 are in series.

$$\frac{1}{dc} = \frac{1}{dc_1} + \frac{1}{dc_2}$$

$$\frac{1}{dc} = \frac{d - x\tan\theta}{\varepsilon_0 k_1 (a\,dx)} + \frac{x\tan\theta}{\varepsilon_0 k_2 (a\,dx)}$$

$$\frac{1}{dc} = \frac{1}{\varepsilon_0 a\,dx}\left(\frac{d - x\tan\theta}{k_1} + \frac{x\tan\theta}{k_2}\right)$$

$$\frac{1}{dc} = \frac{1}{\varepsilon_0 a\,dx}\left(\frac{dk_2 + x\tan\theta\,(k_1 - k_2)}{k_1 k_2}\right)$$

finally, all such capacitances are in parallel.

$$C = \int_{x=0}^{x=a} dc = \int_{x=0}^{x=a} \frac{\varepsilon_0 a k_1 k_2\,dx}{dk_2 + x\tan\theta(k_1 - k_2)}$$

$$= \frac{\varepsilon_0 a k_1 k_2}{(k_1 - k_2)\tan\theta}\left[\ln\left(dk_2 + x\tan\theta(k_1 - k_2)\right)\right]_0^a$$

$$= \frac{\varepsilon_0 a^2 k_1 k_2}{(k_1 - k_2)d}\left[\ln(dk_2 + dk_1 - dk_2) - \ln(dk_2)\right]$$

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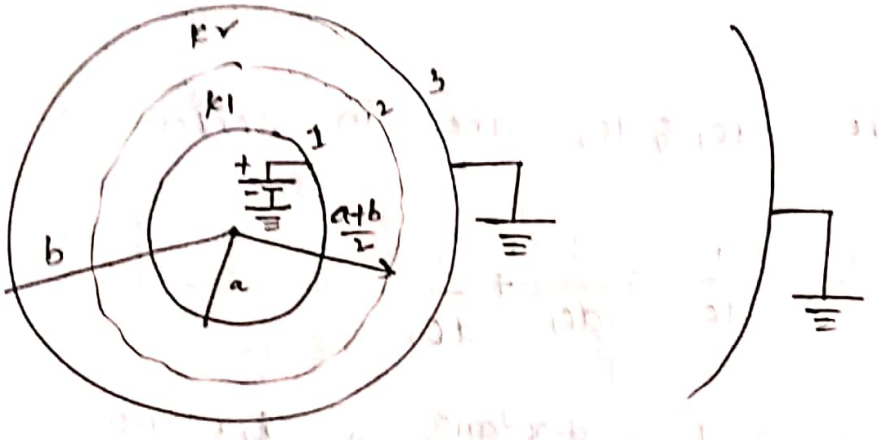
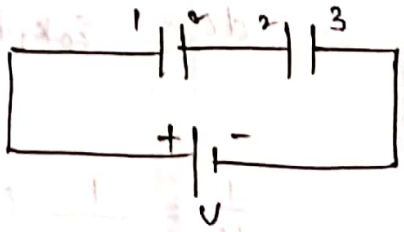
$$C = \frac{\varepsilon_0 a^2 K_1 K_2}{(K_1 - K_2) d} \ln(K_1/K_2)$$

$$K_{eff} = \frac{K_1 K_2 \ln(K_1/K_2)}{K_1 - K_2}$$

Spherical Capacitors





$$C_1 = \frac{4\pi\varepsilon_0 K_1 \left(a\left(\frac{a+b}{2}\right)\right)}{\left(\frac{a+b}{2} - a\right)}$$

$$C_2 = \frac{4\pi\varepsilon_0 K_2 (b)\left(\frac{a+b}{2}\right)}{\left(b - \frac{a+b}{2}\right)}$$

now, $C_{eff} = \frac{C_1 C_2}{C_1 + C_2}$

$$C_{eff} = \frac{4\pi\varepsilon_0 K_{eff}\, ab}{b-a}.$$

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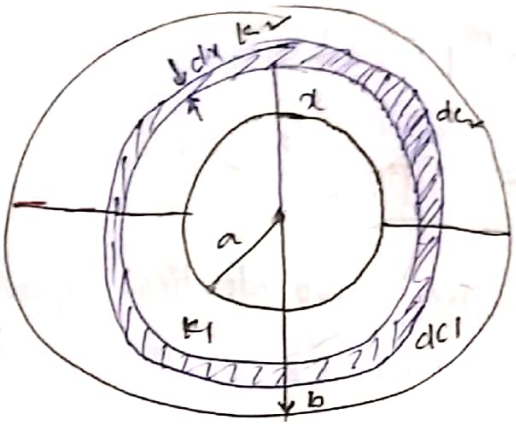
Ex 6



$$dc = \frac{4\pi\varepsilon_0 k x (x+dx)}{x+dx-x}$$

$[dx \ll x]$

$$\boxed{dc = \frac{4\pi\varepsilon_0 k x^2}{dx}}$$

here, $$dc_1 = \frac{4\pi\varepsilon_0 k_1 x^2}{dx}$$

$$dc_2 = \frac{4\pi\varepsilon_0 k_2 x^2}{dx}$$

dc_1, dc_2 are in parallel,

$$dc = dc_1 + dc_2$$

$$= \frac{4\pi\varepsilon_0 x^2}{dx}(k_1+k_2)$$

all such dc are in series combination,

$$\frac{1}{C} = \int_{x=a}^{x=b} \frac{1}{dc}$$

$$\frac{1}{C} = \frac{1}{4\pi\varepsilon_0(k_1+k_2)} \int_{x=a}^{x=b} \frac{1}{x^2}\cdot dx$$

$$\frac{1}{C} = \frac{1}{4\pi\varepsilon_0(k_1+k_2)}\left[\frac{1}{a}-\frac{1}{b}\right]$$

$$\boxed{C = \frac{4\pi\varepsilon_0(k_1+k_2)\,ab}{b-a}}$$

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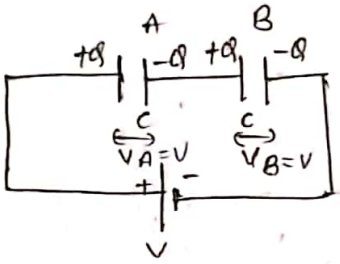
*



A & B are two identical capacitors.

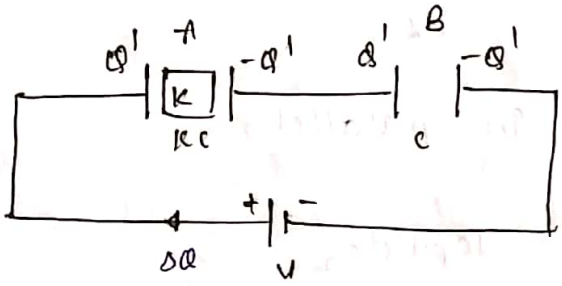
$$V + V = E$$

$$\boxed{V = \frac{E}{2}}$$

i.e, $\boxed{V_A = V_B = \frac{E}{2}}$

& $Q = CV = \frac{CE}{2}$

* A dielectric of (K) is inserted in capacitor A.



$$Q' = C_{eff}\, E$$

$$= \frac{(kC)(C)}{kC+C}(E)$$

$$Q' = \frac{KCE}{K+1} \qquad (K>1)$$

Clearly, $\boxed{Q' > Q}$

* potential difference across A,

$$V_A' = \frac{Q'}{KC} = \frac{KCE}{(1+K)KC} = \frac{E}{1+K} = \frac{2V}{1+K} \qquad (K>1)$$

clearly $\boxed{V_A' < V_B}$

i.e, Potential difference decreases by $\frac{2}{1+K}$.

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* Electric field b/w the plates of 'A'.

$$E_A' = \frac{V_A'}{d} = \frac{2V}{(1+K)d} = \frac{2E_A}{1+K}$$

clearly, $\boxed{E_A' < E_A}$.

i.e, electric field decreases by $\frac{2}{1+K}$.

* potential difference across B,

$$V_B' = \frac{Q'}{C} = \frac{KCE}{(1+K)C} = \frac{KE}{1+K} = \frac{2KV}{1+K} \qquad K>1$$

clearly $\boxed{V_B' > V_B}$

i.e, potential difference increases by $\frac{2K}{1+K}$.

* ~~If~~ electric field b/w the plate of 'B',

$$E_B' = \frac{V_B'}{d} = \frac{2KV}{(1+K)d} = \frac{2KE_B}{1+K}.$$

clearly, $\boxed{E_B' > E_B}$.

* charge flown through battery,

$$\Delta Q = Q' - Q$$

$$= \frac{KCE}{1+K} - \frac{CE}{2}$$

$$= CE\left(\frac{2K-1-K}{1+K}\right)$$

$$\boxed{\Delta Q = \frac{(K-1)}{2(K+1)}CE}$$

* work done by the battery,

$$W_b = \Delta Q \cdot E = \frac{1}{2}CE^2\left(\frac{K-1}{K+1}\right).$$

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* Initial Energy stored in the system,

$$U_i = \frac{1}{2} C_{eff} E^2$$

$$= \frac{1}{2} \times \frac{C}{2} E^2 = \frac{1}{4} C E^2$$

* final Energy stored in the system,

$$U_f = \frac{1}{2} C_{eff} E^2$$

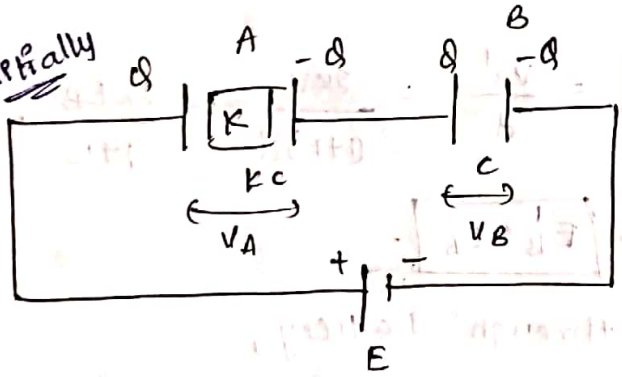
$$U_f = \frac{1}{2}\left(\frac{KC}{K+1}\right) E^2$$

* W-E Theorem

$$U_i + W_b + W_{ext} = U_f + \cancel{H}^{0}$$

$$W_{ext} = U_f - U_i - W_b \Rightarrow \boxed{W_{ext} = \frac{-(K-1) C E^2}{4(K+1)}}$$

Ex6 initially



$$C_{eff} = \frac{KC}{K+1}$$

$$Q = C_{eff} E = \frac{KC}{K+1} \cdot E$$

$$V_A = \frac{Q}{KC} = \frac{E}{1+K}$$

$$V_B = \frac{Q}{C} = \frac{KE}{1+K}$$

clearly $\boxed{V_B > V_A}$.

$$E_A = \frac{V_A}{d}$$

$$E_B = \frac{V_B}{d}$$

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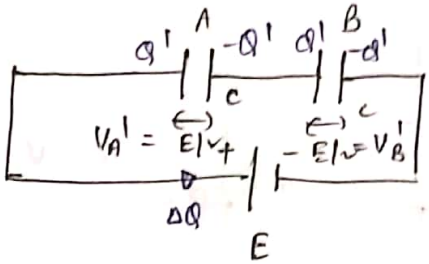
* now dielectric is slowly removed in the initial circuit.

Q is decreased.

flown



$C_{eff} = \frac{C}{2}$

$q' = \frac{C}{2}E$

$V_A' = \frac{q'}{C}$, $V_B' = \frac{E}{2}$

$= \frac{CE}{2C}$

$= \frac{E}{2}$

$W_b = -ve$ charging.

At A, potential ↑ $\Rightarrow \mathcal{E}$ ↑

At B, potential ↓ $\Rightarrow \mathcal{E}$ ↓

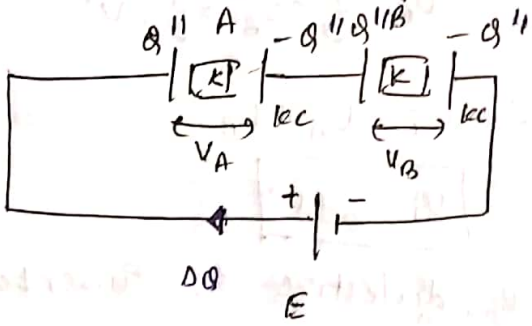
$E_A' = \frac{V_A'}{d}$, $E_B' = \frac{V_B'}{d}$

$V_A' > V_A \Rightarrow E_A' > E_A$

$V_B' < V_B \Rightarrow E_B' < E_B$

& $V_A' = V_B'$

* now another dielectric of dielectric const K is introduced to intial circuit slowly.



$C_{eff} = \frac{KC}{2}$

$q'' = C_{eff}E = \frac{KCE}{2}$

$V_A'' = \frac{q''}{Kc} = \frac{E}{2}$

$V_B = \frac{q''}{Kc} = \frac{E}{2}$

clearly $\boxed{q'' > q}$

$\Delta Q = q'' - q$

$= \frac{KCE}{2} - \frac{KCE}{K+1}$

$= \frac{KCE}{2}\left(\frac{K+1-2}{K+1}\right)$

$\Delta Q = \frac{K(K-1)CE}{2(K+1)}$

$W_b = \Delta Q \cdot E$

$= \frac{K(K-1)CE^2}{2(K+1)}$

$U_i = \frac{1}{2}C_{eff}E^2$

$= \frac{1}{2}\frac{KC}{K+1}E^2 = \frac{KCE^2}{2(K+1)}$

$U_f = \frac{1}{2}C_{eff}E^2$

$= \frac{1}{2}\frac{KC}{2}E^2 = \frac{KCE^2}{4}$

$U_i + W_b + W_{ext} = U_f + \cancel{K}^0 \Rightarrow \boxed{W_{ext} = -\frac{K(K-1)CE^2}{4(K+1)}}$

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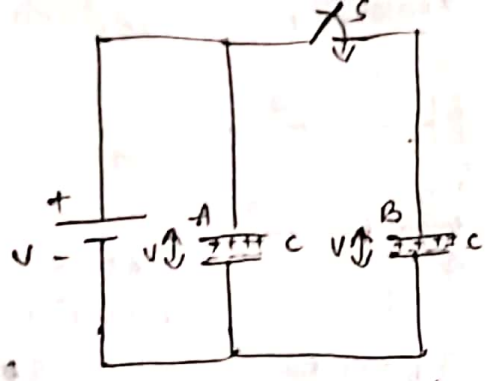
Ex 8



* Initially switch is closed for a long time. (i.e, fully charged)

$$V_A = V_B = V$$

$$E_A = E_B = \frac{V}{d}$$

Initial energy stored,

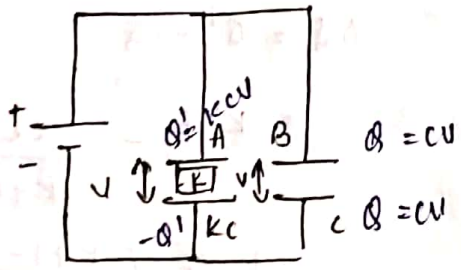
$$U_A = \frac{1}{2} C V_A^2 = \frac{1}{2} C V^2$$

$$U_B = \frac{1}{2} C V_B^2 = \frac{1}{2} C V^2$$

$$\therefore U_i = U_A + U_B$$

$$\boxed{U_i = CV^2}$$

* now dielectric is inserted in A.



$$V_A' = V_B' = V$$

i.e, $V_A' = V_A$, $V_B' = V_B$

$\Rightarrow$ $E_A' = E_A$, $E_B' = E_B$

Charge flown through the battery,

$$\Delta Q = Q' - Q = KCV - CV = (K-1)CV$$

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$$\boxed{\Delta Q = (K-1)CV} \qquad (K>1)$$

* work done by battery,

$$W_b = \Delta Q \cdot V$$

$$= (K-1)CV^2$$

* $U_f = U_A' + U_B'$

$$= \frac{1}{2} KCV^2 + \frac{1}{2} CV^2$$

$$\boxed{U_f = \frac{1}{2}(K+1)CV^2}$$

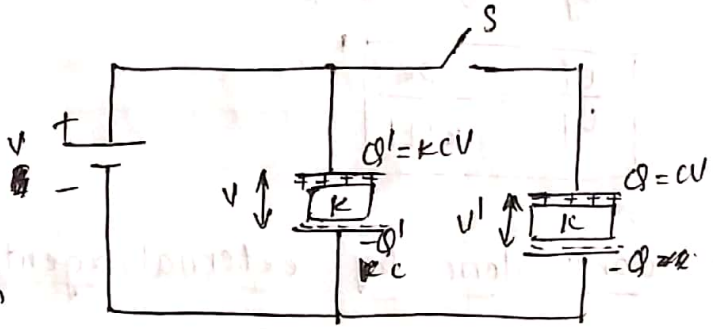
i.e, $U_f > U_i$

* W-E Theorem,

$$U_i + W_b + W_{ext} = U_f + \cancel{A}^{\,0}$$

$$\boxed{W_{ext} = -U_i + U_f - W_b}$$

* Now switch is opened & dielectrics are inserted in both the the capacitors.



* Charge flown through battery,

$$\Delta Q = KCV - CV = (K-1)CV.$$

* work done by battery,

$$W_b = (\Delta Q)V = (K-1)CV^2,$$

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* Initial energy stored,

$$U_A = \frac{1}{2}CV^2, \quad U_B = \frac{1}{2}CV^2$$

$$\therefore (U_i)_{system} = U_A + U_B = CV^2$$

* Final energy stored,

$$U_A' = \frac{1}{2}(KC)V^2 = KU_A$$

clearly $\boxed{U_A' > U_A}$

$$U_B' = \frac{Q^2}{2KC} = \frac{C^2V^2}{2KC} = \frac{1}{2}\frac{CV^2}{K} = \frac{U_B}{K}$$

clearly, $\boxed{U_B' < U_B}$

$$U_f = U_A' + U_B'$$

$$= \frac{1}{2}KCV^2 + \frac{1}{2}\frac{CV^2}{K}$$

$$U_f = \frac{1}{2}CV^2\left(\frac{K^2+1}{K}\right)$$

$$\boxed{\frac{U_i}{U_f} = \frac{2K}{K^2+1}}$$ *

* work done by external agent:

For inserting dielectric in A:

$$U_i + W_b + W_{ext} = U_f + H^{\nearrow 0}$$

$$W_{ext} = (U_f)_A - (U_i)_A - W_b$$

$$= \frac{1}{2}KCV^2 - \frac{1}{2}CV^2 - [K-1]CV^2$$

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$$= \frac{1}{2} K V^2 (K - 1 - 2K + 2)$$

$$W_{ext_1} = \frac{1}{2}(1-K) CV^2$$

$$\boxed{W_{ext_1} = -\frac{(K-1)}{2} CV^2}$$

For inserting dielectric in B!

$$U_i + W_b + W_{ext_2} = U_f + \cancel{W_b}^{0}$$

$$W_{ext_2} = (U_f)_B - (U_i)_B - \cancel{W_b}^{0}$$

$$= \frac{1}{2} \frac{CV^2}{K} - \frac{1}{2} CV^2$$

$$= \frac{1}{2} CV^2 \left(\frac{1}{K} - 1\right)$$

$$= \frac{1}{2} CV^2 \left(\frac{1-K}{K}\right)$$

$$\boxed{W_{ext_2} = -\frac{1}{2} CV^2 \left(\frac{K-1}{K}\right)}$$

Total work done by external agent,

$$W_{ext} = W_{ext_1} + W_{ext_2}$$

$$= -\frac{(K-1) CV^2}{2} - \frac{(K-1) CV^2}{2K}$$

$$= -\frac{CV^2}{2}(K-1)\left(\frac{K+1}{K}\right)$$

$$\boxed{W_{ext} = -\frac{(K^2-1) CV^2}{2K}}$$

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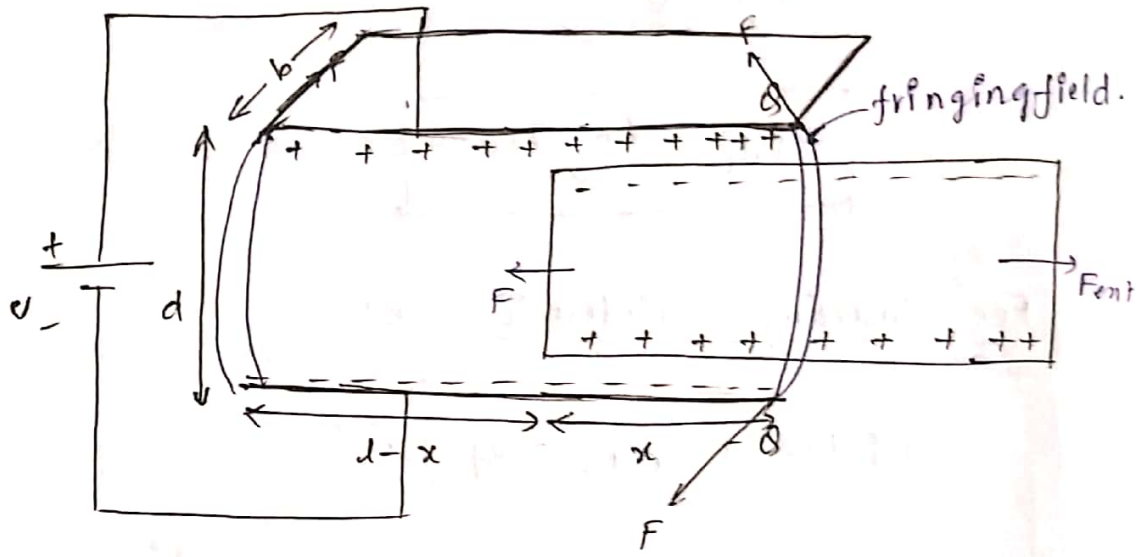
* Force on dielectric -

① when battery is continuing in the circuit -



* effective capacity,

$$C = C_1 + C_2$$

$$C = \frac{\varepsilon_0 b (d-x)}{d} + \frac{\varepsilon_0 k b x}{d}$$

$$= \frac{\varepsilon_0 b}{d} (d - x + kx)$$

$$C_{eff} = \frac{\varepsilon_0 b}{d} (d + (k-1)x) \quad *$$

* energy stored in capacitor,

$$U = \frac{1}{2} C V^2$$

force on the dielectric,

$$F = \left| \frac{dU}{dx} \right|$$

$$= \left| \frac{d}{dx} \left(\frac{1}{2} C V^2 \right) \right|$$

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$$\therefore F = \frac{V^2}{2}\frac{dC}{dx}$$

$$F = \frac{V^2}{2}\frac{\varepsilon_0 b}{d}(K-1)$$

$$\boxed{F = \frac{1}{2}\frac{\varepsilon_0 b}{d}(K-1)V^2}$$ = constant force.

* work done by external agent in inserting the dielectric slowly.

$$W_{ext} = \int \bar{F}_{ext} \cdot d\bar{x} = -\int F_{ext} \cdot dx$$

$$= -\int_{x=0}^{x=x} F \cdot dx$$

$$\boxed{W_{ext} = -\frac{1}{2}\frac{\varepsilon_0 b}{d}(K-1)V^2 x} \quad ***$$

* If dielectric is left itself, it makes "oscillatory motion".

$$l - x = \frac{1}{2}a\left(\frac{T}{4}\right)^2$$

$$\sqrt{\frac{2(l-x)}{a}} = \frac{T}{4}$$

$$\boxed{T = 4\sqrt{\frac{2(l-x)m}{F}}}$$

(9) When battery is removed from circuit :-

Q = const.

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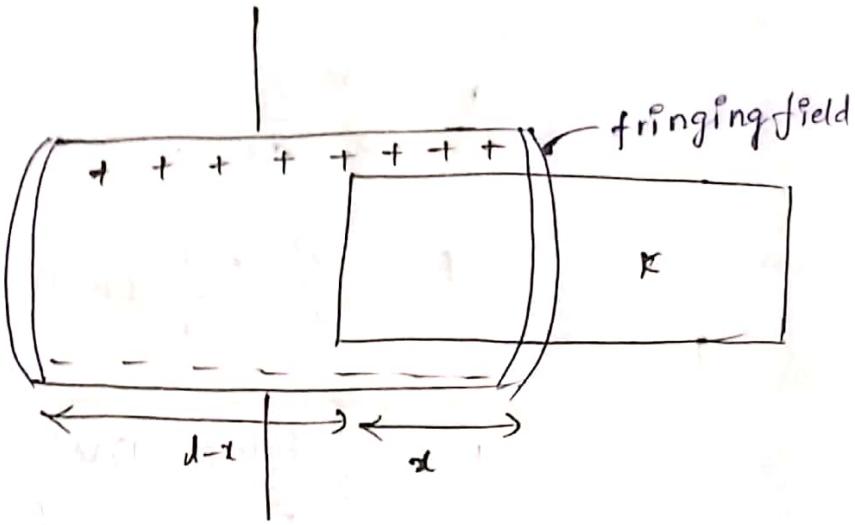

$C = C_1 + C_2$

$$C = \frac{\varepsilon_0 b}{d}\left[d + (k-1)x\right]$$

Energy stored,

$$U = \frac{Q^2}{2C}$$

$$F = \left|\frac{dU}{dx}\right|$$

$$F = \left|\frac{Q^2}{2}\frac{d}{dx}\left(\frac{1}{C}\right)\right|$$

$$F = \frac{Q^2}{2C^2}\frac{dC}{dx}$$ = variable force

$\frac{Q^2}{2C^2}$ — variable; $\frac{dC}{dx}$ — const

To find W_{ext}, we use W.E Theorem,

$$U_i + W_b + W_{ext} = U_f + H \to 0$$

$$W_{ext} = U_f - U_i - W_b$$

(or) $$W_{ext} = \int_{x=0}^{x=x} \bar{F}_{ext} \cdot d\bar{x}$$ → difficult. (really)

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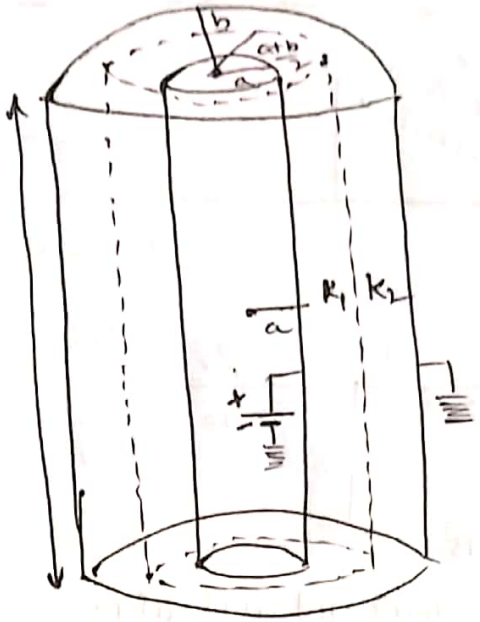
Cylindrical Capacitor :-

①



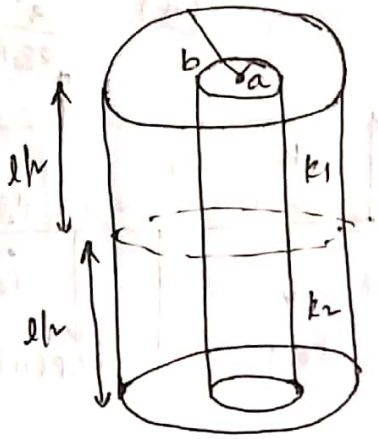
$$C_1 = \frac{2\pi\varepsilon_0 K_1 l}{\ln\left(\frac{a+b}{2a}\right)}$$

$$C_2 = \frac{2\pi\varepsilon_0 K_2 l}{\ln\left(\frac{2b}{a+b}\right)}$$

C_1 & C_2 are in series,

$$\boxed{C = \frac{C_1 C_2}{C_1 + C_2}}$$

②



$$C_1 = \frac{2\pi\varepsilon_0 K_1 (l/2)}{\ln(b/a)}$$

$$C_2 = \frac{2\pi\varepsilon_0 K_2 (l/2)}{\ln(b/a)}$$

here C_1 & C_2 are in parallel

$$\boxed{C = C_1 + C_2}$$

we have,

$$C_{eff} = \frac{2\pi\varepsilon_0 K_{eff} l}{\ln(b/a)}$$

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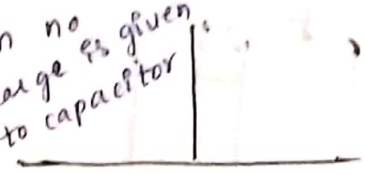
7/05/25 | wednesday ☺

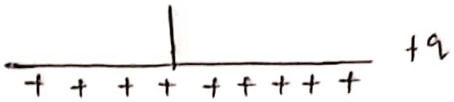
Pressure rise of liquid dielectric in capacitor plates

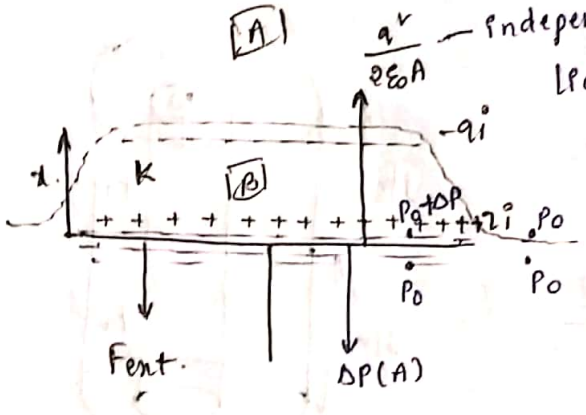
* when no charge is given to capacitor



inner plate is just submerged in liquid.

*





$\frac{q^2}{2\varepsilon_0 A}$ — independent of liquid dielectric

As lower plate is moved in downward direction, x↑.
But there will be no change in region A.

In eqm,

$$F_{ext} = \frac{q^2}{2\varepsilon_0 A} - (\Delta P)A \quad -①$$

* Energy method

$U = U_A + U_B$ → $U_B = \frac{q^2}{2C} = \frac{q^2}{2\varepsilon_0 kA} x$

↓

const

* $dU = F_{ext} \cdot dx$

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If there is battery,

↳ battery is continuing.

$$W_b = (\Delta Q)V$$

$$= (dc)V^2$$

$$dU = F_{ext} \cdot dx + W_b$$

$$= F_{ext} \cdot dx + (dc)V^2$$

$$\boxed{F_{ext} = \frac{dU}{dx} - V^2 \frac{dc}{dx}}$$

$$F_{ext} = \frac{dU}{dx}$$

$$F_{ext} = \frac{d}{dx}\left(\frac{q^2 x}{2\varepsilon_0 K A}\right)$$

$$\boxed{F_{ext} = \frac{q^2}{2\varepsilon_0 K A}} \quad \text{— (2)}$$

From eqns (1) & (2),

$$\frac{q^2}{2\varepsilon_0 K A} = \frac{q^2}{2\varepsilon_0 A} - (\Delta P)A$$

$$(\Delta P)A = \frac{q^2}{2\varepsilon_0 A}\left(1 - \frac{1}{K}\right)$$

$$\boxed{\Delta P = \frac{q^2}{2\varepsilon_0 A^2}\left(1 - \frac{1}{K}\right)}$$

(or)

$$\boxed{\Delta P = \frac{\sigma^2}{2\varepsilon_0}\left(1 - \frac{1}{K}\right)}$$

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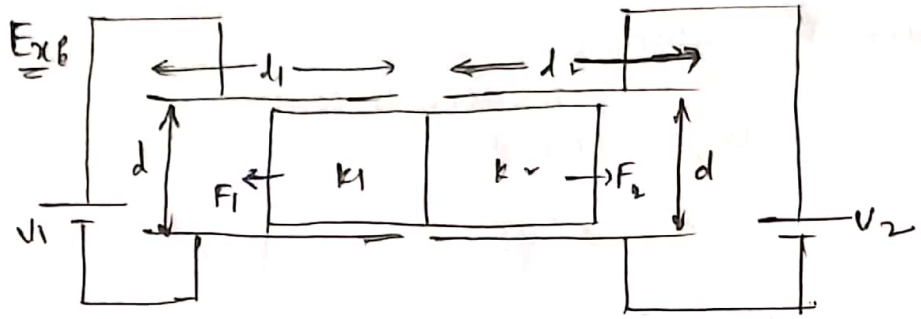
Ex 6



Find the ratio of voltage of the left battery to that of right battery for which dielectric slab may remain in eqm. Given b = width of each plate.

Sol⁶ In eqm,

$F_1 = F_2$ b = width of plate.

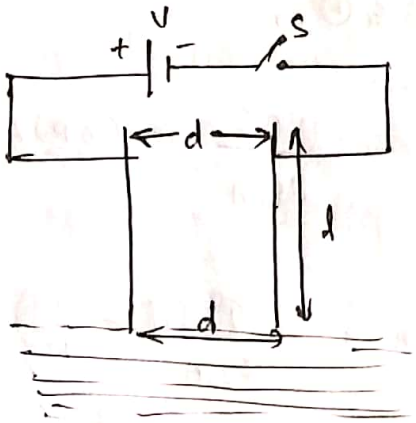
$$\frac{1}{2}\frac{\varepsilon_0 b}{d}(K_1-1)V_1^2 = \frac{1}{2}\frac{\varepsilon_0 b}{d}(K_2-1)V_2^2$$

$$\boxed{\frac{V_1}{V_2} = \sqrt{\frac{K_2-1}{K_1-1}}}$$

Ex 6

b = width of plate

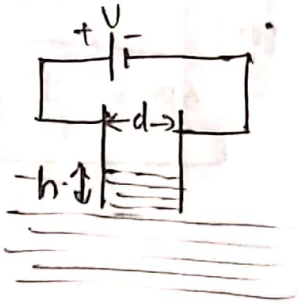
ρ = density of liq of dielectric const k.



when switch is closed find the height b/w the plates to which the liq-level will rise.

Sol⁶ In eqm,

Force on the liquid dielectric = weight of rised liquid



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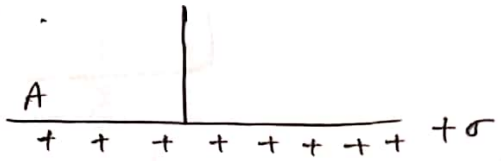
$$\frac{1}{2}\frac{\varepsilon_0 b}{d}(K-1)V^2 = hbdg$$

$$\boxed{h = \frac{\varepsilon_0 (K-1)V^2}{2dgd^2}}$$

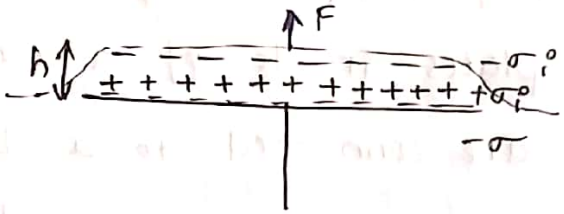
Ex! A horizontal parallel plate capacitor is lowered on a liq surface in such a way that it's lower plate is just submerged in the liq of dielectric constant K. Find the height to which the liq level will be rised blw the plates if the capacitor plates are given a surface charge density σ & $-\sigma$.



Sol^n In eqm,

Force = w

$EQ_0 = mg$ —①



To find net field at the location of $-\sigma_i$ charge density,

$$E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} - \frac{\sigma_i}{2\varepsilon_0}$$

$$= \frac{\sigma}{\varepsilon_0} - \frac{\sigma_i}{2\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{1}{K}\right)[\downarrow]$$

$$= \frac{\sigma}{\varepsilon_0}\left(1 - \frac{1}{2} + \frac{1}{2K}\right)$$

$$\boxed{E = \frac{\sigma}{2\varepsilon_0}\left(1 + \frac{1}{K}\right)[\downarrow]}$$

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$$F = E Q_i \ (\uparrow)$$

$$= \frac{\sigma}{2\varepsilon_0}\left(1+\frac{1}{k}\right)\sigma\left(1-\frac{1}{k}\right)A$$

$$F = \frac{\sigma^2 A}{2\varepsilon_0}\left(1-\frac{1}{k^2}\right)$$

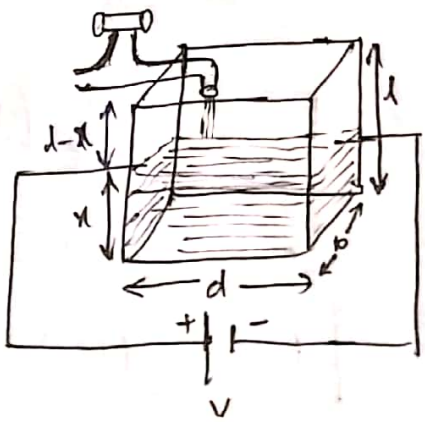
Now,

$$F = mg$$

$$\frac{\sigma^2 A}{2\varepsilon_0}\left(1-\frac{1}{k^2}\right) = Ax h x \rho g$$

$$\boxed{h = \frac{\sigma^2\left(1-\frac{1}{k^2}\right)}{2\varepsilon_0 \rho g}}$$

Exa6 A palallel plate capacitor is made by fixing two plates inside of a container as shown. The plates are connected to a battery of voltage V. If at $t=0$, the tap is opened from which a liq of dielectric constant K starts filling in the container at a rate of $\alpha m^3/s$. Find the current in the connecting wires as a function of time. Neglect resistance in connecting wires.



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Sol^{n} At any instant,

$$C = C_1 + C_2$$

$$= \frac{\varepsilon_0 K x b}{d} + \frac{\varepsilon_0 (l-x) b}{d}$$

$$C = \frac{\varepsilon_0 b}{d} \left(l + (K-1)x \right)$$

charge on the plates,

$$q = CV$$

$$= \frac{\varepsilon_0 b}{d} (l + (K-1)x)(V)$$

current,

$$i = \frac{dq}{dt}$$

$$i = \frac{\varepsilon_0 bV}{d} (K-1) \cdot \frac{dx}{dt}$$

we have,

$$V = x b d$$

$$\frac{dV}{dt} = \frac{dx}{dt} (bd)$$

$$\frac{r}{bd} = \frac{dx}{dt}$$

$$i = \frac{\varepsilon_0 \cancel{b} V}{d} (K-1) \cdot \frac{r}{\cancel{b} d}$$

$$\boxed{i = \frac{\varepsilon_0 (K-1) V r}{d^2}} = \text{const}$$

i is independent of time.

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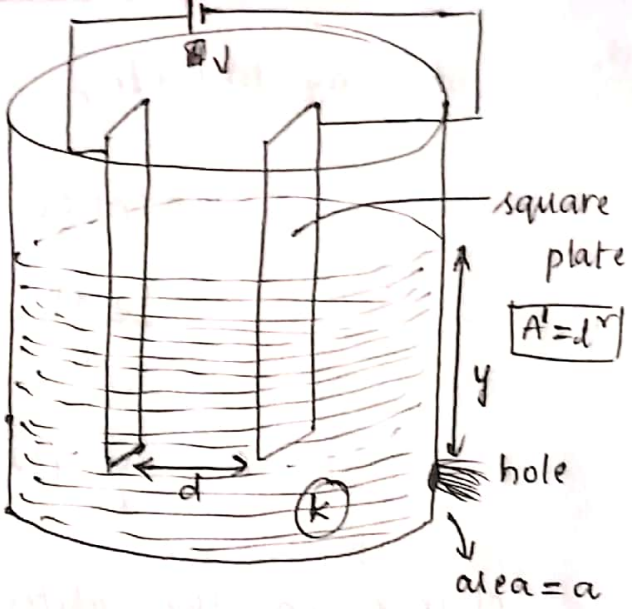
Ex:

Area of cross-section of tank = A

Initial height of liquid (h) = ℓ

Find current flowing in the circuit as a function of time.



Sol:

rate of flow of liquid coming out of orifice,

$$-\frac{dV}{dt} = aV$$

$$-A\frac{dy}{dt} = a\sqrt{2gy}$$

$$\int_{\ell}^{y} \frac{dy}{\sqrt{y}} = \int_{0}^{t} -\frac{a}{A}\sqrt{2g}\cdot dt$$

$$2\left[\sqrt{y}\right]_{\ell}^{y} = -\frac{a}{A}\sqrt{2g}\,(t)$$

$$2\left[\sqrt{y} - \sqrt{\ell}\right] = -\frac{a}{A}\sqrt{2g}\,(t)$$

$$\sqrt{y} = -\frac{a\sqrt{2g}}{2A}t + \sqrt{\ell}$$

$$y = \left(\sqrt{\ell} - \frac{a\sqrt{2g}}{2A}t\right)^2$$

now $$C = \frac{\varepsilon_0(\ell - y)(d)}{d} + \frac{\varepsilon_0 k (y)(\ell)}{d}$$

$$C = \frac{\varepsilon_0 \ell}{d}\left(\ell + (k-1)y\right)$$

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charge, $q = cV$

$$q = \frac{\varepsilon_0 A}{d}\left(d + (K-1)y\right)V$$

now,

$$i = -\frac{dq}{dt}$$

$$= -\frac{\varepsilon_0 AV}{d}(K-1)\frac{dy}{dt}$$

$$\frac{dy}{dt} = 2\left(\sqrt{2}\,-\frac{a\sqrt{2g}}{2A}t\right)\left(-\frac{a\sqrt{2g}}{2A}\right)$$

$$i = \frac{\varepsilon_0 AV}{d}(K-1)\left(\frac{a\sqrt{2g}}{2A}\right)(2)\left(\sqrt{d}-\frac{a\sqrt{2g}}{2A}t\right)$$

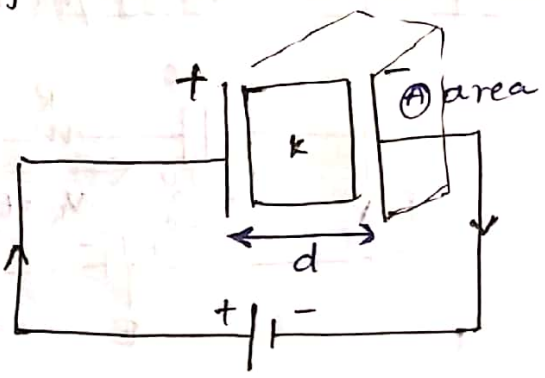
$$= \frac{\varepsilon_0 AV(K-1)}{d}\left(\frac{a}{A}\sqrt{2gd} - \frac{2a^2g}{2A^2}(t)\right)$$

$$\boxed{i = \frac{\varepsilon_0 AV(K-1)}{d}\left(\frac{a}{A}\sqrt{2gd} - \frac{a^2gt}{A^2}\right)}$$

Leakage current through capacitor:

* dielectric is not a perfect insulator. (also air)

* when applied potential reaches to breakdown i value it allows current through it called leakage current.



From ohm's law,

$$i = \frac{V}{R} \quad ; \quad R = \frac{\rho d}{A}$$

ρ - specific resistance of dielectric.

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leakage current)

$$\rho = \frac{VA}{\rho d}$$

$$i = \frac{V \varepsilon_0 K A}{\rho \varepsilon_0 K d}$$

$$\boxed{q = \frac{VC}{\rho \varepsilon_0 K}}$$

ρ → resistivity of dielectric.

we have,

$$R = \frac{\rho d}{A}$$

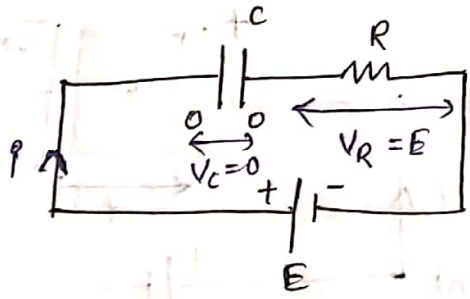
$$C = \frac{\varepsilon_0 K A}{d}$$

$$RC = \frac{\rho d}{A} \times \frac{\varepsilon_0 K A}{d}$$

$$\boxed{RC = \rho \varepsilon_0 K} *$$

R-C circuits:-

a) capacitor is uncharged :



At $t = 0$,

$$V = \frac{q}{C} = 0$$

$$i = \frac{dq}{dt} \neq 0$$

* Capacitor offers 'zero resistance' to the current flow

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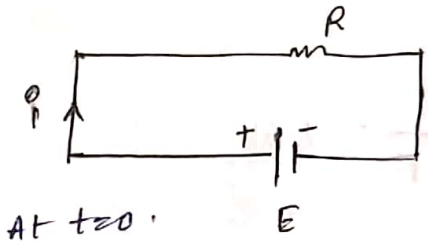
* It behaves like a connecting wire (capacitor is short circuited).

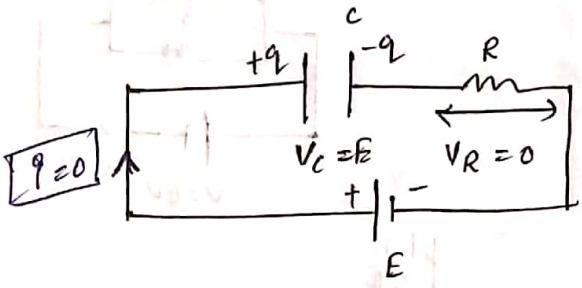
i.e,



$i = \frac{E}{R}$ (max)

At t=0.

B) After a long time, capacitor is fully charged ($t = \infty$) (steady state)



$V_R = 0$

$i_R = 0$

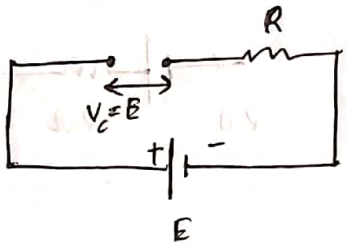
$\boxed{i = 0}$

* Capacitor offers infinite resistance to the current flow. i.e, capacitor is open circuited.

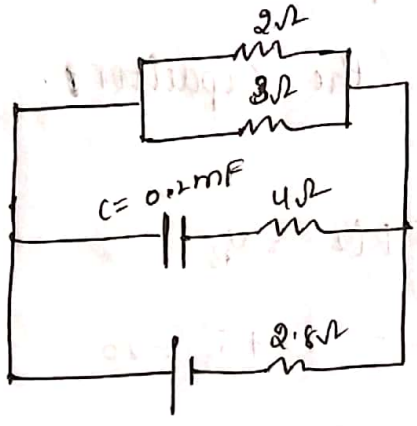
here,

$U_C = \frac{1}{2} C V_C^2$

$q = CV_C$



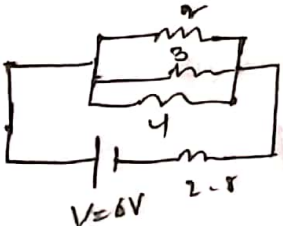
Ex:6



Find current flowing through battery at $t = 0$ & $t = \infty$.

Sol:6

at t=0,



$R_{eq} = \frac{\frac{6}{5} \times 4}{\frac{6}{5} + 4} + 2.8 = \frac{\frac{24}{5}}{\frac{26}{5}} + 2.8$

$= \frac{12}{13} + 2.8$

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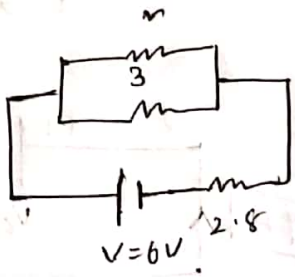
$V = iR$

$i = \frac{V}{R}$

$= \frac{6 \times 13}{12 + 13 \times 2.8} = 1.61$

at $t = \infty$

$R_{eq} = \frac{6}{5} + 2.8 = 1.2 + 2.8 = 4$

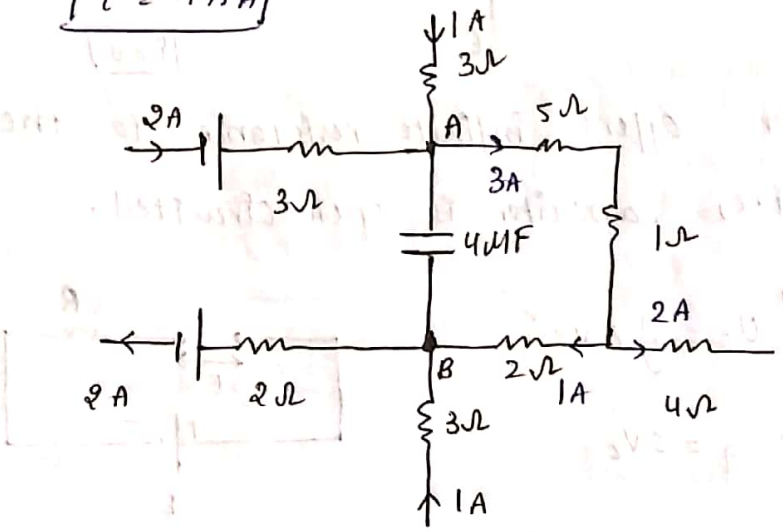


$V = iR$

$\frac{6}{4} = i$

$\boxed{i = 1.5\,A}$

Ex 6



Find energy stored in the capacitor.

Sol6

$V_A + 3(5) - 3(1) - 1(2) = V_B$

$V_A - V_B = 15 + 5 = 20$

$V_C = 20$

$Q = CV_C = 4 \times 20 = 80$

$U = \frac{1}{2} CV_C^2 = \frac{1}{2} \times 4 \times 20 \times 20$

$\boxed{U = 800\,\mu J}$

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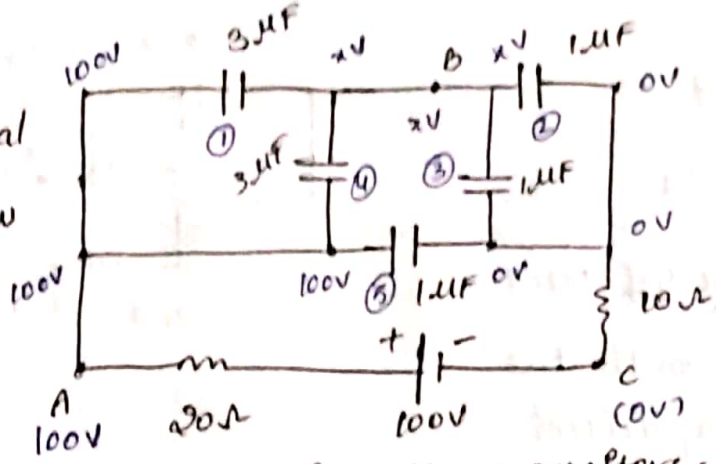
Ex 6

Find potential difference b/w

(i) A & B

(ii) B & C



(iii) Also find energy stored in all capacitors.

Sol

$$(x-100)(3) + (x-100)3 + (x-0)(1) = 0$$

$$6(x-100) + x = 0$$

$$7x = 600$$

$$x = \frac{600}{7}$$

$$\boxed{x = 75}$$

$$V_A - V_B = 100 - 75 = 25$$

$$V_B - V_C = 75 - 0 = 75$$

$$U_1 = \frac{1}{2} \times 3 \times 25 \times 25 = 937.5$$

$$U_2 = \frac{1}{2} \times 1 \times 75 \times 75 = 2812.5$$

$$U_3 = \frac{1}{2} \times 25 \times 25 \times 1 = 312.5$$

$$U_4 = \frac{1}{2} \times 3 \times 25 \times 25 = 937.5$$

$$U_5 = \frac{1}{2} \times 100 \times 100 \times 1 = 5000$$

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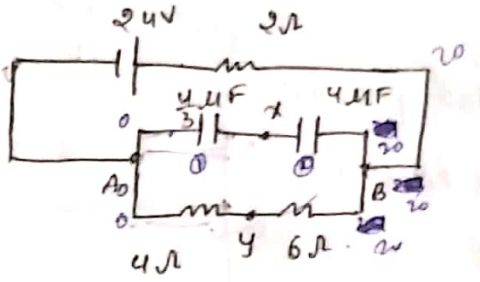
Ex: Find

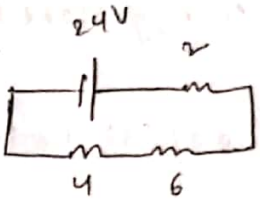
1) $V_x - V_y$

2) Energy stored in both the capacitors



Sol:

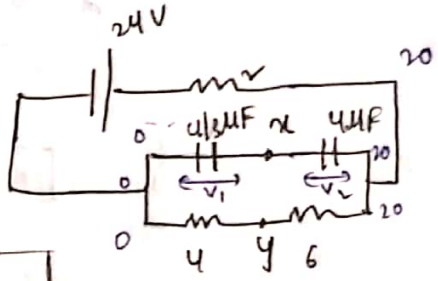
1)



$V = iR$

$24 = i(12)$

$\boxed{i = 2}$



$V_1 : V_2 = C_2 : C_1 = 3:1$

$V_1 = 15$

$V_2 = 5$

$\Rightarrow \boxed{V_x = 15}$

$V_x - 8 \times 2 = V_y$

$V_x - V_y = 12$ ✓

$V_x - V_y = 15 - 4(2)$

$= 15 - 8$

$\boxed{V_x - V_y = 7}$

2) $V_y = 4 \times 2$

$= 8$

$V_x = 12 + 8 = 20$

$U_1 = \frac{1}{2} \times \frac{4}{3} \times 10 \times 20 = \frac{800}{3}\mu J$

$U_2 = \frac{1}{2} \times 4^2 \times 4 \times 4 = 32\mu J$

2) $U_1 = \frac{1}{2} \times \frac{4}{3} \times 15^2$

$= 2 \times 5 \times 15$

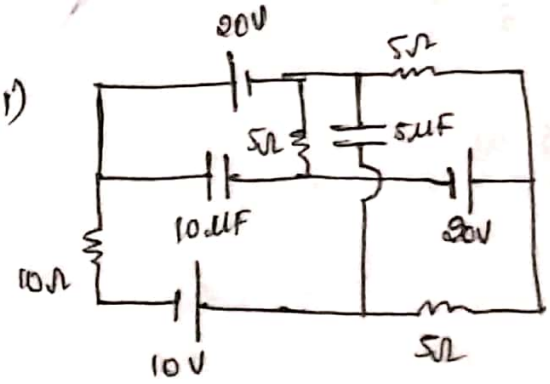
$\boxed{U_1 = 150}$

$U_2 = \frac{1}{2} \times 4 \times 5^2$

$\boxed{U_2 = 50}$

Ex 6

1)



Find the ratio of charge present on 5μF & 10μF capacitor in steddy state.

Ans: $\frac{13}{38}$

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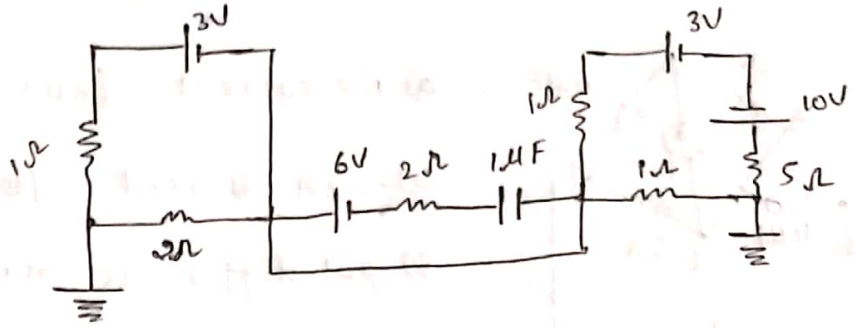
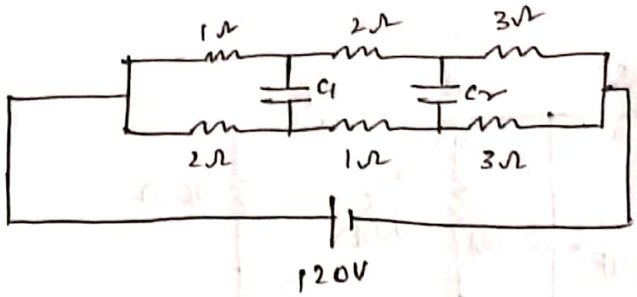
2)



Find the charge on capacitor in steady state.

6μC

3)

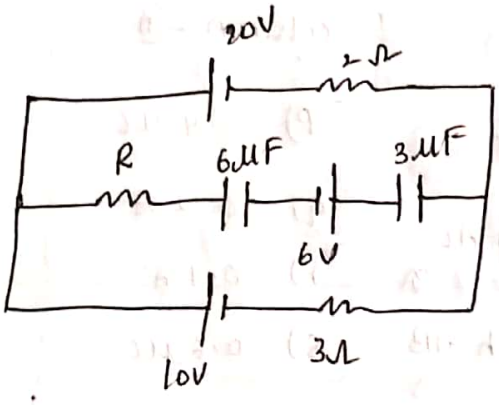


$C_1 = C_2 = 2\mu F$

Find charge on capacitors C_1 & C_2.

40μC zero

4)



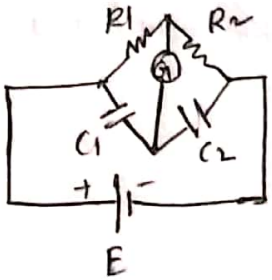
~~a)~~ charge on 6μF is same as charge on 3μF & is equal to 44μC.

~~b)~~ charge on each capacitor is zero.

~~c)~~ current in R resistor is zero.

~~d)~~ charge on capacitor depends on R.

~~Ex~~ 5)



In steady state, if galvanometer shows zero deflection then

1) $\frac{R_1}{R_2} = \frac{C_1}{C_2}$ ~~2)~~ $\frac{R_1}{R_2} = \frac{C_2}{C_1}$ 3) $R_1^2 C_1 = R_2^2 C$

4) $R_1^2 C_1^2 = R C_2^2$

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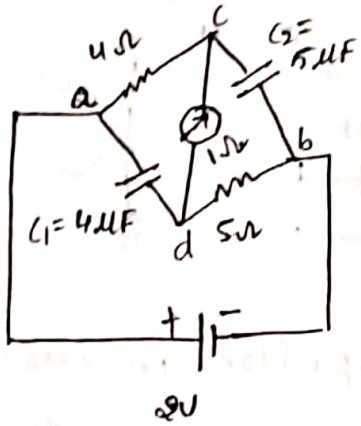
6)



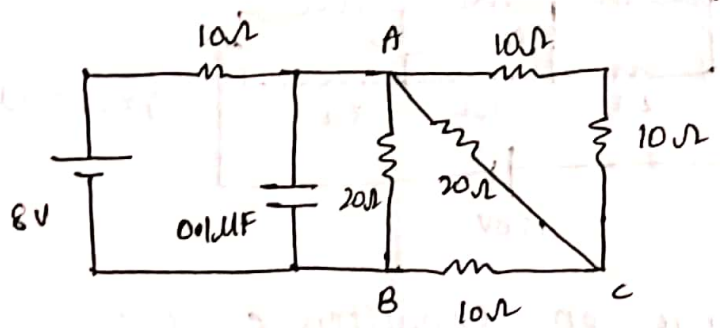
a) No current flows in G_1.

b) 0.2 A current flows in G_1.

c) pot. difference across C_1 is 1V

d) pot. difference across C_2 is 1.2V

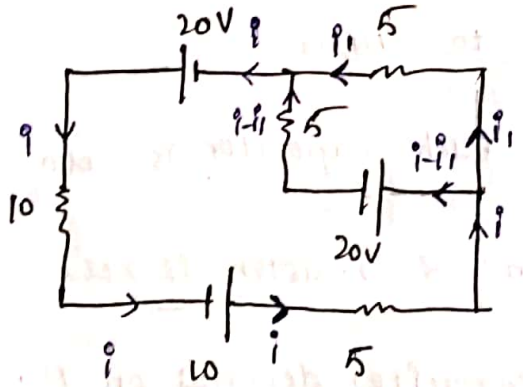
7)



column -I	column -II
a) charge on capacitor q	P) 0.4 μC
b) current in branch AC	q) 0.2 A
c) current in branch AB	r) 0.1 A
	s) 0.6 μC.

Solⁿ

1)



$$10 - 5i - 2i_1 - 5(i - i_1) + 20 - 10i = 0$$

$$10 - 20i + 5i_1 = 0$$

$$20i = 5i_1 + 10$$

$$4i = i_1 + 2 \quad \text{—①}$$

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$$20 - 5i_1 + 5(i - i_1) = 0$$

$$20 = 10i_1 - 5i$$

$$4 + i = 2i_1$$

$$i = 2i_1 - 4$$

↓

put in ①

$$4(2i_1 - 4) = i_1 + 2$$

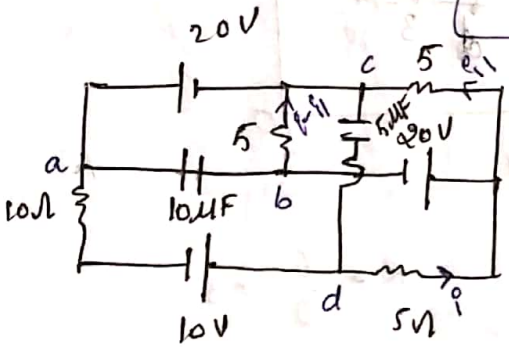
$$8i_1 - 16 = i_1 + 2$$

$$7i_1 = 18$$

$$\boxed{i_1 = 18/7}$$

$$i = \frac{36}{7} - 4 = \frac{36-28}{7}$$

$$\boxed{i = \frac{8}{7}}$$



$$V_a - 20 + 5(i - i_1) = V_b$$

$$V_a - V_b = 20 - 5\left(-\frac{10}{7}\right)$$

$$= \frac{140 + 50}{7}$$

$$= \frac{190}{7}$$

$$q_{10} = \frac{190}{7} \times 10 = \frac{1900}{7}$$

$$V_d - 5i - 5i_1 = V_c$$

$$V_d - V_c = 5i + 5i_1$$

$$= 5\left(\frac{26}{7}\right)$$

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$$q_5 = \frac{5 \times 26 \times 5}{7}$$

$$q_5 : q_{10} = \frac{25 \times 26}{7} : \frac{1900}{7}$$

$$= 26 : 19 \times 4$$

$$\boxed{q_5 : q_{10} = 13 : 38}$$

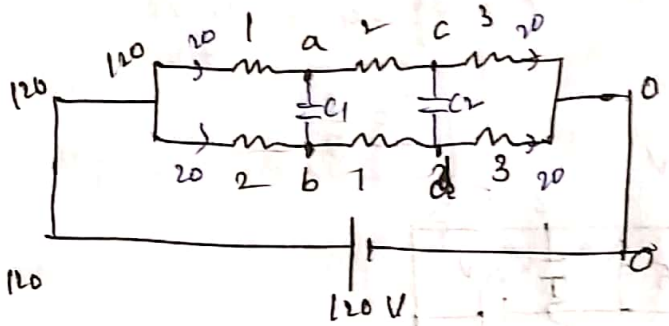
2)

$$q = VC$$

$$= 6 \times 1$$

$$\boxed{q = 6\mu C}$$

3)



$$V = iR \qquad R = \frac{6}{2} = 3$$

$$120 = i(3)$$

$$\boxed{i = 40}$$

$$V_a + 20 - i(20) = V_b$$

$$V_a - V_b = 40 - 20 = 20$$

$$q_{C_1} = (V_a - V_b) C_1$$

$$= 20 \times 2$$

$$\boxed{q_{C_1} = 40}$$

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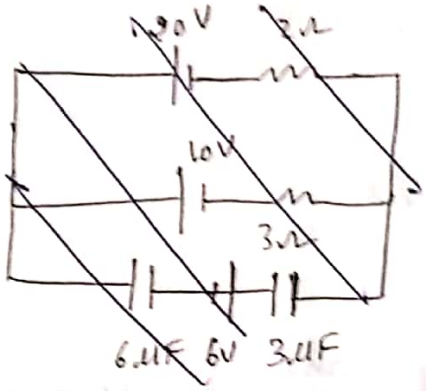
$V_c - 30(2.0) + 3(10) = V_d$

$$\boxed{V_c = V_d}$$

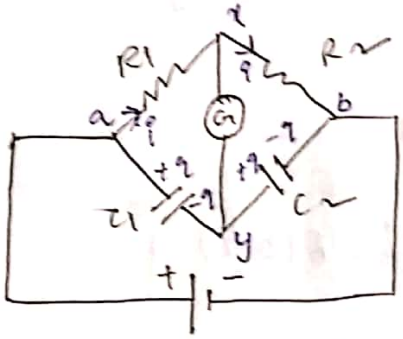
$$q_{c_2} = (V_c - V_d) C_2$$

$$\boxed{q_{c_2} = 0}$$

4)



5)



As galvanometer shows zero deflection.
$V_x = V_y$

$$V_a - V_x = i R_1$$

$$V_x - V_b = i R_2$$

$$V_a - V_y = \frac{q}{C_1}$$

$$V_y - V_b = \frac{q}{C_2}$$

$$i R_1 = \frac{q}{C_1} \qquad i R_2 = \frac{q}{C_2}$$

$$R_1 C_1 = \frac{q}{i} \text{ -①} \qquad R_2 C_2 = \frac{q}{i} \text{ -②}$$

from ① & ② $\Rightarrow R_1 C_1 = R_2 C_2$

$$\boxed{\frac{R_1}{R_2} = \frac{C_2}{C_1}}$$

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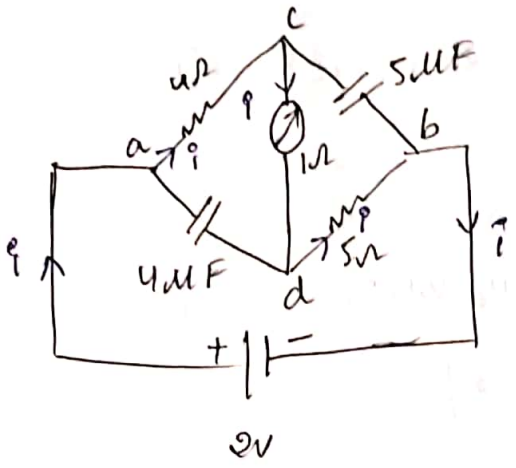
6)



$V = iR$

$2 = i(4+1+5)$

$i = \frac{2}{10}$

$\boxed{i = 0.2\,A}$ ← also through galvanometer.

$V_a - 4i - i = V_d$

$V_a - V_d = 5i = 5(0.2) = 1$

pot drop across $e_1 = 1V$

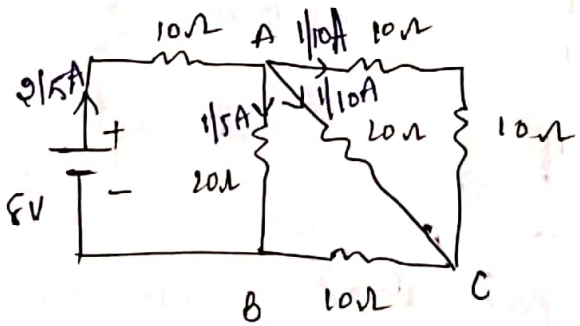
$V_c - i - 5i = V_b$

$V_c - V_b = 6i = 6(0.2)$

$V_c - V_b = 1.2$

pot-drop across $e_2 = 1.2V$

7)



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$V = iR$

$8 = i(20)$

$$\boxed{i = \frac{2}{5}}$$

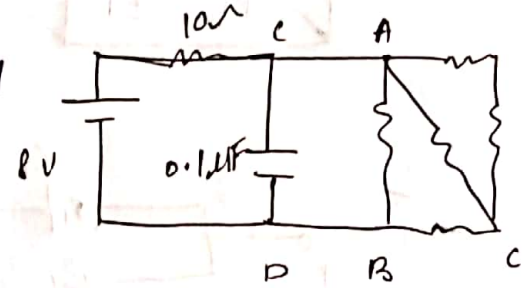
current through $AC = 0.1\,A$

current through $AB = 0.2\,A$

$$V_A - \frac{1}{5}(20) = V_B$$

$$\boxed{V_A - V_B = 4}$$

$$V_A - V_B = V_C - V_D = 4$$



$$q = (V_C - V_D)(0.1)$$

$$= 4 \times 0.1$$

$$\boxed{q = 0.4\,\mu C}$$

4)

→ current on R is zero.

→ charge on capacitor is independent of value of R.

→ charge on the capacitors:

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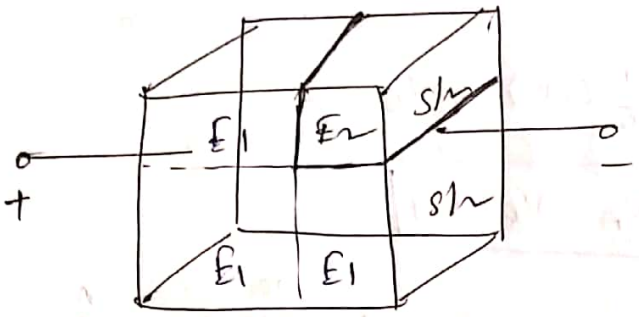
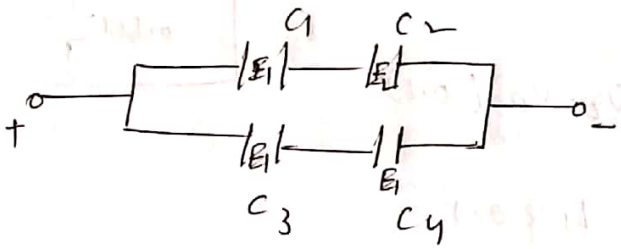
Ex: A parallel plate capacitor having plates of area S and plate seperation d, has capacitance C_1 in air. when two dielectrics of different relative permitivities $E_1 = 2$ & $E_2 = 4$ are introduced between the two plates as shown in the figure. The capacitance becomes C_2. The ratio $\frac{C_2}{C_1}$ is

Sol:



Sol:



$$C_1 = \frac{\varepsilon_0 S/2}{d/2} E_1 = \frac{\varepsilon_0 S}{d} (2)$$

$$C_2 = \frac{\varepsilon_0 S/2}{d/2} E_2 = \frac{\varepsilon_0 S}{d} (4) = 2C_1$$

C_1 & C_2 are in series $\Rightarrow C' = \frac{C_1 C_2}{C_1 + C_2}$ *

$$C_3 = C_4 = \frac{\varepsilon_0 S/2}{d/2} \cdot E_1 = \frac{\varepsilon_0 S}{d} (2)$$

C_3 & C_4 are in series $\Rightarrow C'' = \frac{C_3 C_4}{C_3 + C_4}$

C', C'' are in parallel.

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$$C_{eff} = C' + C''$$

$$= \frac{C_1 C_2}{C_1 + C_2} + \frac{C_3 C_4}{C_3 + C_4}$$

$$= \frac{C_1 \times 2C_1}{3C_1} + \frac{C_3 \times C_3}{2C_3}$$

$$= \frac{2C_1}{3} + \frac{C_3}{2}$$

$$= \frac{2}{3}\frac{\varepsilon_0 S}{d}(2) + \frac{\varepsilon_0 S}{d}$$

$$C_{eff} = \frac{\varepsilon_0 S}{d}\left(\frac{4}{3} + 1\right) = \frac{7}{3}\frac{\varepsilon_0 S}{d}$$

C_1 = C in air = $\frac{\varepsilon_0 S}{d}$

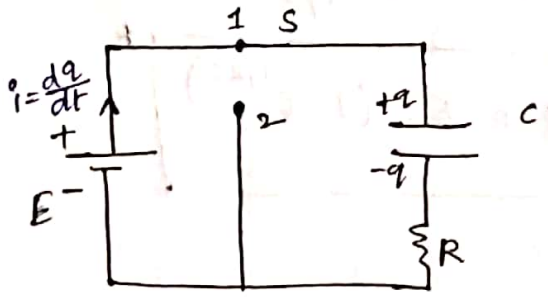
C_2 = C eff = $\frac{7\varepsilon_0 S}{3d}$

$$\boxed{\frac{C_2}{C_1} = \frac{7}{3}}$$

8|05|25 | Thursday

Transient Circuits: (continuously changes (charging & discharging))

a) **Charging of a capacitor:**



* Initially capacitor is uncharged.
* switch is connected to 1.

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* capacitor starts charging.

At anytime t.

Let q = charge on capacitor.

'i' is the current in the circuit.

* From KVL,

$$E - \frac{q}{C} - iR = 0$$

$$\frac{CE - q}{C} = iR$$

$$\frac{Ec - q}{C} = R\left(\frac{dq}{dt}\right)$$

$$\int_{q=0}^{q=q} \frac{dq}{Ec - q} = \int_{t=0}^{t=t} \frac{dt}{RC}$$

$$\left[\ln(Ec - q)\right]_0^q = \frac{-t}{RC}$$

$$\ln(Ec - q) - \ln(Ec) = \frac{-t}{RC}$$

$$\frac{Ec - q}{Ec} = e^{-t/RC}$$

$$1 - \frac{q}{Ec} = e^{-t/RC}$$

$$1 - e^{-t/RC} = q/Ec$$

$$\boxed{q = Ec\left(1 - e^{-t/RC}\right)} \; *$$

points:

* At $t = 0$, $q = Ec(1 - e^0) = 0$

* At $t = \infty$, $q = Ec\left(1 - e^{-\infty}\right) = Ec = q_{(max)}$

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i.e., charge increases exponentially with time.

* $\tau = RC$ = time const of R-C circuit. unit: sec

D.F: [T]

when $T = RC$ sec

$$q = q_0 (1 - e^{-t/RC})$$

$$= q_0 \left(1 - \frac{1}{e}\right)$$

$$= q_0 \left(\frac{e-1}{e}\right)$$

$$= q_0 \left(\frac{2.718 - 1}{2.718}\right)$$

$$q = \frac{1.718}{2.718} q_0$$

$$q = 0.639_0 = 63\% q_0$$

*Imp pts

At $t = 5RC$

$q = q_0 (1 - e^{-5RC/RC})$

$= q_0 (1 - 1/e^5)$

$\boxed{q = 0.998 q_0}$

practically time taken for complete charging of capacitor is $5RC$ (5τ).

Def:

The duration of time for which charge increases from 0 to 63% of max charge is called time constant of RC circuit.

Current in the circuit is-

$$i = \frac{dq}{dt}$$

$$i = \frac{d}{dt}\left(q_0 (1 - e^{-t/RC})\right)$$

$$i = \frac{q_0}{RC} e^{-t/RC}$$

$$i = \frac{CE}{RC} e^{-t/RC}$$

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$$i = \frac{E}{R} e^{-t/RC} = i_0 e^{-t/RC}$$

$$\boxed{i = i_0 e^{-t/RC}} \quad *$$

At $t = 0$, $i = i_0 = \frac{E}{R}$ (max)

At $t = \infty$, $i_0 = 0$

i-e, current decreases exponentially with time.

Potential difference across capacitor:

$$V_C = \frac{q}{C}$$

$$V_C = \frac{q_0(1 - e^{-t/RC})}{C}$$

$$V_C = \frac{EC}{C}(1 - e^{-t/RC})$$

$$\boxed{V_C = E(1 - e^{-t/RC})} \quad *$$

At $t = 0$, $V_C = E(1 - e^0) = 0$

At $t = \infty$, $V_C = E(1 - e^{-\infty}) = E$

clearly, potential drop exponentially increases with time.

Potential drop across resistance:

$$V_R = iR$$

$$= i_0 e^{-t/RC} R \qquad (i_0 = E/R)$$

$$V_R = i_0 R e^{-t/RC}$$

$$\boxed{V_R = E e^{-t/RC}} \quad *$$

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At $t=0$, $V_R = E$ (max)

At $t=\infty$, $V_R = 0$

i.e, p.drop decreases exponentially with time.

Now, $V_C + V_R = E(1-e^{-t/RC}) + E e^{-t/RC}$

$$\boxed{V_C + V_R = E}^{*}$$ = const at all times.

Energy stored in the capacitor:

$$U_C = \frac{q^2}{2C}$$

$$= \frac{q_0^2 (1-e^{-t/RC})^2}{2C} \qquad (q_0 = CE)$$

$$\boxed{U_C = \frac{1}{2} CE^2 (1-e^{-t/RC})^2}$$

At $t=0$, $U_C = 0$

At $t=\infty$, $U_C = \frac{1}{2} CE^2$ (max) $= U_0$

i.e, U_C increases exponentially with time.

Energy dissipated in resistor of resistance R:

$$H = \int_0^t i^2 R\, dt$$

$$= \int_0^t (i_0 e^{-t/RC})^2 R\, dt$$

$$= i_0^2 R \frac{\left[e^{-2t/RC}\right]_0^t}{-2/RC}$$

$$= \frac{-i_0^2 R^2 C}{2}\left[e^{-2t/RC}\right]_0^t = \frac{i_0^2 R^2 C}{2}(1-e^{-2t/RC})$$

$[i_0 = E/R]$

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$$\therefore \boxed{H = \frac{1}{2} CE^2 \left(1 - e^{-2t/RC}\right)}$$

At $t = 0$, $H = 0$

At $t = \infty$, $H = \frac{1}{2} CE^2$ (max)

i.e, H increases exponentially with time.

work done by the battery :

$$W_b = (\Delta Q) E = qE$$

$$= q_0 \left(1 - e^{-t/RC}\right) E$$

$$\boxed{W_b = CE^2 \left(1 - e^{-t/RC}\right)}$$

At $t = 0$, $W_b = 0$

At $t = \infty$, $W_b = CE^2$ (max)

i.e, W_b increases exponentially with time.

now,

$$U_c + H = \frac{1}{2} CE^2 \left(1 - e^{-t/RC}\right)^2 + \frac{1}{2} CE^2 \left(1 - e^{-2t/RC}\right)$$

$$= \frac{1}{2} CE^2 \left[1 + e^{-2t/RC} - 2e^{-t/RC} + 1 - e^{-2t/RC}\right]$$

$$= \frac{1}{2} CE^2 \times 2 \left(1 - e^{-t/RC}\right)$$

$$\boxed{U_c + H = CE^2 \left(1 - e^{-t/RC}\right) = W_b}$$

* Went $= 0$

$U_i = 0$ ($\because$ Initially capacitor is uncharged)

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$$\cancel{v_i^{\to 0}} + w_b + \cancel{w_{ent}^{\to 0}} = v_f + H$$

$$w_b = v_f + H \quad //$$

** In transient circuits,

$$v_f \neq H \neq \frac{w_b}{2}.$$

Power delivered to the capacitor (or) rate of energy stored in the capacitor

$$P_c = \frac{dU_c}{dt} = \frac{d}{dt}\left(\frac{1}{2}CE^2(1-e^{-t/RC})^2\right)$$

$$P_c = \frac{1}{2}CE^2 \cdot 2(1-e^{-t/RC})\left(-\frac{1}{RC}\right)(0-e^{-t/RC})$$

$$= \frac{CE^2}{RC}(1-e^{-t/RC})(e^{-t/RC})$$

$$= \frac{E^2}{R}(1-e^{-t/RC})(e^{-t/RC})$$

$$[E = i_0 R]$$

$$P_c = \frac{i_0^2 R^2}{R}(1-e^{-t/RC})(e^{-t/RC})$$

$$\boxed{P_c = i_0^2 R(1-e^{-t/RC})(e^{-t/RC})}$$

At $t=0$, $P_c = 0$

At $t=\infty$, $P_c = 0$

For max P_c, $\frac{dP_c}{dt} = 0$ (or) $\frac{d^2U_c}{dt^2} = 0$

$$\frac{E^2}{R}\frac{d}{dt}(1-e^{-t/RC})(e^{-t/RC}) = 0$$

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$$\frac{d\phi}{dt}\left(e^{-t/RC} - e^{-2t/RC}\right) = 0$$

$$-\frac{1}{RC}e^{-t/RL} - \frac{-2}{RL}e^{-2t/RL} = 0$$

$$e^{-t/RL} = 2\,e^{-2t/RL}$$

$$\boxed{\frac{1}{2} = e^{-t/RL}}$$

$$e^{t/RL} = 2$$

$$t/RL = \ln 2$$

$$\boxed{t = RL \ln 2}$$

$$P_L = \frac{E^2}{R}\left(1 - \frac{1}{2}\right)\left(1/2\right)$$

$$\boxed{P_L = \frac{E^2}{4R}} \quad *$$

$$\boxed{P_{L\,max} = \frac{i_0^2 R}{4}} \quad *$$

Power dissipated in Resistor :

$$P_R = \frac{dH}{dt} = i^2 R$$

$$\boxed{P_R = i_0^2 R\, e^{-2t/RL}}$$

i.e., P_R is decreases exponentially with time.

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Power delivered by battery to circuit:

$$P_b = E i$$

$$= E i_0 e^{-t/RC}$$

$$\boxed{P_b = i_0^2 R e^{-t/RC}}$$

P_b decreases exponentially with time.

now, $$P_c + P_R = i_0^2 R e^{-t/RC}(1 - e^{-t/RC}) + i_0^2 R e^{-2t/RC}$$

$$= i_0^2 R e^{-t/RC}$$

$$\boxed{P_c + P_R = P_b}$$ *

(or) $$E = iR + \frac{q}{C}$$

$$Ei = i^2 R + \frac{q}{C} i$$

$$Ei = i^2 R + \frac{q}{C}\frac{dq}{dt}$$

$$Ei = i^2 R + \frac{d}{dt}\left(\frac{q^2}{2C}\right)$$

$$\boxed{P_b = P_R + P_c}$$

Graphs:

1) Charge vs time graphs

$$q = q_0(1 - e^{-t/RC})$$

At $t = 0 \Rightarrow q = 0$

At $t = \infty \Rightarrow q = q_0$

slope, $\frac{dq}{dt} = \frac{q_0}{RC} e^{-t/RC}$ — slope is +ve & decreasing continuously

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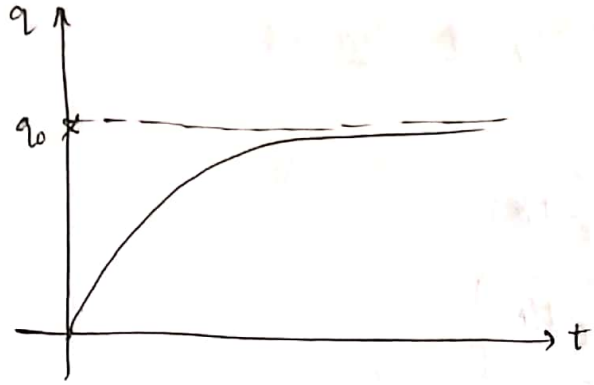


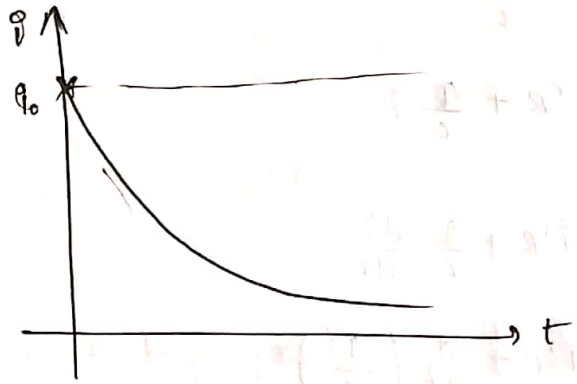
2) current vs time graph:

$$i = i_0 e^{-t/RC}$$

at $t = 0$, $i = i_0$

at $t = \infty$, $i = 0$

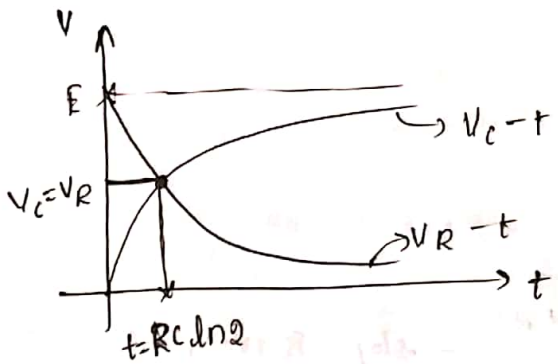
slope, $\frac{di}{dt} = -\frac{i_0}{RC} e^{-t/RC}$ – slope is -ve & decreasing continuously



3) potential drop vs time graph:

$$V_C = E(1 - e^{-t/RC})$$

$$V_R = E(e^{-t/RC})$$



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$$V_C = V_R$$

$$I = 2e^{-t/RC}$$

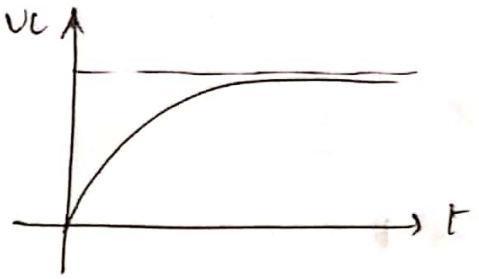
$$\frac{1}{2} = \frac{1}{e^{t/RC}}$$

$$e^{t/RC} = 2$$

$$\boxed{t = RC \ln 2}$$

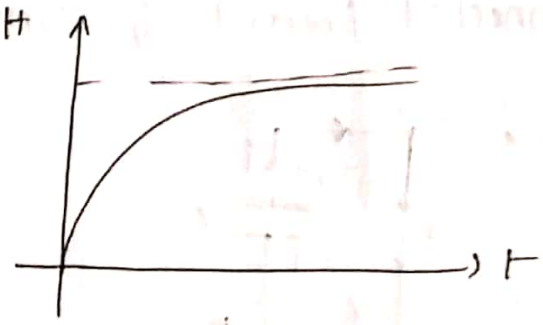
4) Energy stored in capacitor vs time :

$$U_C = \frac{1}{2} C E^2 (1 - e^{-t/RC})^2$$



5) Energy dissipated in R vs time :

$$H = \frac{1}{2} C E^2 (1 - e^{-2t/RC})$$



6) Power delivered to the capacitor vs time :

$$P_C = I_0^2 R e^{-t/RC} (1 - e^{-t/RC})$$

at $t = 0$, $P_C = 0$, at $t = \infty$, $P_C = 0$.

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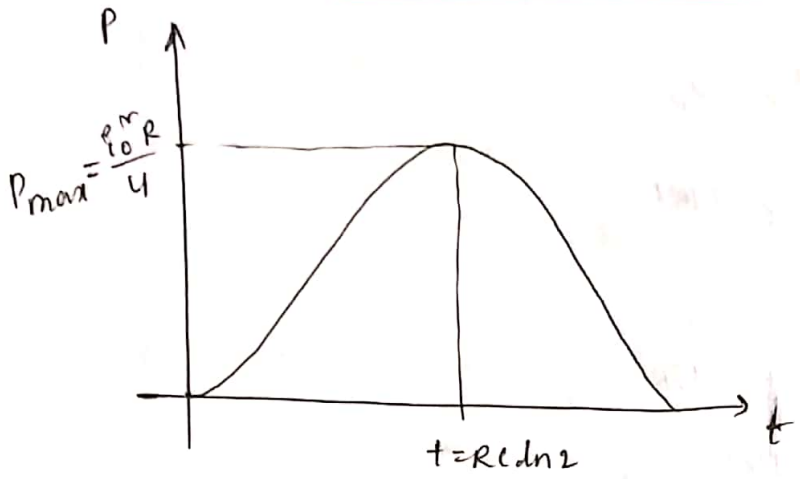
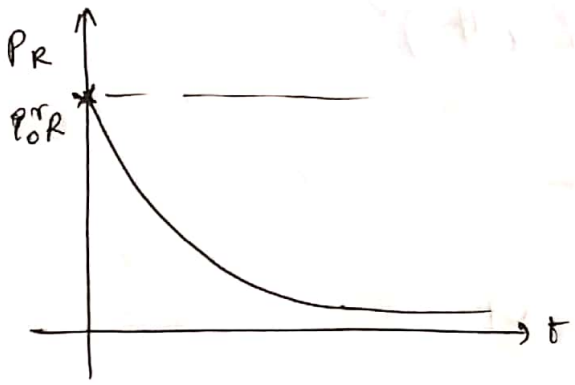


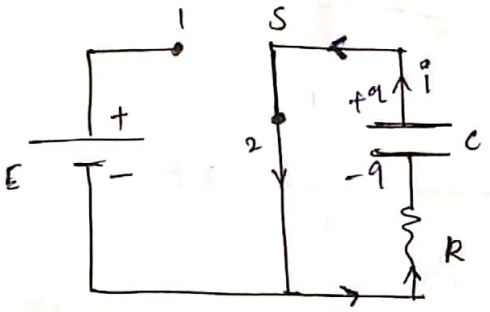
7) Power dissipated in R vs time :-

$$P_R = i_0^2 R e^{-2t/RC}$$



(8) Decay of Charge :-

* when capacitor is fully charged, (i.e, $q_c = q_0 = CE$)
* switch is disconnected from 1 & connected to 2.



At any time t,

q = charge on plates, i = current in circuit.

$$i = \frac{d}{dt}(CE - q)$$

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$$i = -\frac{dq}{dt}$$

From KVL,

$$\frac{q}{C} - iR = 0$$

$$q = iRC$$

$$\frac{q}{RC} = -\frac{dq}{dt}$$

$$\int_0^t -\frac{dt}{RC} = \int_{q_0}^{q} \frac{dq}{q}$$

$$[\ln q]_{q_0}^{q} = \frac{-t}{RC}$$

$$\ln q/q_0 = -\frac{t}{RC}$$

$$\boxed{q = q_0 e^{-t/RC}}$$

i.e, charge decreases exponentially with time.

* current,

$$i = -\frac{dq}{dt}$$

$$= \frac{-d}{dt} q_0 e^{-t/RC}$$

$$= \frac{q_0}{RC} e^{-t/RC}$$

$$= \frac{dE}{Rt} e^{-t/RC}$$

$$\boxed{i = i_0 e^{-t/RC}}$$

i.e, decreases exponentially with time.

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* potential diff across C &

$$V_c = \frac{q}{c} = \frac{q_0}{c} e^{-t/RC}$$

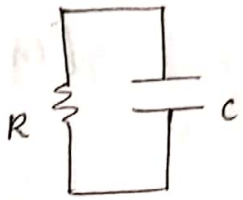
$$\boxed{V_c = E\, e^{-t/RC}}$$

* potential drop across R &

$$V_R = i_0 R$$

$$= i_0 (e^{-t/RC}) R$$

$$\boxed{V_R = i_0 R\, (e^{-t/RC})}$$



capacitor acts as battery.

Clearly $\boxed{V_R = V_c}$ at all times.

* Energy stored in C &

$$U_c = \frac{q^2}{2C}$$

$$= \frac{q_0^2}{2C} e^{-2t/RC}$$

$$\boxed{U_c = \frac{1}{2} C E^2 e^{-2t/RC}}$$

* Heat energy produced in R &

$$H = \int_0^t i^2 R\, dt$$

$$= i_0^2 R \int_0^t e^{-2t/RC} \cdot dt$$

$$= i_0^2 R \left(-\frac{RC}{2}\right) \left[e^{-2t/RC}\right]_0^t$$

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$$H = \frac{-i_0^2 R^2 C}{2}\left(e^{-2t/RC} - 1\right)$$

$$H = \frac{i_0^2 R^2 C}{2}\left(1 - e^{-2t/RC}\right) \Rightarrow \boxed{H = \frac{1}{2}CE^2\left(1 - e^{-2t/RC}\right)}$$

* Increase in Energy stored in capacitor:

$$\Delta U = U_i - U_{at\ t}$$

$$= \frac{1}{2}CE^2 - \frac{1}{2}CE^2\left(e^{-\frac{2t}{RC}}\right)$$

$$\Delta U = \frac{1}{2}CE^2\left(1 - e^{-2t/RC}\right)$$

= heat dissipated in R.

* Rate of energy stored in C:

$$P_c = \frac{dU_c}{dt}$$

$$= -\frac{1}{2}(2)E^2 e^{-2t/RC}\left(\frac{2}{R}\right)$$

$$P_c = -\frac{E^2}{R}e^{-2t/RC}$$

$$\boxed{P_c = -i_0^2 R\, e^{-2t/RC}}$$

* Rate of heat dissipated in R:

$$P_R = \frac{dH}{dt}$$

$$= \frac{1}{2}(2)CE^2\left(-e^{-2t/RC}\right)\left(-\frac{2}{RC}\right)$$

$$\boxed{P_R = i_0^2 R\, e^{-2t/RC}}$$

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Finally,

$$q = q_0 e^{-t/RC}$$

when $t = RC = \tau$

$$q = q_0 e^{-1}$$

$$= \frac{q_0}{e}$$

$$q = \frac{q_0}{2.718}$$

$$q = 0.379_0 = 37\% q_0$$

During discharging, charge decays from q_0 to 37% q_0 (i.e, 37% max value) in time RC.

↳ Time constant.

Calculation of Time constant:

a) During charging:

Steps:

1) Short circuit the battery.
2) Find R_{eff} across C.
3) Calculate time constant using $\tau = R_{eff} C_{eff}$.

b) During discharging:

Steps:

1) Remove the battery from the circuit.
2) Calculate R_{eff} across C.

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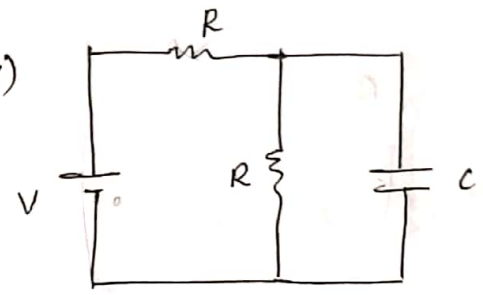
3) Calculate Time constant using $\tau = R_{eff} C_{eff}$

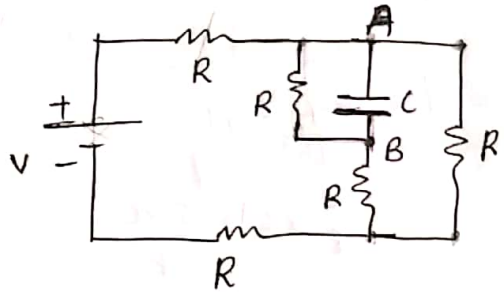
Ex:6

1)



Calculate Time constant during charging and discharging.

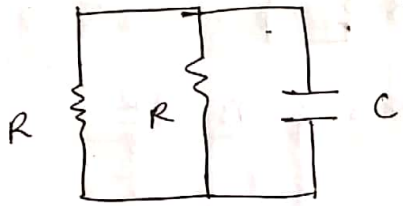
2)



calculate Time constant during charging and discharging.

Sol:6

1) During charging,

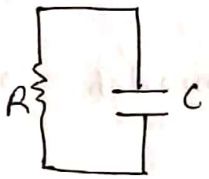


$\tau = R_{eff} C_{eff}$

$= \frac{R \times C}{2}$

$\boxed{\tau = \frac{RC}{2}}$

During discharging,



$\tau = R_{eff} C$

$\boxed{\tau = RC}$

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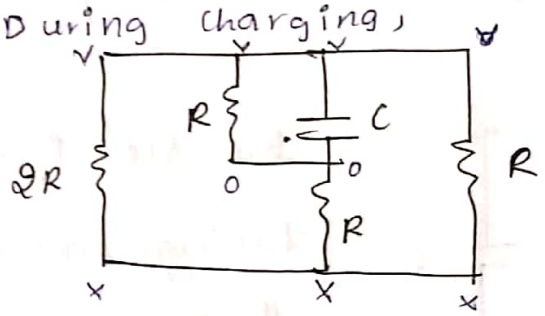
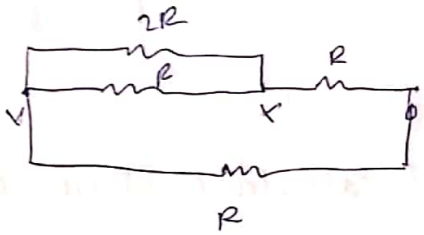
2) During charging,



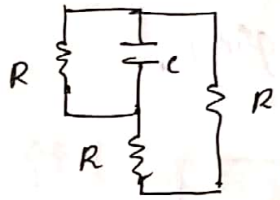


Reff across C is $\frac{5R}{8}$

$T = R_{eff} C_{eff}$

$= \frac{5R}{8} \times C = \frac{5RC}{8}$

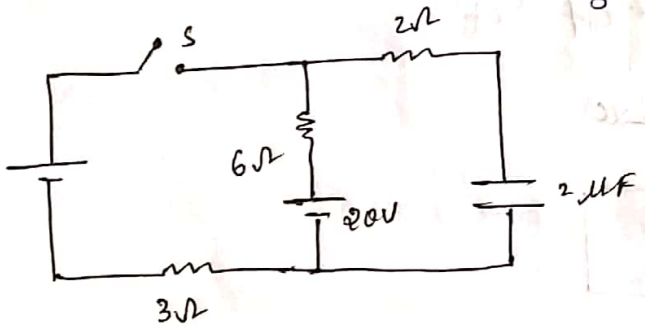
During discharging,



Reff across C is $\frac{2R}{3}$

$T = R_{eff} C = \frac{2RC}{3}$.

Ex 6



1) Find the ratio of time constant before and after charging the switch (during charging).

2) Find current flowing through 2Ω at i) $t = 0$ when ii) $t = \infty$

switch is closed.

3) Find charge on the capacitor as a function of time

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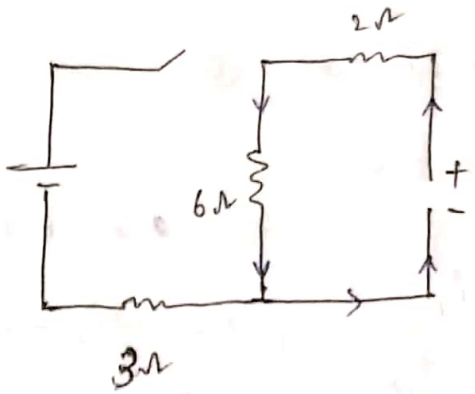
after the switch is closed.

Sol: 1)

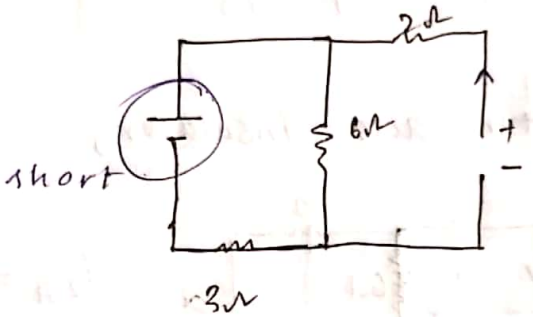


Before switch is closed.

$$R_{eff} = 6+2 = 8\,\Omega$$

$$\tau_1 = R_{eff}\,C = 8 \times 2 = 16\,\mu sec.$$

After switch is closed

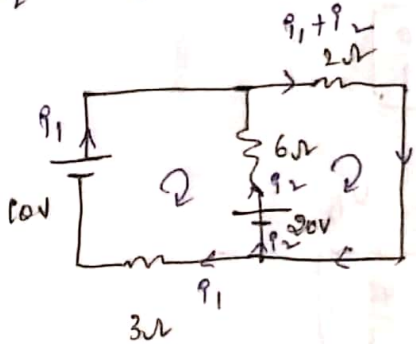


$$R_{eff} = \frac{6\times 3}{6+3} + 2 = 4$$

$$\tau_2 = R_{eff} \times C = 4 \times 2 = 8\,\mu sec.$$

$$\frac{\tau_1}{\tau_2} = \frac{16}{8} = 2$$

2)



At t=0,

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$10 + 6i_2 - 20 - 3i_1 = 0$ — ①

$10 - 2(i_1 + i_2) - 3i_1 = 0$ — ②

$10 = 6i_2 - 3i_1$

$\frac{10}{3} = 2i_2 - i_1$

$i_1 = 2i_2 - \frac{10}{3}$

$= \frac{40}{9} - \frac{10}{3}$

$i_1 = \frac{10}{9}$

$10 = 6i_1 + 2i_2$

$10 = 10i_2 - \frac{50}{3} + 2i_2$

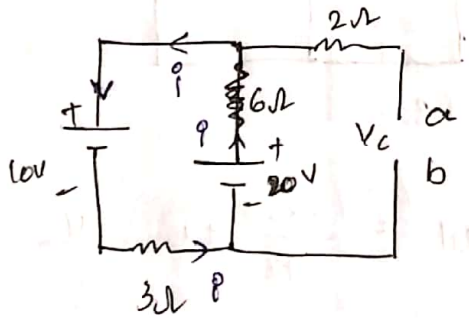
$\frac{20}{3} = 12 i_2$

$\frac{30}{3}$

$\boxed{i_2 = \frac{20}{9}}$

$i_1 + i_2 = \frac{10+20}{9} = \frac{30}{9} = \frac{10}{3}$

At $t = \infty$, capacitor acts insulator,



$i_{2\Omega} = 0$

$-3i + 20 - 6i - 10 = 0$

$10 = 9i$

$\boxed{i = \frac{10}{9}}$

$V_a + \frac{10}{9}(6) - 20 = V_b$

$\boxed{V_a - V_b = \frac{40}{3} V}$

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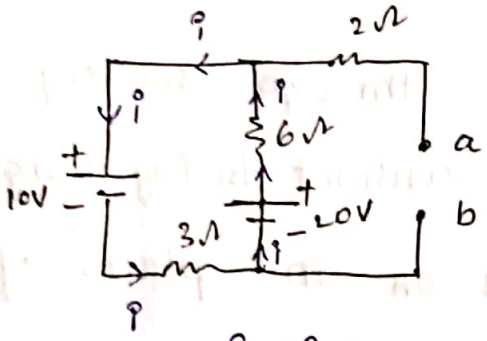
charge on (q) = $(V_a - V_b)(C) = \frac{40}{3}(2) = \frac{80}{3}\mu C$.

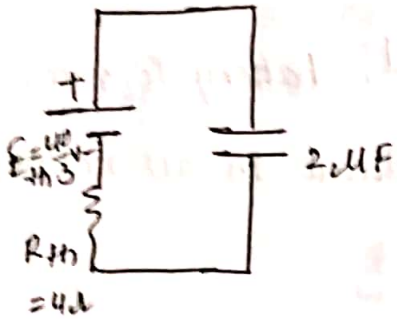
* To find charge on the plates as a function of t (use thevinin's theorem)

To find E_{th}:

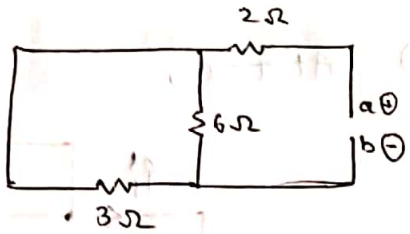


$i = \frac{10}{9}$ A

$V_a - V_b = \frac{40}{3}$ V



To find R_{th}:-



$R_{th} = 4\Omega$

$q = q_0 (1 - e^{-t/RC})$

charge on the plates,

$q = q_0 (1 - e^{-t/\tau})$

$= C\mathcal{E}_{th} (1 - e^{-t/R_{th}C})$

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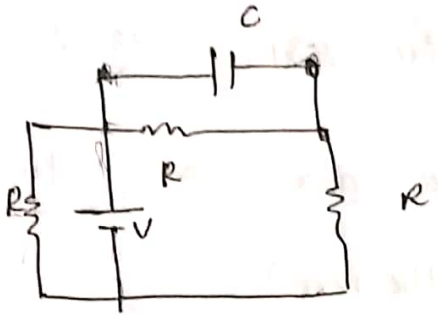
Ex 6



Find

1) current through battery at $t=0$ & $t=\infty$

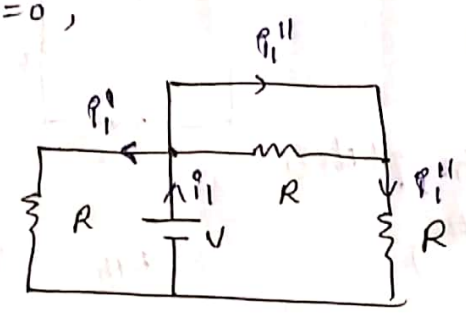
2) Time constant during charging and discharging.

3) charge on the plates of capacitor as a function of time.

4) when steady state is reached, battery is removed from circuit. Find heat produced in all resistors after this instant.

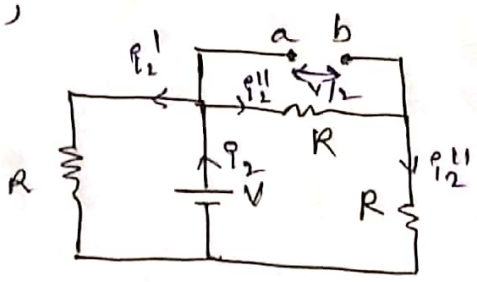
Sol 6

1) At $t=0$,



$$q_1 = q_1' + q_1'' = \frac{V}{R} + \frac{V}{R} = \frac{2V}{R}$$

At $t=\infty$,



$$q_2' = \frac{V}{R} \qquad q_2'' = \frac{V}{2R}$$

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$$i_2 = i_2' + i_2'' = \frac{3V}{2R}$$

here, $V_c = V_a - V_b$

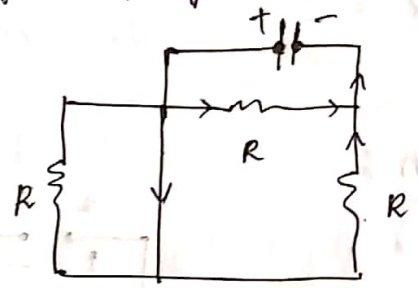
$$V_a - i_2'' R = V_b$$

$$V_a - V_b = i_2'' R = \frac{V}{2}$$

charge $q = \frac{CV}{2}$.

3) Time const,

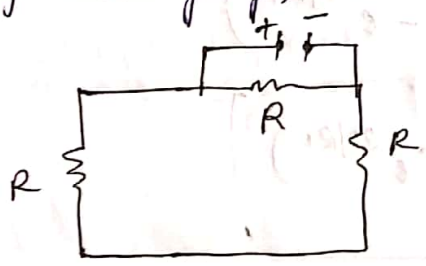
during charging,



$$R_{eff} = \frac{R}{2}$$

$$\boxed{T = \frac{RC}{2}}$$

during discharging,



$$R_{eff} = \frac{2R}{3}$$

$$\boxed{T = \frac{2RC}{3}}$$

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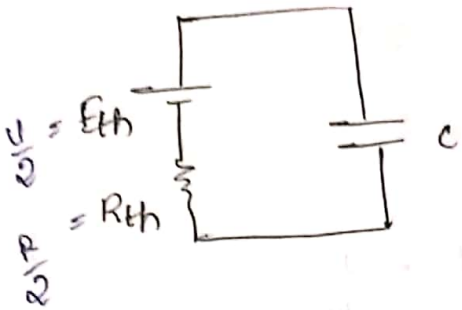
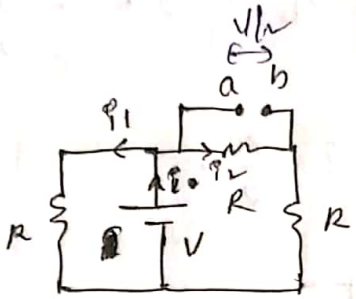
3) use thevinin's theorem.



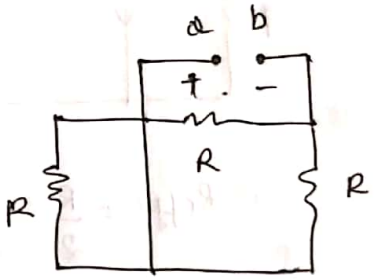
To find $\mathcal{E}_{th}$ &



$$q_2 = \frac{V}{2R}$$

$$V_a - V_b = q_2 R = \frac{V}{2}$$

To find R_{th} &



$$R_{eff} = R/2$$

$$q = q_0 (1 - e^{-t/RC})$$

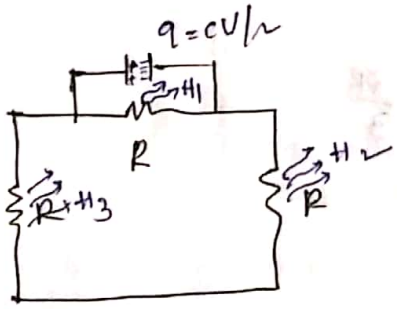
$$= \frac{CV}{2} (1 - e^{-t/RC/2})$$

$$\tau = R_{th} C$$

$$= \frac{RC}{2}$$

$$q = \frac{CV}{2} (1 - e^{-2t/RC})$$

4)



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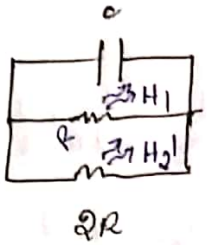
$$U = \frac{q^2}{2C} = \frac{q^2V^2}{4(2q)} \Rightarrow \boxed{U = \frac{CV^2}{8}}$$

$$U = H_1 + H_2 + H_3$$



$$H_2' = H_2 + H_3$$

$$H = \frac{V^2}{R}t$$

$$H_1 : H_2' = 2R : R = 2:1$$

$$H_1 + H_2' = U$$

$$H_1 = \frac{2}{3}U = \frac{2}{3} \times \frac{CV^2}{8} = \frac{CV^2}{12}$$

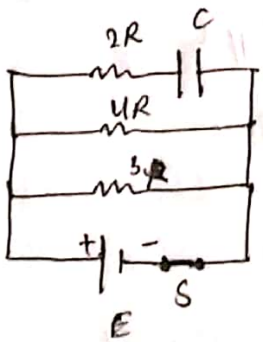
$$H_2' = \frac{1}{3}U = \frac{CV^2}{24}$$

$$H_2 + H_3 = \frac{CV^2}{24}$$

$$H_2 = H_3 \qquad H_2 = H_3 = \frac{CV^2}{48}$$

$$\therefore H_1 = \frac{CV^2}{12}, \; H_2 = H_3 = \frac{CV^2}{48}.$$

Ex 6



Find

1) current through battery at $t=0$ & $t=\infty$.
2) Time constant during charging and discharging.
3) Charge on capacitor as a function of time.

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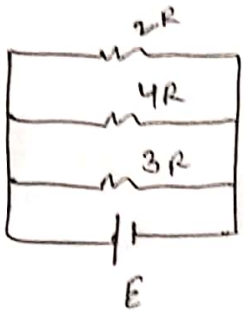
4) when steady state is attained, switch is opened. Find heat energy produced in all resistors.

Sol 6

1) at $t = 0$



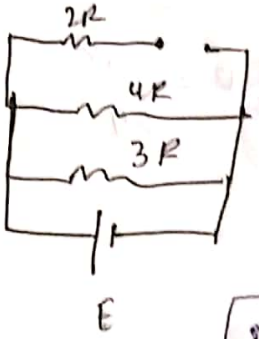
$$\text{Reff} \Rightarrow \frac{1}{R_e} = \frac{1}{2R} + \frac{1}{3R} + \frac{1}{4R}$$

$$R_e = \frac{12R}{13}$$

$$E = i R_{eff}$$

$$\Rightarrow i = \frac{E}{12R/13} = \frac{13E}{12R}$$

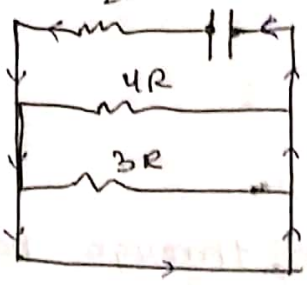
at $t = \infty$.



$$R_{eff} = \frac{3 \times 4}{3+4} R = \frac{12R}{7}$$

$$\boxed{i = \frac{7E}{12R}}$$

Q) during charging,



$$T = R_{eff}\, C$$

$$= 2R C$$

during discharging,

remove battery from circuit.

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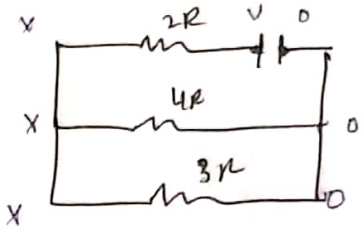
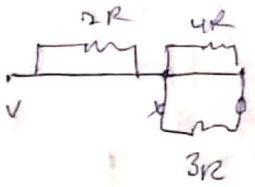




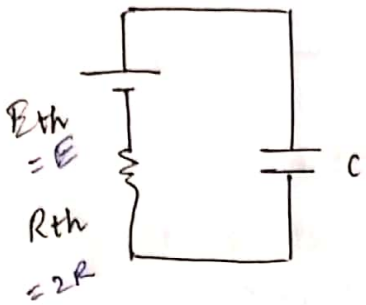
$R_{eff} = \frac{12R}{7} + 2R$

$= \frac{26R}{7}$.

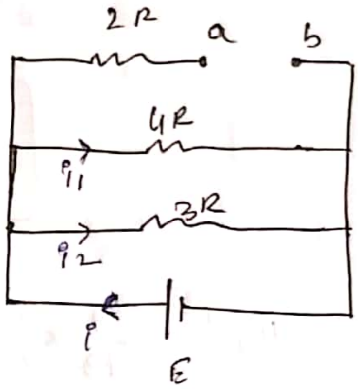
$T = R_{eff} C$

$= \frac{26RC}{7}$

3)



To find E_{Th}:



$i = \frac{7E}{12R}$

$i_1 : i_2 = 3:4$

$i_1 = \frac{3}{7} \times \frac{7E}{12R} = \frac{E}{4R}$

$i_2 = \frac{4}{7} \times \frac{7E}{12R} = \frac{E}{3R}$

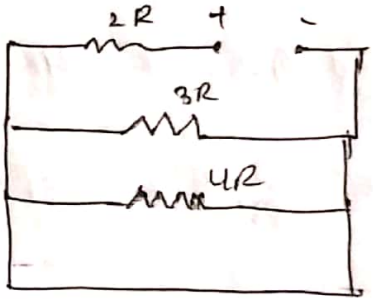
$V_a - 4R i_1 = V_b$

$V_a - V_b = 4R \times \frac{E}{4R}$

$V_a - V_b = E$

$\boxed{V_c = E}$

To find R_{Th}:



$T = 2RC$

$q = I_0(1 - e^{-t/T})$

$\boxed{q = CE\ (1 - e^{-t/2RC})}$

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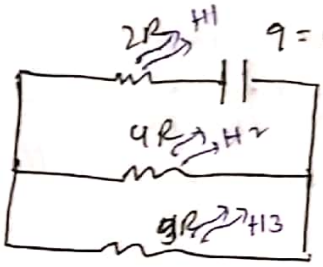
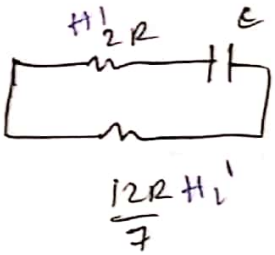
4)



$$U = \frac{q^2}{2c} = \frac{1}{2}\frac{C^2E^2}{C} = \frac{1}{2}CE^2$$



$$H_1 : H_2' = \frac{12}{7} : 2$$

$$= 6 : 7$$

$$H_1 = \frac{6}{13} \times \frac{1}{2} CE^2$$

$$\boxed{H_1 = \frac{3CE^2}{13}}$$

$$H_2' = \frac{7}{13} \times \frac{1}{2} CE^2 = \frac{7}{26} CE^2$$

$$H_2 : H_3 = 3 : 4$$

$$H_2 = \frac{3}{7} H_2' = \frac{3}{26} CE^2$$

$$H_3 = \frac{4}{7} H_2' = \frac{4}{26} CE^2$$

∴ heat produced in 2R, 4R, 3R resistors are $\frac{3}{13} CE^2$, $\frac{3}{26} CE^2$, $\frac{4}{26} CE^2$ respectively.

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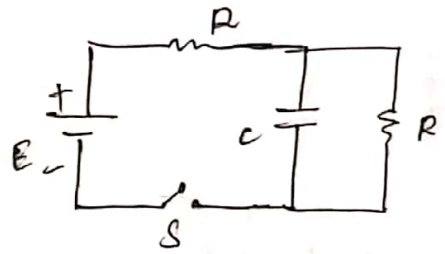
Ex6

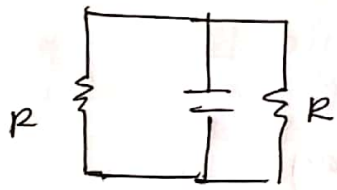


when switch is closed, Find

1) Time constant during charging and discharging
2) Current through battery at $t=0$ & $t=\infty$
3) charge on capacitor as a function of time.
4) Find the max value of rate of energy stored in the capacitor.

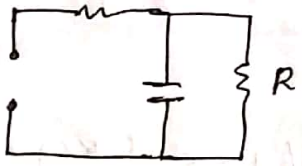
Sol6

1) during charging,



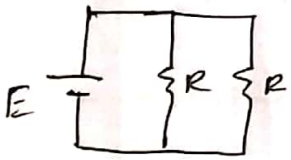
$T = R_{eff}\, C = \frac{R}{2} C$

during discharging,



$T = R_{eff}\, C = RC$

2) at $t = 0$



$q = \frac{E}{R/2}$

$$\boxed{q = \frac{2E}{R}}$$

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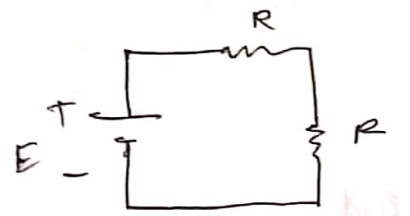
at $t = \infty$



$R_{eff} = 2R$

$$\boxed{i = \frac{E}{2R}}$$

3)

$$q = q_0 (1 - e^{-t/T})$$

$$= q_0 (1 - e^{-t/RC/2})$$

$$\boxed{q = q_0 (1 - e^{-2t/RC})}$$

4)

$$U_C = \frac{q^2}{2C}$$

$$= \frac{q_0^2 (1 - e^{-2t/RC})^2}{2C}$$

$$= \frac{q_0^2}{2C} (1 - e^{-2t/RC})^2$$

$$\frac{dU_C}{dt} = \frac{q_0^2}{2C} (2)(1 - e^{-2t/RC})(-e^{-2t/RC} \cdot -\frac{2}{RC})$$

$$= \frac{q_0^2}{2C} \times \frac{4}{RC} (e^{-2t/RC} - e^{-4t/RC})$$

$$\frac{d^2U_C}{dt^2} = 0$$

$$\frac{q_0^2}{2C} \times \frac{4}{RC} \times \left[-\frac{2}{RC} e^{-2t/RC} + e^{-4t/RC} \cdot -\frac{4}{RC} \right] = 0$$

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$$e^{-2t/RC} \cdot \frac{2}{RC} = e^{-4t/RC} \cdot \frac{4}{RC}$$

$$e^{2t/RC} = 2$$

$$P = \frac{dU_C}{dt} = \frac{q_0^2}{2C} \times \frac{4}{RC}\left(e^{-2t/RC} - e^{-4t/RC}\right)$$

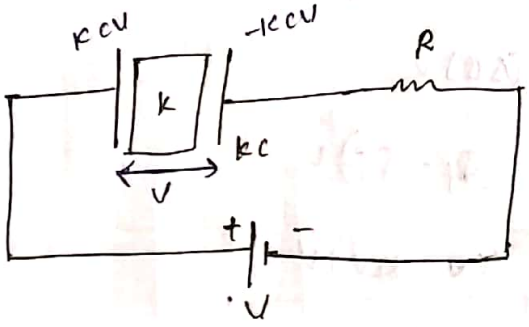
$$= \frac{2q_0^2}{RC^2}\left(\frac{1}{2} - \frac{1}{4}\right)$$

$$= \frac{2q_0^2}{RC^2}\left(\frac{1}{4_2}\right) = \frac{q_0^2}{RC^2}\left(\frac{1}{2}\right) = \frac{E^2}{2R}$$

$$= \left(\frac{i_0 R}{2}\right)^2 \times \frac{1}{2R} \Rightarrow \boxed{P = \frac{i_0^2 R}{8}}$$

Ex: An air filled capacitor of capacitance C is completely filled with a dielectric material of dielectric const K & then capacitor is connected to any ideal battery of emf V in series with resistor of resistance R. after steady state is reached the dielectric material inside the capacitor is suddenly removed. After this instant, Find total heat produced in the resistor.

Sol:



$$U = \frac{1}{2} KC V^2$$

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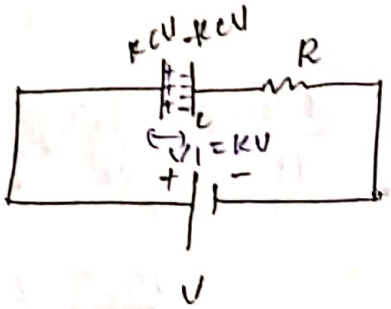
Now, dielectric is removed quickly.

* due to this charge on the plates will not change but capacity of capacitor changes from kc to c · bco3

$$i = \frac{dq}{dt \to 0}$$



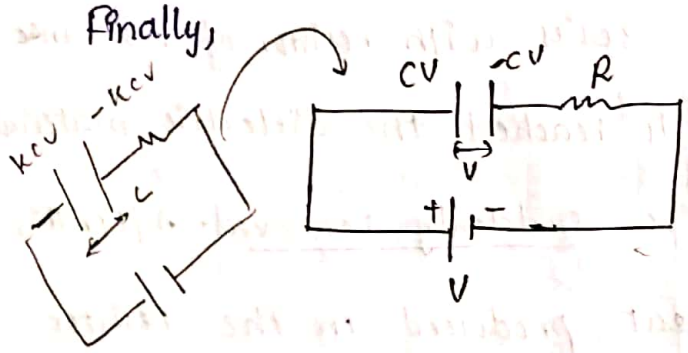
$$U_i = \frac{1}{2} C (KV)^2$$

$$U_i = \frac{1}{2} k^2 C V^2$$

$$\boxed{U_i^\circ = kU}$$

here, W_{ent} = +ve while removing dielectric quickly so, Pot-energy stored in capacitor increases from $\frac{1}{2}kcv^2$ to $\frac{1}{2}k^2cv^2$.

Finally,



$$U_f = \frac{1}{2} CV^2$$

* work done by battery,

$$W_b = (\Delta Q) V$$

$$= (Q_f - Q_i^\circ) V$$

$$= (CV - KCV) V$$

$$\boxed{W_b = -(K-1) CV^2}$$

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we have,

$$U_i + W_b + W_{ext}^{0} = U_f + H$$

$$\frac{1}{2}k^2CV^2 + -(k-1)CV^2 = \frac{1}{2}CV^2 + H$$

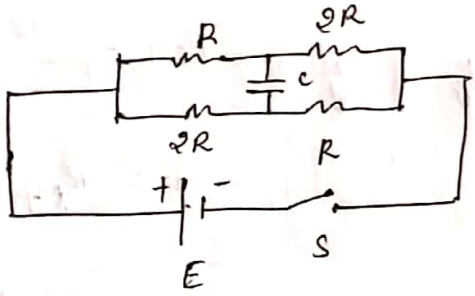
$$\frac{1}{2}k^2CV^2 - (k-1)CV^2 - \frac{1}{2}CV^2 = H$$

$$\frac{1}{2}CV^2\left(k^2 - 2k + 1 - 1\right) = H$$

$$\frac{(k-1)^2CV^2}{2} = H$$

$$\boxed{H = \frac{1}{2}CV^2(k-1)^2}$$

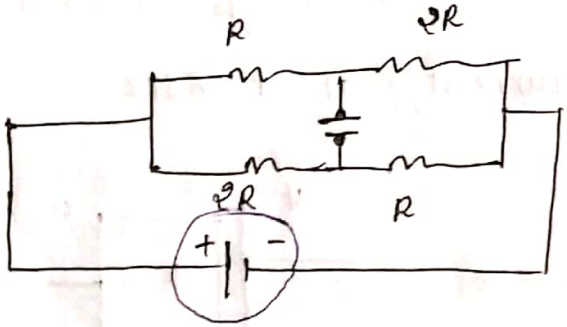
Ex6



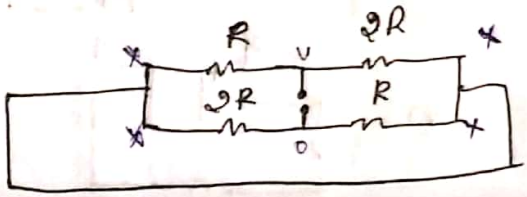
Find time constant of the given circuit when switch is closed & switch is opened.

Sol6

* when switch is closed – charging.



=>



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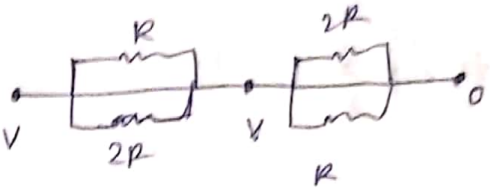
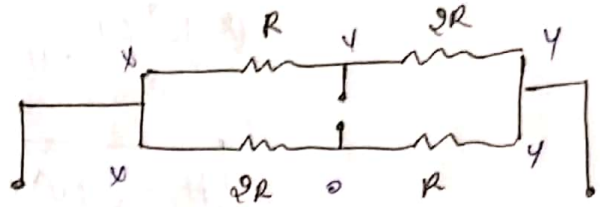


$$R_{eff} = \frac{2R}{3} + \frac{2R}{3} = \frac{4R}{3}$$

$$\tau = R_{eff}\, C = \frac{4RC}{3}.$$

- when switch is opened - discharging



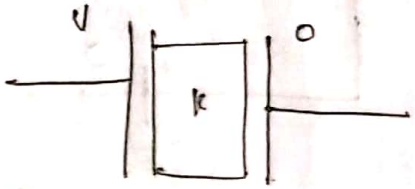
$$R_{eff} = \frac{3R}{2}$$

$$\tau = R_{eff}\, C = \frac{3RC}{2}.$$

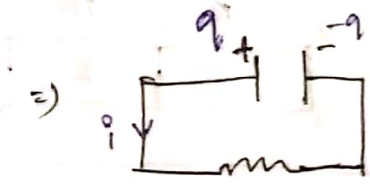
9/05/25 | Friday.

Ex: A leakage parallel plate capacitor is completely filled with a material of dielectric const $K = 5$ and electric conductivity $\sigma = 7.4 \times 10^{-12}\,\Omega^{-1}m^{-1}$. If the charge on the plates at $t = 0$ is $q_0 = 8.85\,\mu C$. Find value of leakage current at $t = 5$ sec.

Sol:



$K = 5$.



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we have, $i = \frac{V}{R} = \frac{q}{CR} = \frac{q_0(e^{-t/RC})}{CR}$

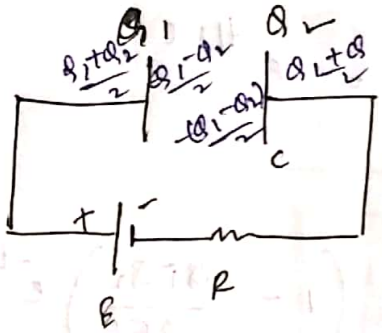
$C = \frac{\varepsilon_0 K A}{d}$, $R = \frac{\rho d}{A}$

$RC = \varepsilon_0 K \rho = \frac{\varepsilon_0 K}{\sigma} = \frac{8.85 \times 10^{-12} \times 5}{7.4 \times 10^{-12}}$

$i = \frac{8.85 \times 10^{-6}}{\frac{8.85 \times 10^{-12} \times 5}{7.4 \times 10^{-12}}} \left[e^{-5/RC}\right]$

$i =$

Ex:

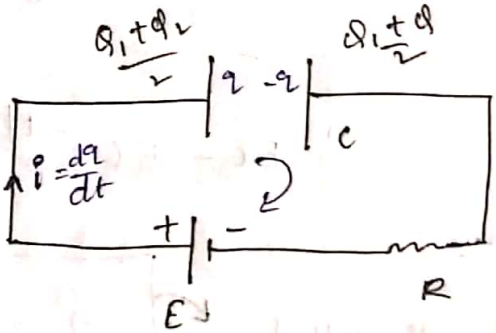


Q_1 & Q_2 are the initial charges on plates. Now these plates are connected to battery through resistance 'R'. Find charge on the plate as a function of time.

Sol:

$V_i = \frac{Q_1 - Q_2}{2C}$

Suppose $E \to V_i$



$i = \frac{d}{dt}\left[q - \left(\frac{Q_1 - Q_2}{2}\right)\right]$

$\boxed{i = \frac{dq}{dt}}$

$E - \frac{q}{C} - iR = 0$

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$$\frac{EC-q}{C} = R\frac{dq}{dt}$$

$$\frac{dq}{dt} = \frac{EC-q}{CR}$$

$$\int_{\frac{q_1-q_2}{2}}^{q} \frac{dq}{EC-q} = \int_{t=0}^{t=t} \frac{dt}{CR}$$

$$\left[\ln(EC-q)\right]_{\frac{q_1-q_2}{2}}^{q} = \frac{-t}{CR}$$

$$\ln\left(\frac{EC-q}{EC-\left(\frac{q_1-q_2}{2}\right)}\right) = \frac{-t}{RC}$$

$$\frac{EC-q}{EC-\left(\frac{q_1-q_2}{2}\right)} = e^{-t/RC}$$

$$1-\frac{q}{EC} = \left(1-\frac{q_1-q_2}{2EC}\right)e^{-t/RC}$$

$$1-\left(1-\frac{q_1-q_2}{2EC}\right)e^{-t/RC} = \frac{q}{EC}$$

$$EC - EC\,e^{-t/RC} + \left(\frac{q_1-q_2}{2}\right)e^{-t/RC} = q$$

$$\boxed{q = CE\left(1-e^{-t/RC}\right) + \left(\frac{q_1-q_2}{2}\right)e^{-t/RC}}$$

This is called principle of Superposition

$CE\left(1-e^{-t/RC}\right)$ ⇓ charging the empty capacitor with battery

$\left(\frac{q_1-q_2}{2}\right)e^{-t/RC}$ ⇓ decay of capacitor through resistance without battery

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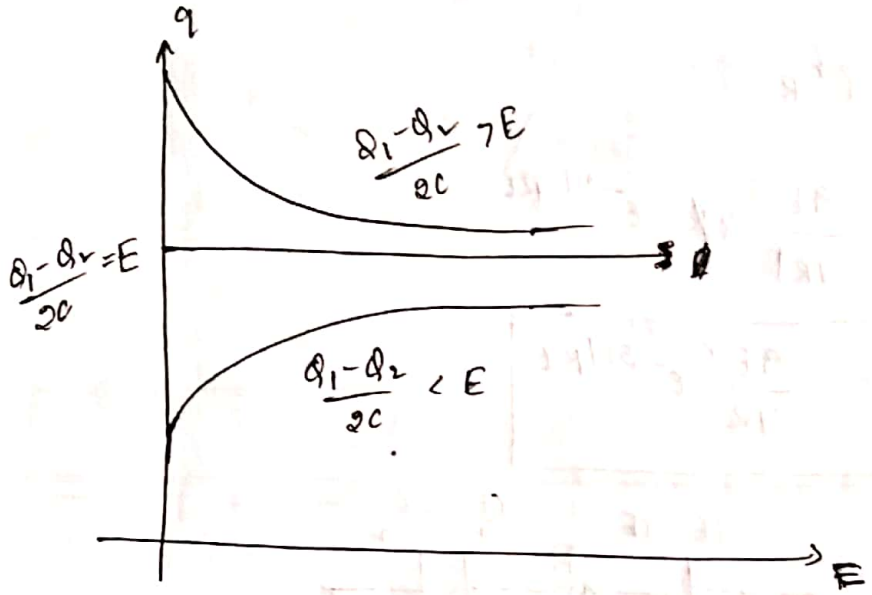
Points

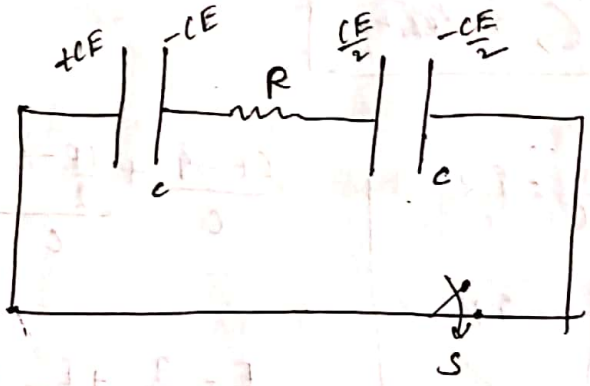
* If $\frac{Q_1 - Q_2}{2C} < E$, charge flows from battery to capacitor plate.

* If $\frac{Q_1 - Q_2}{2C} > E$, charge flows from capacitor to battery.

* If $\frac{Q_1 - Q_2}{2C} = E$, there is no charge flow.



Exe

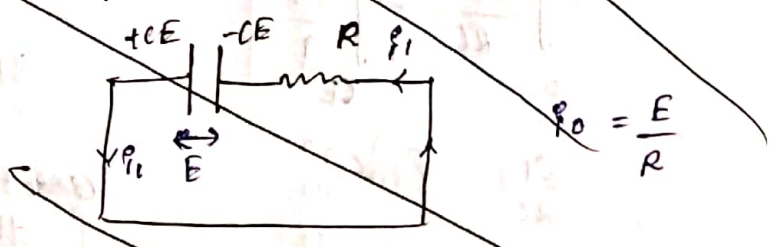


Switch is closed at $t=0$. Find power dissipated in R as a function of time.

Sol

M-I :

Principle of Superposition



$$i = i_0 (e^{-t/RC}) = \frac{E}{R}(e^{-t/RC})$$

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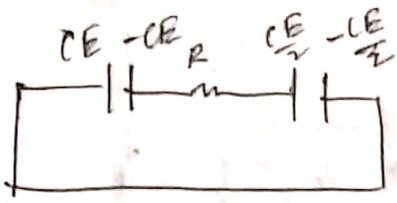
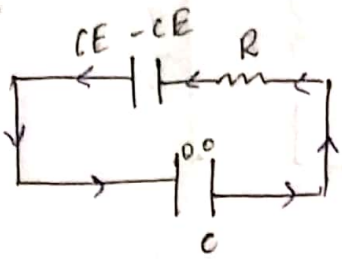
$$\frac{-t}{2RC} = \left[\ln(3EC - 4q)\right]_{CE}^{CE-q}$$

$$\frac{-t}{2RC} = \ln\left(\frac{3EC - 4(CE-q)}{CE}\right)$$

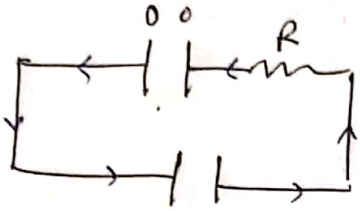
Sol:



M-I:



$$i_1 = \frac{E}{R} e^{-2t/RC}$$



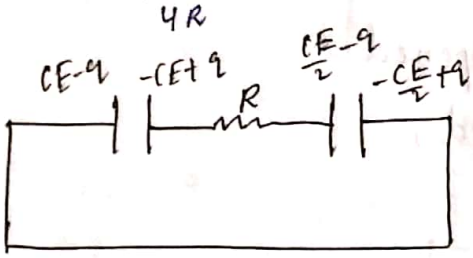
$$i_2 = \frac{E}{2R} e^{-2t/RC}$$

$-\frac{CE}{2}$ $+\frac{CE}{2}$

$$i = i_1 + i_2 \Rightarrow \boxed{i = \frac{3E}{2R} e^{-2t/RC}}$$

$$P = i^2 R = \frac{9E^2}{4R} e^{-4t/RC}$$

M-II:



$$\frac{CE-q}{C} + \frac{\frac{CE}{2}-q}{C} - iR = 0$$

$$\frac{3E}{2} - \frac{2q}{C} = iR$$

$$\frac{3EC-4q}{2CR} = \frac{dq}{dt}$$

$$\int_0^q \frac{dq}{3EC-4q} = \int_0^t \frac{dt}{2CR} \Rightarrow q = \frac{3CE}{4}(1 - e^{-2t/RC})$$

$$\Rightarrow i = \frac{dq}{dt} \Rightarrow i = \frac{3E}{2R} e^{-2t/RC}$$

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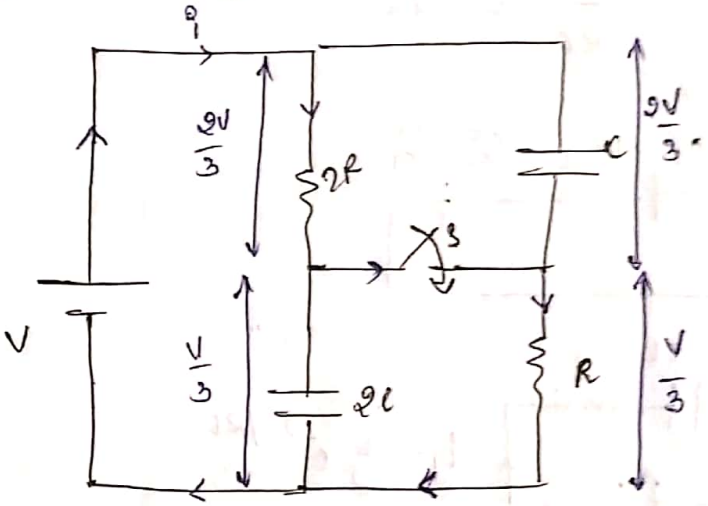
Ex: Switch is closed for a long time. Now switch is opened at $t=0$. Find current through battery as a function of time.



Sol: when switch is closed,

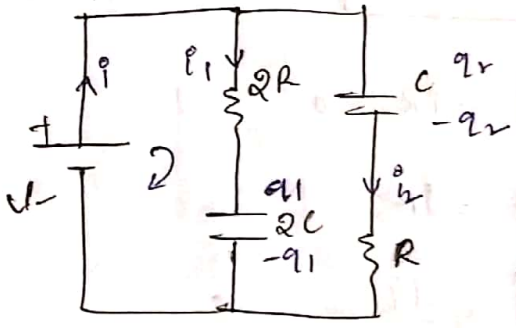
$$i = \frac{V}{3R}$$

$$V_{2R} = \frac{V}{3R}(2R) = \frac{2V}{3}$$

$$V_R = \frac{V}{3R}(R) = \frac{V}{3}$$

At $t=0$, switch is opened.

when switch is opened,



$$V - i_1(2R) - \frac{q_1}{2C} = 0$$

on differentiation,

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$$0 - 2R\frac{di_1}{dt} - \frac{1}{2C}\frac{dq_1}{dt} = 0$$

$$+2R\frac{di_1}{dt} = -\frac{i_1}{2C}$$

$$\int_{V/3R}^{i_1} \frac{di_1}{i_1} = -\int_0^t \frac{dt}{4RC}$$

Initial current,

$$V - i_1(2R) - \frac{Q_0}{3(2C)} = 0$$

$$V - \frac{V}{3} = i_1(2R)$$

$$\boxed{i_1 = \frac{V}{3R}}$$

$$\ln\left(i_1 / V/3R\right) = \frac{-t}{4RC}$$

$$\boxed{i_1 = \frac{V}{3R}e^{-t/4RC}}$$

M-II:

Loop $V - i_1(2R) - \frac{q_1}{2C} = 0$

$$\frac{2CV - q_1}{2C} = i_1(2R)$$

$$\boxed{\frac{2CV - q_1}{2C} = 2R\cdot\frac{dq}{dt}}$$

$$\frac{2CV - q_1}{2C} = 2R\frac{dq}{dt}$$

$$\int_0^t \frac{dt}{4RC} = \int_{\frac{2CV}{3}}^{q_1} \frac{dq}{2CV - q_1}$$

$$\frac{-t}{4RC} = \ln\left(\frac{2CV - q_1}{2CV - \frac{2CV}{3}}\right)$$

$$e^{-t/4RC} = \frac{2CV - q}{\frac{4CV}{3}}$$

$$q = 2CV - \frac{4CV}{3}e^{-t/4RC}$$

$$i_1 = \frac{dq}{dt} = +\frac{4CV}{3} \times e^{-t/4RC} \cdot \left(+\frac{1}{4RC}\right)$$

$$\boxed{i_1 = \frac{V}{3R}e^{-t/4RC}}$$

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$$V - i_2 R - \frac{q_2}{C} = 0$$

$$\frac{CV - q_2}{C} = i_2 R$$

$$\frac{CV - q_2}{CR} = \frac{dq_2}{dt}$$

$$\int_0^t \frac{dt}{CR} = \int_{\frac{CV}{3}}^{q_2} \frac{dq_2}{CV - q_2}$$

$$\frac{-t}{CR} = \ln\left(\frac{CV - q_2}{CV - \frac{CV}{3}}\right)$$

$$e^{-t/CR} = \frac{CV - q_2}{\frac{2CV}{3}}$$

$$\frac{2CV}{3} e^{-t/CR} = CV - q_2$$

$$q_2 = CV - \frac{2CV}{3} \cdot e^{-t/CR}$$

$$i_2 = \frac{dq_2}{dt} = +\frac{2V}{3} \times e^{-t/CR} \cdot \frac{+1}{R}$$

$$\boxed{i_2 = \frac{2V}{3R} e^{-t/CR}}$$

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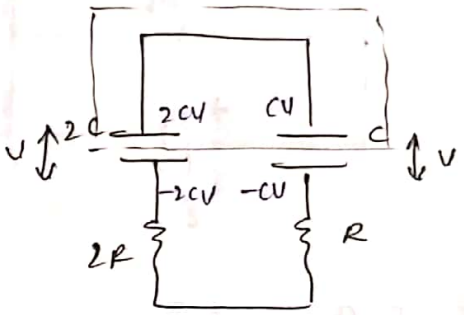
** when steady state is reached, remove the battery then



Total charge = 2CV + CV

= 3CV

common potential,

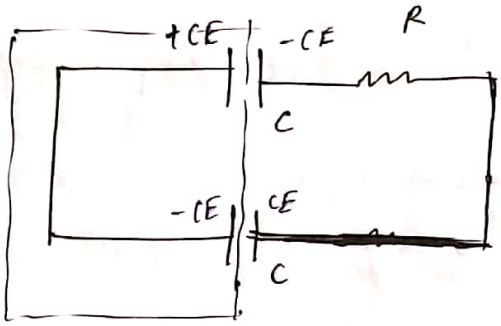
$$V = \frac{\text{total charge}}{\text{total capacitance}}$$

$$= \frac{3CV}{3C} = V$$

∴ charge flowing through circuit = 0.

Heat produced in resistors = 0.

Ex6



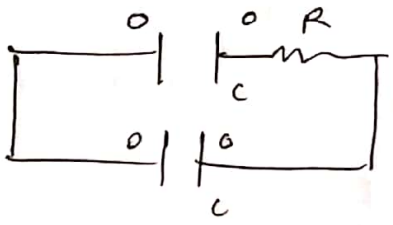
Find total heat produced in resistance.

Sol6

Total charge = CE − CE = 0

common potential = 0

$$U_i = \frac{1}{2}CE^2 + \frac{1}{2}CE^2 = CE^2$$



$U_f = 0$

$$U_i + \cancel{W_b}^{0} + \cancel{W_{ext}}^{0} = \cancel{U_f}^{0} + H \Rightarrow \boxed{H = U_i = CE^2}$$

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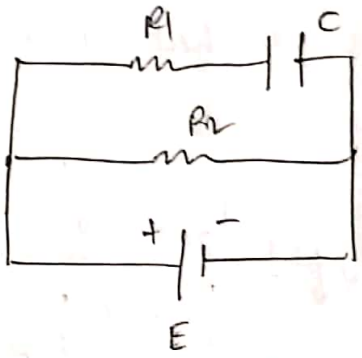
Ex6



At $t=0$,

current through battery

$$i = \frac{E}{R_1 R_2}(R_1+R_2)$$

At $t=\infty$,

$$i = \frac{E}{R_2}$$

here,

$$q_C = CE$$

& $U_C = \frac{1}{2}CE^2$.

At any time t,

$q_1 = \frac{E}{R_1} e^{-t/R_1 C}$, decreases with time.

& $i_2 = \frac{E}{R_2}$ = const at all times.

$$i = i_1 + i_2$$

$$= \frac{E}{R_1} e^{-t/R_1 C} + \frac{E}{R_2}$$

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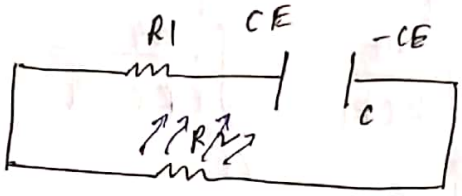
a) when steady state is reached, battery is removed from circuit,



charge on plates,

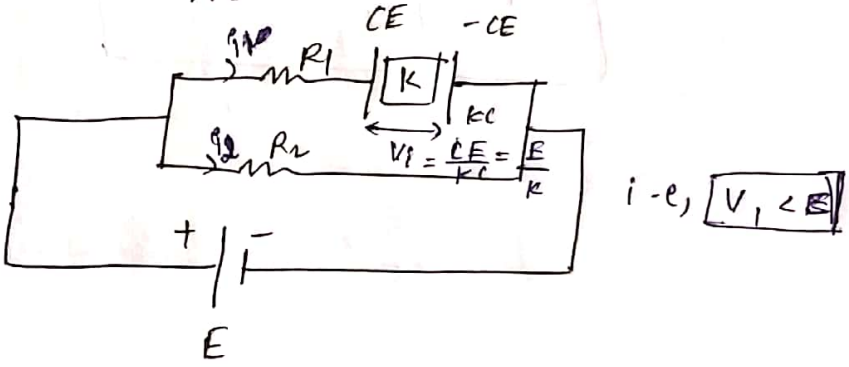
$$q = CE\, e^{-t/(R_1+R_2)C}$$

$$U = H_1 + H_2 = \frac{1}{2} CE^2$$

$$\frac{H_1}{H_2} = \frac{i^2 R_1 t}{i^2 R_2 t} = \frac{R_1}{R_2}$$

solve for H_1 & H_2.

b) without removing the battery if dielectric is quickly inserted b/w the plates then, charge on plates remains same as before but capacity change from c to Kc.



i-e, $V_1 < E$

In this case,

$$i_2 = \frac{E}{R_2} = \text{const}$$

At any time t,

$$E - i_1 R_1 - \frac{q}{KC} = 0$$

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$$\frac{EKC - q}{KCR_1} = \frac{dq}{dt}$$

$$\int_0^t \frac{dt}{KCR_1} = \int_{CE}^{q} \frac{dq}{EKC - q}$$

$$\frac{-t}{KCR_1} = \ln\left(\frac{EKC - q}{EC(K-1)}\right)$$

$$e^{-t/KCR_1} = \frac{EKC}{EC(K-1)} - \frac{q}{EC(K-1)}$$

$$\frac{q}{EC(K-1)} = \frac{K}{K-1} - e^{-t/KCR_1}$$

$$q = KEC - EC(K-1)\, e^{-t/KCR_1}$$

$$q = EC\left(K - (K-1)\, e^{-t/KCR_1}\right)$$

$$i = \frac{dq}{dt} = EC\left(+(K-1)\, e^{-t/KCR_1}\right)\left(+\frac{1}{KCR_1}\right)$$

$$\boxed{i = \frac{(K-1)E}{KR_1}\, e^{-t/KCR_1}}$$

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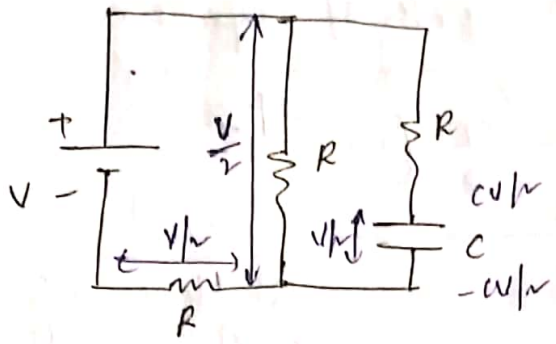
Ex6



At $t=0$,

current through battery,

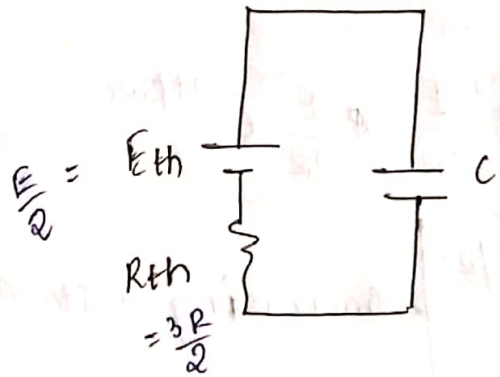
$$i = \frac{V}{R + R/2} = \frac{2V}{3R}$$

At $t=\infty$,

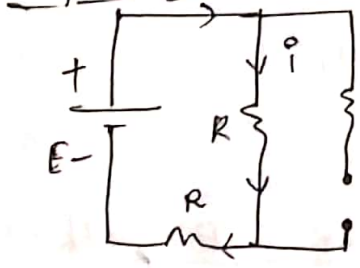
current through battery,

$$\boxed{i = \frac{V}{2R}}$$

At any time t,



To find Eth:



$$i = \frac{E}{2R}$$

$$V_a - V_b = E_{th} = \frac{E}{2R} \times (R) = \frac{E}{2}$$

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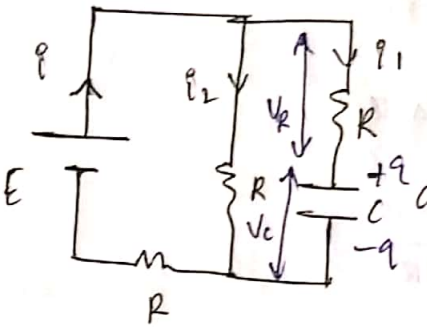
now, charge on the plates of capacitor,

$$q = \frac{CE}{2}\left(1 - e^{-2t/3RC}\right)$$

Original circuit,



$$q = \frac{CE}{2}\left(1 - e^{-2t/3RC}\right)$$

$$i = i_1 + i_2$$

$$i_1 = \frac{dq}{dt} = \frac{CE}{2} \cdot \left(+e^{-2t/3RC}\right)\left(+\frac{2}{3RC}\right)$$

$$= \frac{E}{3R}\left(e^{-2t/3RC}\right)$$

$$V_R = i_1 R = \frac{E}{3} e^{-2t/3RC}$$

$$V_C = \frac{q}{C} = \frac{E}{2}\left(1 - e^{-2t/3RC}\right)$$

$$i_2 R = V_R + V_C$$

$$i_2 = \frac{E}{3R} e^{-2t/3RC} + \frac{E}{2R} - \frac{E}{2R} e^{-2t/3RC}$$

$$i_2 = \frac{E}{2R} - \frac{E}{6R} e^{-2t/RC}$$ increasing with time.

now, current through battery,

$$i = i_1 + i_2$$

$$= \frac{E}{3R}\left(e^{-2t/3RC}\right) + \frac{E}{2R} - \frac{E}{6R} e^{-2t/3RC}$$

Here $\frac{dq}{dt}$ ≠ current through battery

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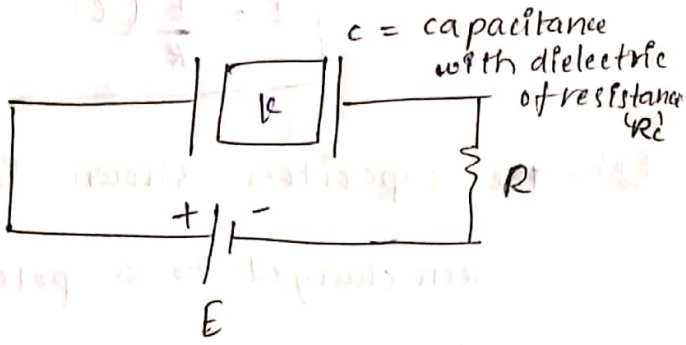
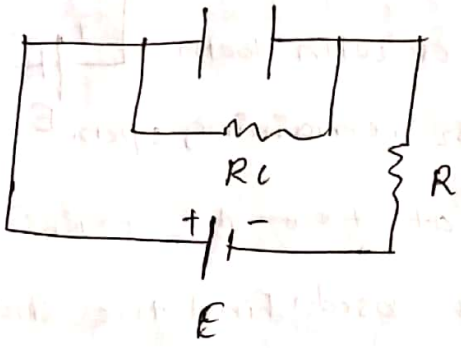
$$i = \frac{E}{6R} e^{-2t/3R_c} + \frac{E}{2R}$$ ← decrease with time.

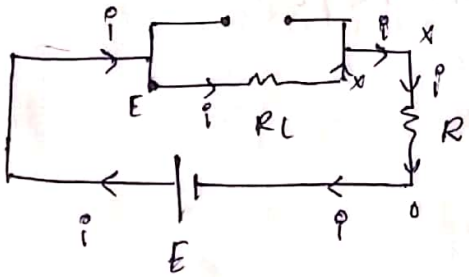
Ex: Find 1) charge on the plates of capacitor as a function of time.

2) current through battery as a function of time

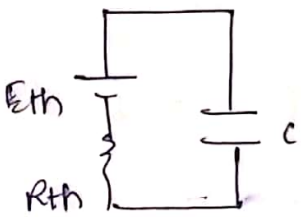


Sol:





$$i = \frac{E}{R+R_c}$$



$$E_{th} = \frac{ER_c}{R+R_c}$$

$$R_{th} = \frac{R \cdot R_c}{R+R_c}$$

$$q = q_0 (1 - e^{-t/R_{th}C})$$

$$= E_{th} C (1 - e^{-t/R_{th}C})$$

$$\frac{ERC}{R+R_c} (1 - e^{-t/(R+R_c)C})$$

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$$q = \frac{ERC}{R_c+R}\left(1 - e^{-\frac{(R_c+R)t}{R_cRC}}\right)$$

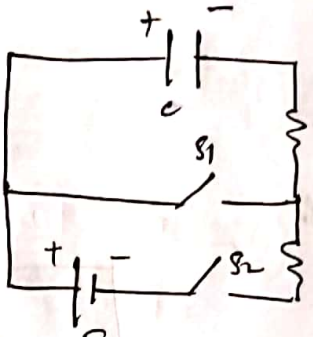
$$i = \frac{dq}{dt}$$

$$= \frac{ERC}{R_c+R}\left(-e^{-\frac{(R_c+R)t}{RR_cC}}\right)\left(-\frac{R_c+R}{RR_cC}\right)$$

$$\boxed{i = \frac{E}{R}\left(e^{-\frac{(R_c+R)t}{RR_cC}}\right)}$$

10|05|25|Saturday

Ex: The capacitors shown in figure has been charged to a potential difference of V volts, so that it carries a charge CV with both the switches S_1 & S_2 remaining open.



Switch S_1 is closed at $t=0$. At $t=R_1C$ switch S_1 is opened and S_2 is closed. Find the charge on the capacitor at $t = 2R_1C + R_2C$.

Sol:

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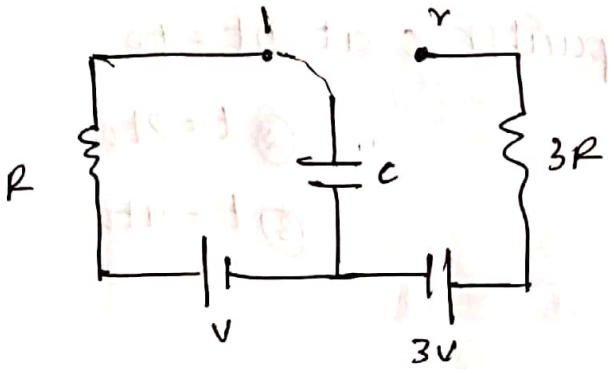
Ex 1



In the circuit shown initially switch S is at 1 till C is fully charged. Now S is transformed to 2. then Find (at t=0)

1) what is initial current in the circuit. $\frac{2V}{3R}$

2) what is steady state charge on capacitor. $3CV$

3) work done by 3V battery. $6CV^2$

4) what is the heat developed in the circuit. $2CV^2$.

5) current in circuit as a function of time t.

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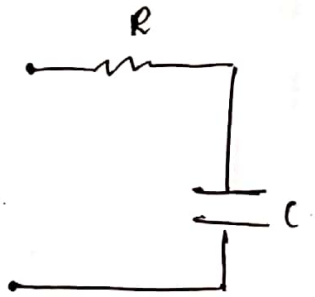
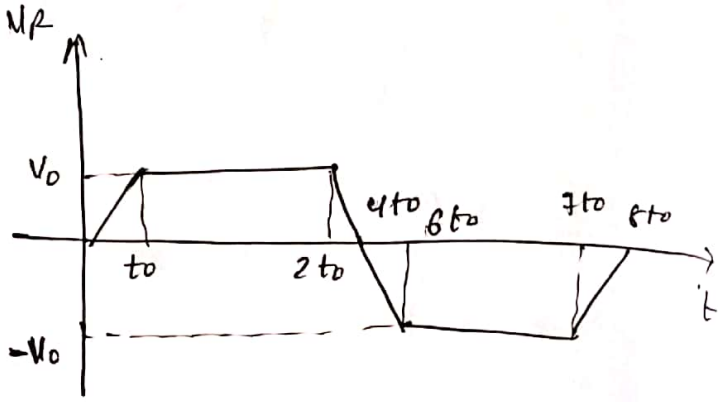
Ex:





Find charge on capacitor c at ① $t = t_0$

② $t = 2t_0$

③ $t = 4t_0$

Sol:

* b/w $t = 0$ to $t_0 \Rightarrow y = mx$

$$V = \frac{V_0}{t_0} t$$

$$iR = \frac{V_0}{t_0} t$$

$$i = \frac{V_0}{Rt_0} t$$

$$\frac{dq}{dt} = \frac{V_0}{Rt_0} t$$

$$\int_0^q dq = \frac{V_0}{Rt_0} \int_0^t t \cdot dt$$

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$$q = \frac{V_0}{Rt_0}\frac{t^2}{2}$$

$$\boxed{q = \frac{V_0}{2Rt_0}t^2}$$ parabola.

At $t = t_0$

$$q = \frac{V_0}{2R_0 t_0}t_0^2 = \frac{V_0 t_0}{2R}$$

* b/w $t = t_0$ to $2t_0$

$$V_R = V_0$$

$$iR = V_0$$

$$i = \frac{V_0}{R}$$

$$\frac{dq}{dt} = \frac{V_0}{R}$$

$$\int_{\frac{V_0 t_0}{2R}}^{q} dq = \frac{V_0}{R}\int_{t=t_0}^{t=t} dt$$

$$q - \frac{V_0 t_0}{2R} = \frac{V_0}{R}(t - t_0)$$

$$q = \frac{V_0 t}{R} - \frac{V_0 t_0}{R} + \frac{V_0 t_0}{2R}$$

$$\boxed{q = \frac{V_0 t}{R} - \frac{V_0 t_0}{2R}}$$

at $t = 2t_0$

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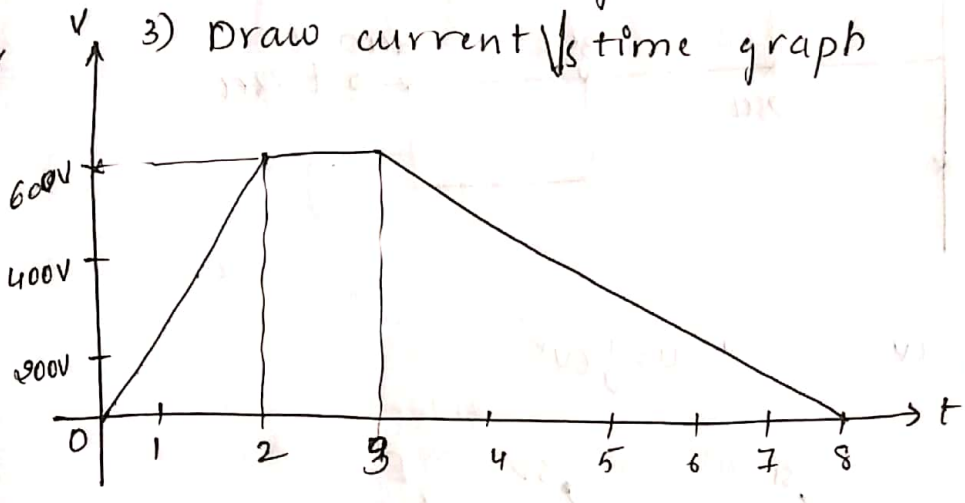
Ex! The voltage applied across a capacitor having a capacitance of 10μF is varies as shown. Calculate –

1) the charge

2) Energy stored in the capacitor when the terminal voltage is 600V.

3) Draw current Vs time graph



Sol^n we have,

$$q = cv$$

$$i = \frac{dq}{dt}$$

$$\boxed{i = c\frac{dv}{dt}}$$

b/w $0 \le t \le 2$ sec

$$\frac{dv}{dt} = \frac{600}{2} = 300 = \text{const}$$

$$i = 10 \times 300 = 3000\mu A = 3mA$$

b/w $2 \le t \le 3$

$$\frac{dv}{dt} = 0 \qquad i = c\frac{dv}{dt} = 0$$

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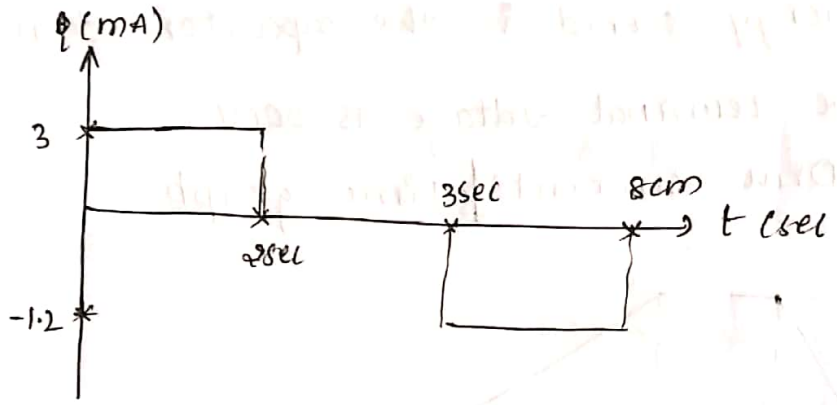
b/w $3 \le t \le 8$ sec

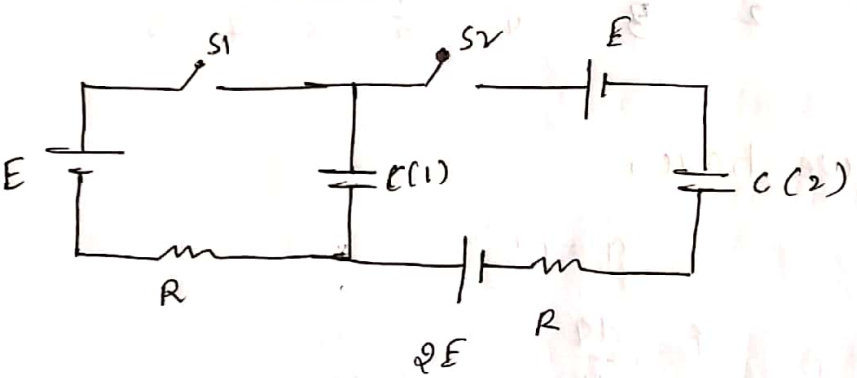
$$\frac{dv}{dt} = -\frac{600}{5} = -120$$

$$i = -10 \times 120 = -1200\,\mu A = -1.2\,mA$$



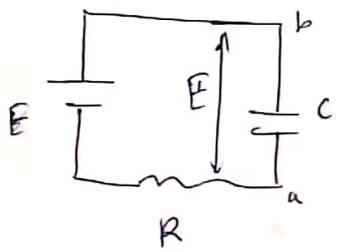
* $q = CV$ * $U = \frac{1}{2}CV^2$

Ex!



First S_1 is closed and when steady state is reached for C(1), S_1 is opened and S_2 is closed at $t = 0$.

1) Just after closing switch (S_2), current through S_2 is



$V_a - iR + E = V_b$

$V_a - V_b = -E + iR$

$V_b - V_a = E - iR$

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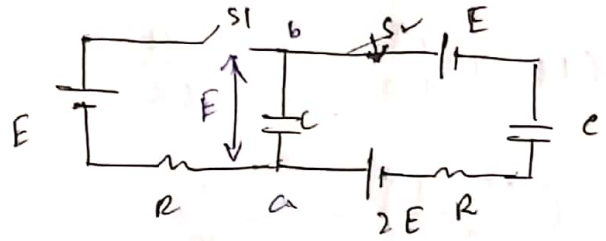


$$V_a + \cancel{E} - \cancel{E} - iR + 2E = V_a$$

$$iR = 2E$$

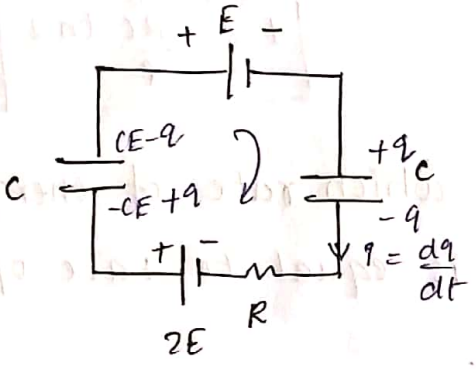
$$\boxed{i = \frac{2E}{R}}$$

2) Time constant.

$$C_{eff} = \frac{C}{2}$$

$$\boxed{T = \frac{RC}{2}}$$

3) The time at which charge on both capacitors is equal.



$$-E - \frac{q}{C} - iR + 2E + \frac{CE-q}{C} = 0.$$

$$E - \frac{2q}{C} + E = iR$$

$$\frac{2(CE-q)}{CR} = i$$

$$\frac{dq}{dt} = \frac{2(CE-q)}{CR}$$

$$\int_0^q \frac{dq}{2[E-q]} = \int_0^t \frac{dt}{CR}$$

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$$\left[\ln (CE-q)\right]_0^q = -\frac{2t}{CR}$$

$$\ln\left(\frac{CE-q}{CE}\right) = -\frac{2t}{CR}$$

$$1-\frac{q}{CE} = e^{-2t/RC}$$

$$\boxed{CE\left(1-e^{-2t/RC}\right) = q}$$

$CE-q = q$

$CE = 2q$

$$1-e^{-2t/RC} = \frac{1}{2}$$

$$\frac{1}{2} = e^{-2t/RC}$$

$$e^{2t/RC} = 2$$

$$2t/RC = \ln 2$$

$$\boxed{t = \frac{RC\ln 2}{2}}$$

4) The time at which rate of energy dissipation in q_3 is equal to rate of energy stored in $C(2)$.

$$P = V_C i$$

$$\frac{CE-q}{C} i = \frac{q}{C} i$$

$$CE = 2q$$

$$\boxed{t = \frac{RC}{2}\ln 2}$$

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